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Entangling Unravellings for Quantum State Engineering and Quantum Metrology

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A coloro che mi hanno insegnato a non mollare mai

True happiness is to enjoy the present, without anxious dependence upon the future, not to amuse ourselves with either hopes or fears but to rest satisfied with what we have, which is sufficient, for he that is so wants nothing. The greatest blessings of mankind are within us and within our reach. A wise man is content with his lot, whatever it may be, without wishing for what he has not.
- Seneca

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Introduction

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1

Introduction to Quantum Mechanics

1.1 From classical physics to quantum mechanics

By the end of the nineteenth century, classical physics seemed to offer a complete and convincing description of the natural world. Analytical mechanics, Maxwell's electrodynamics, and classical statistical physics formed a remarkably successful framework, capable of explaining a vast range of phenomena. Within this picture, the state of a physical system at a given time was assumed to determine, at least in principle, its future evolution. Observables were regarded as quantities possessing definite values, while the role of theory was to describe their behaviour through deterministic laws.

This confidence began to weaken when physicists turned their attention to microscopic phenomena. A number of experimental results, such as black-body radiation [1] and the photoelectric effect [2] could not be satisfactorily explained within the classical framework. These were not minor discrepancies that could be fixed by adjusting a few assumptions. Rather, they suggested that the classical picture itself was no longer adequate at the atomic scale.

The first attempts to deal with this crisis already required a significant break from classical intuition. Planck's introduction of energy quanta, Einstein's interpretation of the photoelectric effect, and Bohr's model of the hydrogen atom all pointed in the same direction. They represented an intermediate stage, in which classical concepts were still partially retained while new quantization rules were introduced in a rather phenomenological way [3].

A coherent formulation emerged only in the second half of the 1920s, with the development of matrix mechanics [4, 5] and wave mechanics [6]. Although these approaches looked very different at first, they were soon recognized as mathematically equivalent descriptions of the same physical theory. This marked the real birth of quantum mechanics as a general framework. From that point, the issue was no longer how to repair classical physics but to confirm that microscopic systems required a different language.

In classical physics, the state of a system is ideally viewed as a complete specification of its physical properties while in quantum mechanics, this idea has to be revised. The state no longer provides, in general, definite values for all observables before a measurement is performed. Instead, it encodes the possible outcomes of a measurement and the statistical structure relating them.

In classical statistical physics, probabilities usually arise because one lacks complete in-

formation about the system. The probabilistic description is therefore associated with ignorance about an underlying state of affairs that is assumed to be well defined. It is totally different in quantum mechanics, where the probabilistic content is not merely a practical limitation of the observer but is built into the theory itself [7].
The conceptual consequences become even more striking when composite systems are considered. In the quantum framework, the state of a global system cannot always be reduced to independent states of its constituents. This gives rise to entanglement, one of the most distinctive features of the quantum realm. Quantum correlations are not simply stronger versions of classical correlations but they reflect a different logical structure, in which the whole cannot always be understood as a collection of independently defined parts.
What originally emerged from the study of atomic phenomena eventually became the foundation of several modern research areas. Quantum optics, many-body physics, quantum information, and quantum metrology all rely on genuinely non-classical features such as coherence, interference, and entanglement.
This perspective is especially relevant for the present thesis. The analysis of collective quantum properties in systems of many two-level atoms, and in particular the study of non-classical correlations and spin-squeezed states, belongs to the conceptual framework opened by quantum mechanics.

1.2 Postulates of quantum mechanics for closed and isolated systems

Having introduced the conceptual transition from classical physics to quantum mechanics, we now turn to the standard formulation of the theory[8, 9]. In its textbook form, quantum mechanics is first presented as a theory for closed and isolated systems. This setting is idealized, yet it provides the natural starting point to define the basic notions of state, observable, measurement, composition of systems, and time evolution.
This formulation is not the most general one. In particular, it is based on pure states, projective measurements, and unitary dynamics. These assumptions are fully adequate for introducing the theory in its simplest and most transparent form, but they will later need to be generalized when discussing realistic situations in which a system interacts with external degrees of freedom. For the purposes of the present chapter, however, the standard formulation is exactly the one we need. It allows us to establish the conceptual backbone of quantum mechanics before moving, in the next chapter, to the theory of open quantum systems.

Postulate 1 : States

To every closed physical system there corresponds a Hilbert space $\mathcal{H}$. The pure states of the system are represented by normalized vectors $|\psi\rangle \in \mathcal{H}$, namely

$$\langle\psi|\psi\rangle = 1. \tag{1.1}$$

More precisely, a physical pure state is not identified with a single vector, but with a ray

in Hilbert space, since two vectors that differ only by an overall phase describe the same physical situation.
Indeed, if $|\psi\rangle \in \mathcal{H}$ is a normalized state vector, then for any real phase $\alpha \in \mathbb{R}$ the vector

$$|\psi'\rangle = e^{i\alpha} |\psi\rangle \tag{1.2}$$

represents the same physical pure state. This is the invariance of pure states under the action of the group $U(1)$ of global phase transformations.
The reason is that all measurable predictions remain unchanged under the replacement $|\psi\rangle \to e^{i\alpha} |\psi\rangle$.
An equivalent way to state the same fact is to observe that the rank-one projector associated with the pure state is invariant under a global phase transformation:

$$|\psi'\rangle\langle\psi'| = e^{i\alpha} |\psi\rangle \langle\psi| e^{-i\alpha} = |\psi\rangle\langle\psi| \,. \tag{1.3}$$

For this reason, pure states are properly identified with rays in Hilbert space, or equivalently with rank-one projectors.
If $|\psi_1\rangle$ and $|\psi_2\rangle$ are admissible states, then any normalized linear combination of them is again an admissible state. This introduces one of the most characteristic features of quantum theory, namely the superposition principle. This has no direct analogue in the classical description. It is the mathematical origin of interference phenomena and, more generally, of the non-classical structure of the theory. Quantum mechanics is insensitive to a common $U(1)$ phase multiplying the whole vector, but it is sensitive to relative phases between different components of a superposition.

Postulate 2 : Observables
Every observable quantity is associated with a self-adjoint operator acting on the Hilbert space of the system $\mathcal{H}$. If the observable is denoted by $\hat{X}$, its possible measurement outcomes are the eigenvalues appearing in its spectral decomposition. For the case of a discrete spectrum, one writes

$$\hat{X} = \sum_x x\, \hat{P}_x, \tag{1.4}$$

where the real numbers x are the possible outcomes and the operators $\hat{P}_x$ are the orthogonal projectors onto the corresponding eigenspaces.
This postulate establishes the link between physical quantities and operator theory: the fact that observables are represented by self-adjoint operators ensures, on one hand, that the possible outcomes are real, and on the other hand that the spectral theorem may be used to connect the formal object $\hat{X}$ with experimentally accessible values. For this reason, even when one speaks of the measurement of an observable, the truly relevant objects are not only its eigenvalues, but also the corresponding projectors.

Postulate 3 : Measurements and conditional states
If the system is prepared in the pure state $|\psi\rangle$ and one measures the observable $\hat{X}$, quantum mechanics does not in general predict a single certain result. Instead, it assigns probabilities to the possible outcomes. For a discrete observable, the probability of obtaining the value x is given by the Born rule

$$p(x) = \langle\psi|\hat{P}_x|\psi\rangle \,. \tag{1.5}$$

Correspondingly, the expectation value of the observable reads

$$\left\langle \hat{X} \right\rangle = \langle\psi|\hat{X}|\psi\rangle \, . \tag{1.6}$$

The Born rule is one of the central ingredients of the entire formalism. It is the rule that connects the mathematical description of the state with the statistical predictions of the theory. Once the state and the observable are assigned, the Born rule determines the probability distribution of the measurement outcomes. In this sense, it is the point at which the formalism becomes experimentally meaningful.
The standard formulation also includes a rule for the state change induced by an ideal projective measurement. If the outcome x is obtained, the state immediately after the measurement is given by

$$|\psi\rangle \longrightarrow |\psi_x\rangle = \frac{\hat{P}_x |\psi\rangle}{\sqrt{\langle\psi|\hat{P}_x|\psi\rangle}}. \tag{1.7}$$

It expresses the fact that the state after the measurement is the normalized projection of the initial state onto the eigenspace corresponding to the observed result.

Postulate 4 : Dynamics
In the absence of noise and measurements, the evolution of a closed and isolated quantum system is unitary. If the state of the system at time t_0 is $|\psi(t_0)\rangle$, then at time t it is given by

$$|\psi(t)\rangle = \hat{U}(t, t_0) \, |\psi(t_0)\rangle \, , \tag{1.8}$$

where $\hat{U}(t, t_0)$ is a unitary operator, namely

$$\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0) = \hat{U}(t, t_0)\hat{U}^\dagger(t, t_0) = \mathbb{1}. \tag{1.9}$$

If the dynamics is generated by a Hamiltonian $\hat{H}$, the same evolution may be written in differential form through the Schrödinger equation

$$i\frac{d}{dt} |\psi(t)\rangle = \hat{H} \, |\psi(t)\rangle \, . \tag{1.10}$$

Once the initial state and the Hamiltonian are given, the state at any later time is uniquely determined. In this respect, quantum mechanics preserves a form of deterministic evolution, but only between measurements. The probabilistic character of the theory enters when one connects the evolving state to the outcomes of observations through the Born rule.
It is also worth emphasizing that unitarity guarantees that the normalization of the state is preserved in time, and thus that total probability remains equal to one throughout the evolution of an isolated system.

Postulate 5 : Composite systems
If a physical system is composed of two subsystems, described respectively by the Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, then the composite system is described on the tensor-product space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \tag{1.11}$$

More generally, a multipartite system is described by the tensor product of the Hilbert spaces associated with all its constituents.
This postulate is of fundamental importance, it is precisely this structure that makes entanglement possible. Indeed, not every state in the global Hilbert space can be written as a product of states of the individual subsystems. Therefore, the description of the whole cannot in general be reduced to independent descriptions of its parts.
In the classical picture, correlations may be interpreted as reflecting incomplete information about local properties that each subsystem already possesses. In the quantum case, the tensor-product structure allows for correlations that are genuinely non-classical. For this reason, the postulate on composite systems is not just a formal extension of the single-system case. It is the starting point for the entire theory of quantum correlations. For the whole thesis, this postulate will be important. Since our interest lies in collective properties of many two-level systems, all the relevant notions of multipartite state, collective observable, and entanglement rely directly on the tensor-product description.

These assumptions are entirely appropriate for a first formulation of the theory and will be sufficient for the developments of the present chapter. However, they are not general enough to describe realistic situations in which the system interacts with an external environment, or in which measurements are not ideal projective measurements. For this reason, they should be regarded here as the natural starting point of the discussion. In Chapter 2, we will move to a more general formulation based on density operators, generalized measurements, and open-system dynamics.

1.3 States and observables

The postulates introduced in the previous section provide the conceptual skeleton of quantum mechanics. We now discuss in more detail the mathematical objects entering that formulation. In particular, we clarify the notions of physical system, quantum state, and observable, and we introduce the Dirac notation that will be used throughout the thesis [9, 10].

1.3.1 Hilbert spaces and Dirac notation

In the standard formulation adopted in this chapter, a physical system is associated with a Hilbert space $\mathcal{H}$. For the purposes of the present discussion, and consistently with the framework that will be used in the following sections, we mainly refer to finite-dimensional Hilbert spaces and to observables with discrete spectrum. This is the natural setting for the description of qubits and, more generally, of finite collections of two-level systems.
A Hilbert space is a complex vector space endowed with an inner product. Therefore, if $|\psi\rangle, |\phi\rangle \in \mathcal{H}$ and $\alpha, \beta \in \mathbb{C}$, then $\alpha |\psi\rangle + \beta |\phi\rangle \in \mathcal{H}$.
We adopt the Dirac notation throughout. A vector in $\mathcal{H}$ is denoted by a ket $|\psi\rangle$ and the corresponding dual vector is denoted by a bra $\langle\psi| = (|\psi\rangle)^\dagger$. The inner product between two vectors $|\phi\rangle$ and $|\psi\rangle$ is written as $\langle\phi|\psi\rangle \in \mathbb{C}$. In the physics convention, the inner

product is linear in the second entry and antilinear in the first one, namely

$$\langle\phi|\alpha\psi+\beta\chi\rangle=\alpha\langle\phi|\psi\rangle+\beta\langle\phi|\chi\rangle\,, \tag{1.12}$$

while

$$\langle\alpha\phi+\beta\chi|\psi\rangle=\alpha^*\langle\phi|\psi\rangle+\beta^*\langle\chi|\psi\rangle\,. \tag{1.13}$$

Moreover, one has $\langle\phi|\psi\rangle=\langle\psi|\phi\rangle^*$ and $\langle\psi|\psi\rangle\geq 0$, with equality if and only if $|\psi\rangle=0$. The norm induced by the inner product is $\|\psi\|=\sqrt{\langle\psi|\psi\rangle}$. Physical pure states are represented by normalized vectors, namely vectors satisfying $\langle\psi|\psi\rangle = 1$. As already discussed, the physical content does not change under multiplication by a global phase. Hene, the pure state is properly associated with the ray generated by $|\psi\rangle$ rather than with the vector itself.

Let now $\{|e_n\rangle\}_{n=1}^{d}$ be an orthonormal basis of $\mathcal{H}$. By orthonormality,

$$\langle e_m|e_n\rangle=\delta_{mn}, \tag{1.14}$$

where δ_{mn} is the Kroneckr delta. The completeness of the basis is expressed by the resolution of the identity

$$\sum_{n=1}^{d}|e_n\rangle\langle e_n|=\mathbb{1}. \tag{1.15}$$

Using Eq. (1.15), any state vector may be expanded as

$$|\psi\rangle=\sum_{n=1}^{d}|e_n\rangle\langle e_n|\psi\rangle=\sum_{n=1}^{d}c_n|e_n\rangle\,, \tag{1.16}$$

where the expansion coefficients are $c_n=\langle e_n|\psi\rangle$, and therefore $\langle\psi|=\sum_{n=1}^{d}c_n^*\langle e_n|$. If $|\psi\rangle$ is normalized, then Eqs. (1.14) and (1.16) imply

$$1=\langle\psi|\psi\rangle=\sum_{n=1}^{d}|c_n|^2. \tag{1.17}$$

This expression already shows the probabilistic interpretation of the coefficients once a basis has been fixed. In particular, when the chosen basis is the eigenbasis of an observable, the moduli squared of the coefficients become the probabilities of the corresponding measurement outcomes through the Born rule.

1.3.2 Linear operators, outer products, and adjoints

Besides vectors, quantum mechanics makes use of linear operators acting on $\mathcal{H}$. If $\hat{A}$ is a linear operator, then $\hat{A}\left(\alpha|\psi\rangle+\beta|\phi\rangle\right)=\alpha\hat{A}|\psi\rangle+\beta\hat{A}|\phi\rangle$. Given two vectors $|\phi\rangle$ and $|\psi\rangle$, one may construct the outer product $|\phi\rangle\langle\psi|$, which is itself a linear operator and acts on an arbitrary vector $|\chi\rangle$ as

$$\left(|\phi\rangle\langle\psi|\right)|\chi\rangle=|\phi\rangle\langle\psi|\chi\rangle\,. \tag{1.18}$$

In particular, the operator $|\psi\rangle\langle\psi|$ is the rank-one projector associated with the pure state $|\psi\rangle$. For a normalized state, it satisfies $\left(|\psi\rangle\langle\psi|\right)^2 = |\psi\rangle\langle\psi|$ and $\mathrm{Tr}(|\psi\rangle\langle\psi|) = 1$.

This operator representation will become especially useful in the next chapter, where the density-operator formalism will be introduced in full generality.
Let us now consider the matrix representation of an operator. Once a basis $\{|e_n\rangle\}_{n=1}^{d}$ is fixed, the matrix elements of $\hat{A}$ are defined as $A_{mn} = \langle e_m|\hat{A}|e_n\rangle$. Using the completeness relation twice, the operator may be reconstructed as

$$\hat{A} = \sum_{m,n=1}^{d} A_{mn}\, |e_m\rangle\langle e_n| \,. \tag{1.19}$$

Therefore, once a basis is chosen, vectors and operators admit the familiar column-vector and matrix representations of linear algebra. Dirac notation is simply a more flexible and basis-independent way of encoding the same information.
The adjoint of an operator $\hat{A}$ is defined by the relation

$$\langle\phi|\hat{A}|\psi\rangle = \left\langle \hat{A}^\dagger\phi \middle| \psi \right\rangle, \tag{1.20}$$

or equivalently by

$$\langle\phi|\hat{A}|\psi\rangle = \langle\psi|\hat{A}^\dagger|\phi\rangle^* \,. \tag{1.21}$$

An operator is said to be self-adjoint if

$$\hat{A}^\dagger = \hat{A}. \tag{1.22}$$

In the finite-dimensional setting considered in this chapter, the terms self-adjoint and Hermitian may be used interchangeably.
Self-adjoint operators play a distinguished role in quantum mechanics because observables are represented precisely by operators of this kind. A first important property is that their expectation values are always real. Indeed, if $\hat{A}^\dagger = \hat{A}$, then

$$\langle\psi|\hat{A}|\psi\rangle^* = \langle\psi|\hat{A}^\dagger|\psi\rangle = \langle\psi|\hat{A}|\psi\rangle \,, \tag{1.23}$$

hence $\langle\psi|\hat{A}|\psi\rangle \in \mathbb{R}$.
A second key property is that the eigenvalues of a self-adjoint operator are real. If $\hat{A}\,|a\rangle = a\,|a\rangle$, then

$$a\,\langle a|a\rangle = \langle a|\hat{A}|a\rangle = \langle a|\hat{A}|a\rangle^* = a^*\,\langle a|a\rangle \,, \tag{1.24}$$

and therefore $a = a^*$.
Finally, eigenvectors corresponding to distinct eigenvalues are orthogonal. If $\hat{A}\,|a\rangle = a\,|a\rangle$ and $\hat{A}\,|b\rangle = b\,|b\rangle$ with $a \neq b$, then

$$a\,\langle a|b\rangle = \left\langle \hat{A}a \middle| b \right\rangle = \left\langle a \middle| \hat{A}b \right\rangle = b\,\langle a|b\rangle \,, \tag{1.25}$$

so that $(a-b)\,\langle a|b\rangle = 0$, namely $\langle a|b\rangle = 0$.
In finite dimension, these properties are summarized by the spectral theorem: every self-adjoint operator is diagonalizable in an orthonormal basis and may be written as

$$\hat{A} = \sum_a a\,\hat{P}_a, \tag{1.26}$$

where the coefficients $a \in \mathbb{R}$ are the eigenvalues and the operators $\hat{P}_a$ are the orthogonal projectors onto the corresponding eigenspaces. In the nondegenerate case, one simply has

$$\hat{A} = \sum_a a \, |a\rangle\langle a| \, . \tag{1.27}$$

For this reason, self-adjoint operators provide exactly the mathematical structure needed to represent observables.

1.3.3 Observables and spectral decomposition

We are now in the position to discuss observables more concretely. Let $\hat{X}$ be a self-adjoint operator acting on $\mathcal{H}$. In the finite-dimensional case, the spectral theorem guarantees that $\hat{X}$ may be diagonalized and written in the form

$$\hat{X} = \sum_x x \, \hat{P}_x, \tag{1.28}$$

where $x \in \mathbb{R}$ are the eigenvalues of the observable and $\hat{P}_x$ are the orthogonal projectors onto the corresponding eigenspaces. These projectors satisfy $\hat{P}_x \hat{P}_{x'} = \delta_{xx'} \hat{P}_x$ and $\sum_x \hat{P}_x = \mathbb{1}$. If the spectrum is nondegenerate, each eigenspace is one-dimensional and one may write

$$\hat{P}_x = |x\rangle\langle x| \, , \qquad \hat{X} = \sum_x x \, |x\rangle\langle x| \, , \tag{1.29}$$

with $\hat{X} \, |x\rangle = x \, |x\rangle$.
The expectation value of $\hat{X}$ in the pure state $|\psi\rangle$ is

$$\left\langle \hat{X} \right\rangle_\psi = \langle \psi | \hat{X} | \psi \rangle \, . \tag{1.30}$$

By inserting the spectral decomposition (1.28), one finds

$$\left\langle \hat{X} \right\rangle_\psi = \sum_x x \, \langle \psi | \hat{P}_x | \psi \rangle \, . \tag{1.31}$$

The coefficients $p(x) = \langle \psi | \hat{P}_x | \psi \rangle$ are non-negative and sum to one, and therefore define a probability distribution over the possible outcomes. This is the precise way in which the state vector determines the statistics of the observable.
A second quantity of central importance is the variance of $\hat{X}$ in the state $|\psi\rangle$,

$$\left(\Delta^2 \hat{X} \right)_\psi = \left\langle \hat{X}^2 \right\rangle_\psi - \left\langle \hat{X} \right\rangle_\psi^2 . \tag{1.32}$$

The corresponding standard deviation is simply $\left(\Delta \hat{X} \right)_\psi = \sqrt{\left(\Delta^2 \hat{X} \right)_\psi}$. These quantities characterize the spread of the outcomes around the mean value and will become particularly relevant later on, when discussing collective observables and the reduction of quantum fluctuations below suitable reference levels.

It is also useful to comment briefly on compatibility of observables. If two observables $\hat{X}$ and $\hat{Y}$ commute, namely

$$[\hat{X}, \hat{Y}] = 0, \tag{1.33}$$

then, in the finite-dimensional setting, they admit a common eigenbasis, up to the usual care required in the presence of degeneracies. In that case, the two observables may be simultaneously diagonalized and can be jointly sharp in the same states. When the commutator does not vanish, such a common diagonalization is in general not possible. This is the algebraic origin of quantum incompatibility and, more generally, of the fact that different experimental arrangements may probe genuinely different aspects of the same physical system.

1.4 Time evolution in quantum mechanics

We now discuss in more detail the dynamical postulate introduced in the previous section. For a closed and isolated quantum system, the time evolution is described by a family of unitary operators acting on the Hilbert space of the system. This is the natural quantum analogue of reversible evolution in the absence of external interactions.

1.4.1 Time-evolution operator and its general properties

Let $|\psi(t_0)\rangle$ be the state of the system at time t_0. Its state at time t is written as

$$|\psi(t)\rangle = \hat{U}(t, t_0)\, |\psi(t_0)\rangle\,, \tag{1.34}$$

where $\hat{U}(t, t_0)$ is the time-evolution operator from t_0 to t. Since the evolution is unitary, one has

$$\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0) = \hat{U}(t, t_0)\hat{U}^\dagger(t, t_0) = \mathbb{1}. \tag{1.35}$$

As a consequence, the norm of the state is preserved:

$$\langle\psi(t)|\psi(t)\rangle = \langle\psi(t_0)|\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0)|\psi(t_0)\rangle = \langle\psi(t_0)|\psi(t_0)\rangle\,. \tag{1.36}$$

More generally,

$$\langle\phi(t)|\psi(t)\rangle = \langle\phi(t_0)|\hat{U}^\dagger(t, t_0)\hat{U}(t, t_0)|\psi(t_0)\rangle = \langle\phi(t_0)|\psi(t_0)\rangle\,. \tag{1.37}$$

Hence, transition probabilities are invariant under the evolution.
The propagator $\hat{U}(t, t_0)$ must satisfy a consistency condition. If $t_0 \le t_1 \le t_2$, evolving first from t_0 to t_1 and ten from t_1 to t_2 must be equivalent to evolving directly from t_0 to t_2. Therefore,

$$\hat{U}(t_2, t_1)\hat{U}(t_1, t_0) = \hat{U}(t_2, t_0). \tag{1.38}$$

Moreover, $\hat{U}(t_0, t_0) = \mathbb{1}$ and, by combining Eqs. (1.35) and (1.38), one obtains $\hat{U}^{-1}(t, t_0) = \hat{U}(t_0, t) = \hat{U}^\dagger(t, t_0)$.

1.4.2 Infinitesimal evolution and Hamiltonian generator

The local structure of the dynamics is obtained by considering an infinitesimal time step. For a sufficiently regular propagator, one may write

$$\hat{U}(t+\mathrm{d}t, t) = \mathbb{1} - i\,\hat{G}(t)\,\mathrm{d}t + \mathcal{O}(\mathrm{d}t^2), \tag{1.39}$$

where $\hat{G}(t)$ is the generator of the evolution. Unitarity imposes a constraint on $\hat{G}(t)$. Indeed, from

$$\hat{U}^\dagger(t+\mathrm{d}t, t)\hat{U}(t+\mathrm{d}t, t) = \mathbb{1} \tag{1.40}$$

and Eq. (1.39), one finds

$$\left(\mathbb{1} + i\,\hat{G}^\dagger(t)\,\mathrm{d}t\right)\left(\mathbb{1} - i\,\hat{G}(t)\,\mathrm{d}t\right) = \mathbb{1} + i\left[\hat{G}^\dagger(t) - \hat{G}(t)\right]\mathrm{d}t + \mathcal{O}(\mathrm{d}t^2). \tag{1.41}$$

Therefore,

$$\hat{G}^\dagger(t) = \hat{G}(t). \tag{1.42}$$

The generator of a unitary evolution must thus be self-adjoint. It is therefore identified with the Hamiltonian of the system [11], namely $\hat{G}(t) = \hat{H}(t)$.
The same result may be written in a more global form. By differentiating the identity

$$\hat{U}(t+\mathrm{d}t, t_0) = \hat{U}(t+\mathrm{d}t, t)\hat{U}(t, t_0) \tag{1.43}$$

and using Eq. (1.39), one obtains

$$\hat{U}(t+\mathrm{d}t, t_0) = \left(\mathbb{1} - i\,\hat{H}(t)\,\mathrm{d}t\right)\hat{U}(t, t_0) + \mathcal{O}(\mathrm{d}t^2). \tag{1.44}$$

Subtracting $\hat{U}(t, t_0)$ from both sides, dividing by $\mathrm{d}t$, and taking the limit $\mathrm{d}t \to 0$, one finds

$$i\frac{\partial}{\partial t}\hat{U}(t, t_0) = \hat{H}(t)\hat{U}(t, t_0), \tag{1.45}$$

with initial condition

$$\hat{U}(t_0, t_0) = \mathbb{1}. \tag{1.46}$$

This is the evolution equation for the time-evolution operator. By applying it to the initial state $|\psi(t_0)\rangle$, one immediately obtains the Schrödinger equation, already introduced in Eq.(1.10)

$$i\frac{\mathrm{d}}{\mathrm{d}t}\left|\psi(t)\right\rangle = \hat{H}(t)\left|\psi(t)\right\rangle. \tag{1.47}$$

When the Hamiltonian is time independent, namely $\hat{H}(t) = \hat{H}$, the propagator depends only on the time difference, $\hat{U}(t, t_0) = \hat{U}(t - t_0)$. By introducing $\tau = t - t_0$, one has

$$\hat{U}(\tau_1)\hat{U}(\tau_2) = \hat{U}(\tau_1 + \tau_2), \tag{1.48}$$

together with $\hat{U}(0) = \mathbb{1}$. Hence, the time evolution defines a one-parameter unitary group.

At this point one may invoke Stone's theorem [12]. In the present setting, the theorem states that every strongly continuous one-parameter unitary group is generated by a unique self-adjoint operator $\hat{H}$ such that

$$\hat{U}(\tau) = e^{-i\hat{H}\tau}. \tag{1.49}$$

Conversely, every self-adjoint operator $\hat{H}$ defines a strongly continuous one-parameter unitary group through Eq. (1.49). This gives a precise mathematical meaning to the statement that the Hamiltonian is the generator of time translations.
Equation (1.49) is written in units such that $\hbar = 1$. If $\hbar$ is kept explicit, one has

$$\hat{U}(\tau) = e^{-i\hat{H}\tau/\hbar}. \tag{1.50}$$

Throughout the following we adopt the natural units: unless otherwise stated, we set $\hbar = 1$.
The self-adjointness of $\hat{H}$ guarantees the unitarity of the evolution, since $\hat{U}^\dagger(\tau) = \left(e^{-i\hat{H}\tau}\right)^\dagger = e^{i\hat{H}\tau}$ and therefore $\hat{U}^\dagger(\tau)\hat{U}(\tau) = \mathbb{1}$. In the time-independent case, Eq. (1.45) can be solved explicitly:

$$\hat{U}(t, t_0) = e^{-i\hat{H}(t-t_0)}, \tag{1.51}$$

so that

$$|\psi(t)\rangle = e^{-i\hat{H}(t-t_0)} |\psi(t_0)\rangle . \tag{1.52}$$

1.4.3 Time-dependent Hamiltonians and time ordering

The time-dependent case is slightly more delicate. Formally, Eq. (1.45) suggests the expression

$$\hat{U}(t, t_0) \stackrel{?}{=} \exp\left[-i\int_{t_0}^{t} \hat{H}(s)\,\mathrm{d}s\right]. \tag{1.53}$$

However, this formula is valid only when

$$[\hat{H}(t_1), \hat{H}(t_2)] = 0 \qquad \forall\, t_1, t_2. \tag{1.54}$$

Indeed, if Hamiltonians at different times do not commute, the exponential of the time integral is not sufficient to reproduce the correct ordered product of infinitesimal propagators.
In the general case, the formal solution is written as the time-ordered exponential

$$\hat{U}(t, t_0) = \mathcal{T} \exp\left[-i\int_{t_0}^{t} \hat{H}(s)\,\mathrm{d}s\right], \tag{1.55}$$

where $\mathcal{T}$ denotes chronological ordering. More precisely, for a product of Hamiltonians evaluated at different times, the action of $\mathcal{T}$ is defined by arranging the operators from left to right in decreasing time order. For instance,

$$\mathcal{T}\Big[\hat{H}(t_1)\hat{H}(t_2)\Big] = \begin{cases} \hat{H}(t_1)\hat{H}(t_2), & t_1 > t_2, \\ \hat{H}(t_2)\hat{H}(t_1), & t_2 > t_1, \end{cases} \tag{1.56}$$

and, more generally,

$$\mathcal{T}\Big[\hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n)\Big] = \hat{H}(t_{\pi(1)})\hat{H}(t_{\pi(2)})\cdots\hat{H}(t_{\pi(n)}), \tag{1.57}$$

where the permutation π is such that

$$t_{\pi(1)} \geq t_{\pi(2)} \geq \cdots \geq t_{\pi(n)}. \tag{1.58}$$

Equation (1.55) may be expanded as the Dyson series [13],

$$\hat{U}(t,t_0) = \mathbb{1} + \sum_{n=1}^{\infty}(-i)^n \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2 \cdots \int_{t_0}^{t_{n-1}} \mathrm{d}t_n\, \hat{H}(t_1)\hat{H}(t_2)\cdots\hat{H}(t_n). \tag{1.59}$$

The first terms are therefore

$$\hat{U}(t,t_0) = \mathbb{1} - i\int_{t_0}^{t} \hat{H}(t_1)\,\mathrm{d}t_1 + (-i)^2 \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2\, \hat{H}(t_1)\hat{H}(t_2) + \cdots. \tag{1.60}$$

This ordered structure ensures that the solution correctly reproduces the composition of infinitesimal propagators even when the Hamiltonian at different times does not commute with itself.

1.5 Statistical ensembles and mixed states

The vector-state formalism is fully adequate when the preparation procedure singles out one definite pure state. In that case, the state of the system is completely described by a normalized vector in Hilbert space, or equivalently by the associated rank-one projector. In many physically relevant situations, however, this description is no longer sufficient. The preparation may be only partially controlled, or it may produce different pure states with given classical probabilities. One is then naturally led to consider statistical ensembles rather than individual state vectors.

More precisely, suppose that a preparation device produces the pure state $|\psi_j\rangle$ with probability p_j, where $p_j \geq 0$ and $\sum_j p_j = 1$. One then speaks of a statistical ensemble of pure states,

$$\mathcal{E} = \{p_j, |\psi_j\rangle\}_j\,. \tag{1.61}$$

The physical meaning of this description is : before any measurement is performed, one does not know which of the vectors $|\psi_j\rangle$ has actually been prepared in a given run, but one does know the probabilities with which the different alternatives occurs.

If $\hat{X}$ is an observable, the expectation value associated with the ensemble is obtained by averaging over the possible preparations:

$$\left\langle \hat{X} \right\rangle = \sum_j p_j \langle\psi_j|\hat{X}|\psi_j\rangle\,. \tag{1.62}$$

At this stage, an important structural fact emerges. Although the ensemble is defined as a collection of vectors and probabilities, all measurable predictions depend on it only through a single operator. Indeed, by linearity one may rewrite Eq. (1.62) as

$$\left\langle \hat{X} \right\rangle = \mathrm{Tr}\Big(\hat{\rho}\hat{X}\Big), \tag{1.63}$$

provided one introduces the density operator

$$\hat{\rho} = \sum_j p_j \, |\psi_j\rangle\langle\psi_j| \, . \tag{1.64}$$

This is the natural generalization of the projector associated with a pure state. In the special case in which one of the probabilities is equal to one, say $p_{j_0} = 1$, Eq. (1.64) reduces to $\hat{\rho} = |\psi_{j_0}\rangle\langle\psi_{j_0}|$, and the pure-state formalism is recovered.
The density operator is therefore not introduced as an abstract mathematical object, but as the unique operator encoding all statistical predictions associated with a preparation procedure. In particular, if two distinct ensembles define the same operator $\hat{\rho}$, then they yield the same expectation values for every observable and are therefore physically indistinguishable as far as measurements on the system are concerned.
The operator $\hat{\rho}$ satisfies three basic properties. First, it is self-adjoint. Second, it is positive. Third, it has unit trace. Explicitly,

$$\hat{\rho}^\dagger = \hat{\rho}, \qquad \langle\phi|\hat{\rho}|\phi\rangle \geq 0 \;\; \forall \; |\phi\rangle \in \mathcal{H}, \qquad \mathrm{Tr}(\hat{\rho}) = 1. \tag{1.65}$$

The proof follows immediately from Eq. (1.64). Self-adjointness is obvious, while positivity follows from

$$\langle\phi|\hat{\rho}|\phi\rangle = \sum_j p_j \, \langle\phi||\psi_j\rangle\langle\psi_j||\phi\rangle = \sum_j p_j |\langle\psi_j|\phi\rangle\,|^2 \geq 0. \tag{1.66}$$

Finally,

$$\mathrm{Tr}(\hat{\rho}) = \sum_j p_j \, \mathrm{Tr}(|\psi_j\rangle\langle\psi_j|) = \sum_j p_j = 1. \tag{1.67}$$

Conversely, every positive trace-one operator defines a quantum state. In this sense, the density-operator formalism is not an optional extension of the vector formalism, but its proper completion. Pure states appear as a special case, while mixed states arise whenever the preparation cannot be associated with a single ray in Hilbert space.
The set of quantum states has a convex structure. If $\hat{\rho}_1$ and $\hat{\rho}_2$ are density operators and $0 \leq p \leq 1$, then

$$\hat{\rho} = p\hat{\rho}_1 + (1-p)\hat{\rho}_2 \tag{1.68}$$

is again a density operator. This reflects the elementary fact that one may probabilistically mix two preparation procedures. Pure states are precisely the extremal points of this convex set.
If the system is in the pure state $|\psi\rangle$, then $\hat{\rho} = |\psi\rangle\langle\psi|$ and therefore $\hat{\rho}^2 = \hat{\rho}$. More generally,

$$\hat{\rho} \text{ pure} \iff \hat{\rho}^2 = \hat{\rho} \iff \mathrm{Tr}\big(\hat{\rho}^2\big) = 1. \tag{1.69}$$

If instead $\mathrm{Tr}(\hat{\rho}^2) < 1$, the state is mixed. The quantity

$$\gamma(\hat{\rho}) = \mathrm{Tr}\big(\hat{\rho}^2\big) \tag{1.70}$$

is called the purity. It provides a simple measure of the degree of mixedness of the state.

A key conceptual point is that the ensemble decomposition of a mixed state is, in general, not unique. Different ensembles may define the same density operator and therefore the same statistical predictions for all observables. For instance, by considering two generic orthonormal basis for a two-level system we have:

$$\frac{1}{2}\left|v_1\right\rangle\!\left\langle v_1\right| + \frac{1}{2}\left|v_2\right\rangle\!\left\langle v_2\right| = \frac{1}{2}\left|w_1\right\rangle\!\left\langle w_1\right| + \frac{1}{2}\left|w_2\right\rangle\!\left\langle w_2\right| = \frac{\mathbb{1}}{2}. \tag{1.71}$$

The two ensembles correspond to different preparation procedures, but they define the same statistical operator. Hence, from the point of view of measurement statistics, they represent the same state.

This observation is of fundamental importance. It shows that the density operator is the physically relevant object, whereas the ensemble decomposition is not unique and should not be regarded as part of the intrinsic definition of the state. This point will become even more important in the next chapter, where mixed states will arise not only from classical uncertainty in the preparation, but also from the reduced description of subsystems entangled with external degrees of freedom.

1.5.1 Two-level systems (qubits)

We now specialize to a two-level system, or qubit. Its Hilbert space is $\mathbb{C}^2$. We choose as computational basis the orthonormal basis

$$\mathcal{B}_z = \{\left|0\right\rangle, \left|1\right\rangle\}, \qquad \langle 0|0\rangle = \langle 1|1\rangle = 1, \qquad \langle 0|1\rangle = 0. \tag{1.72}$$

With the convention adopted throughout, $\sigma_z\left|0\right\rangle = \left|0\right\rangle$ and $\sigma_z\left|1\right\rangle = -\left|1\right\rangle$, so that, in the ordered basis $\{\left|0\right\rangle, \left|1\right\rangle\}$,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1.73}$$

The states introduced above form the eigenbasis of σ_x, namely $\sigma_x\left|\pm\right\rangle = \pm\left|\pm\right\rangle$. We therefore denote $\mathcal{B}_x = \{\left|+\right\rangle, \left|-\right\rangle\}$ as the σ_x eigenbasis.

A generic pure qubit state may be written as

$$\left|\psi\right\rangle = \alpha\left|0\right\rangle + \beta\left|1\right\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{1.74}$$

Up to a global phase, it can always be parametrized as

$$\left|\psi\right\rangle = \cos\frac{\theta}{2}\left|0\right\rangle + e^{i\phi}\sin\frac{\theta}{2}\left|1\right\rangle, \tag{1.75}$$

with $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$. The corresponding pure-state density operator is simply $\hat{\rho}_\psi = \left|\psi\right\rangle\!\left\langle\psi\right|$.

Since $\{\mathbb{1}, \sigma_x, \sigma_y, \sigma_z\}$ is a basis for the space of 2×2 matrices, every qubit density operator admits the decomposition

$$\hat{\rho} = \frac{1}{2}\left(\mathbb{1} + \vec{r} \cdot \vec{\sigma}\right), \tag{1.76}$$

where $\vec{r} = (r_x, r_y, r_z) \in \mathbb{R}^3$ is the Bloch vector and $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$. Explicitly,

$$\hat{\rho} = \frac{1}{2}\begin{pmatrix} 1 + r_z & r_x - ir_y \\ r_x + ir_y & 1 - r_z \end{pmatrix}. \tag{1.77}$$

The components of $\vec{r}$ are obtained as expectation values of the Pauli operators,

$$r_k = \mathrm{Tr}(\hat{\rho}\sigma_k), \qquad k \in \{x, y, z\}, \tag{1.78}$$

since $\mathrm{Tr}(\sigma_k) = 0$ and $\mathrm{Tr}(\sigma_j \sigma_k) = 2\delta_{jk}$.
For the pure state in Eq. (1.75), one finds

$$r_x = \sin\theta\cos\phi, \qquad r_y = \sin\theta\sin\phi, \qquad r_z = \cos\theta. \tag{1.79}$$

Thus, pure qubit states are in one-to-one correspondence with points on the unit sphere in $\mathbb{R}^3$, the Bloch sphere.
The positivity of $\hat{\rho}$ imposes a restriction on $\vec{r}$. Using $(\vec{r} \cdot \vec{\sigma})^2 = \|\vec{r}\|^2 \mathbb{1}$, one finds that the eigenvalues of $\hat{\rho}$ are

$$\lambda_\pm = \frac{1 \pm \|\vec{r}\|}{2}. \tag{1.80}$$

Therefore,

$$\hat{\rho} \geq 0 \iff \|\vec{r}\| \leq 1. \tag{1.81}$$

This means that the physically allowed states fill the Bloch ball. Pure states lie on its surface, $\|\vec{r}\| = 1$, whereas mixed states lie in its interior, $\|\vec{r}\| < 1$. The maximally mixed state is the center of the ball, namely $\hat{\rho}_{\mathrm{mm}} = \mathbb{1}/2$ and $\vec{r} = 0$.
The relation between the Bloch vector and the purity is immediate:

$$\mathrm{Tr}\left(\hat{\rho}^2\right) = \frac{1 + \|\vec{r}\|^2}{2}. \tag{1.82}$$

Hence,

$$\mathrm{Tr}\left(\hat{\rho}^2\right) = 1 \iff \|\vec{r}\| = 1, \qquad \mathrm{Tr}\left(\hat{\rho}^2\right) < 1 \iff \|\vec{r}\| < 1. \tag{1.83}$$

Thus, the length of the Bloch vector directly quantifies the degree of purity of the state.
Finally, any qubit observable may be written as

$$\hat{X} = a_0 \mathbb{1} + \vec{a} \cdot \vec{\sigma}, \tag{1.84}$$

with $a_0 \in \mathbb{R}$ and $\vec{a} \in \mathbb{R}^3$. Its expectation value in the state $\hat{\rho}$ is

$$\left\langle \hat{X} \right\rangle = \mathrm{Tr}\left(\hat{\rho}\hat{X}\right) = a_0 + \vec{a} \cdot \vec{r}. \tag{1.85}$$

In particular, one has $\langle \sigma_x \rangle = r_x$, $\langle \sigma_y \rangle = r_y$, and $\langle \sigma_z \rangle = r_z$.

1.6 Bipartite and multipartite systems

We now discuss in more detail the structure of composite quantum systems. This is the natural setting in which quantum correlations, and in particular entanglement, arise. Since the present thesis is ultimately concerned with collective properties of many two-level systems, the tensor-product structure of the Hilbert space plays a central role from the very beginning.

1.6.1 Tensor-product structure and product bases

According to the composition postulate of quantum mechanics, if two subsystems A and B are associated with Hilbert spaces $\mathcal{H}_A$ and $\mathcal{H}_B$, then the composite system is described on the tensor-product space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \tag{1.86}$$

If $\{|a_i\rangle\}_{i=1}^{d_A}$ and $\{|b_j\rangle\}_{j=1}^{d_B}$ are orthonormal bases of $\mathcal{H}_A$ and $\mathcal{H}_B$, then the set

$$\mathcal{B}_{AB} = \{|a_i\rangle \otimes |b_j\rangle\}_{i,j} \tag{1.87}$$

is an orthonormal basis of $\mathcal{H}_{AB}$, since

$$(\langle a_i| \otimes \langle b_j|)\,(|a_{i'}\rangle \otimes |b_{j'}\rangle) = \langle a_i|a_{i'}\rangle\,\langle b_j|b_{j'}\rangle = \delta_{ii'}\delta_{jj'}. \tag{1.88}$$

Therefore, any pure bipartite state may be expanded as

$$|\Psi\rangle_{AB} = \sum_{i=1}^{d_A}\sum_{j=1}^{d_B} c_{ij}\;|a_i\rangle \otimes |b_j\rangle\,, \tag{1.89}$$

with normalization condition

$$\sum_{i,j}|c_{ij}|^2 = 1. \tag{1.90}$$

Operators on the tensor-product space are defined accordingly. If $\hat{A}$ acts on $\mathcal{H}_A$ and $\hat{B}$ acts on $\mathcal{H}_B$, then $\hat{A} \otimes \hat{B}$ is defined by

$$(\hat{A} \otimes \hat{B})\,(|\phi\rangle_A \otimes |\chi\rangle_B) = (\hat{A}\,|\phi\rangle_A) \otimes (\hat{B}\,|\chi\rangle_B), \tag{1.91}$$

and extended by linearity. In the product basis, its matrix elements read

$$\langle a_i, b_j|\hat{A} \otimes \hat{B}|a_{i'}, b_{j'}\rangle = \langle a_i|\hat{A}|a_{i'}\rangle\;\langle b_j|\hat{B}|b_{j'}\rangle\,. \tag{1.92}$$

Important special cases are the local operators $\hat{A}\otimes\mathbb{1}_B$ and $\mathbb{1}_A\otimes\hat{B}$, which act nontrivially only on one subsystem. For instance,

$$(\hat{A} \otimes \mathbb{1}_B)\,(|\phi\rangle_A \otimes |\chi\rangle_B) = (\hat{A}\,|\phi\rangle_A) \otimes |\chi\rangle_B\,. \tag{1.93}$$

If the global state is described by the density operator $\hat{\rho}_{AB}$, the expectation value of a local observable acting on A only is

$$\left\langle \hat{A} \otimes \mathbb{1}_B \right\rangle = \mathrm{Tr}_{AB}\left[\hat{\rho}_{AB}(\hat{A} \otimes \mathbb{1}_B)\right]. \tag{1.94}$$

This motivates the introduction of reduced states and partial trace.

1.6.2 Local operators, reduced states, and partial trace

Let $\{|b_j\rangle\}$ be an orthonormal basis of $\mathcal{H}_B$. The partial trace over subsystem B is the map

$$\mathrm{Tr}_B : \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B) \to \mathcal{B}(\mathcal{H}_A) \tag{1.95}$$

defined by

$$\mathrm{Tr}_B(\hat{X}_{AB}) = \sum_j (\mathbb{1}_A \otimes \langle b_j|)\, \hat{X}_{AB}\, (\mathbb{1}_A \otimes |b_j\rangle). \tag{1.96}$$

The reduced states are then

$$\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB}), \qquad \hat{\rho}_B = \mathrm{Tr}_A(\hat{\rho}_{AB}). \tag{1.97}$$

It is important to verify that the reduced state reproduces all local expectation values. Writing

$$\hat{\rho}_{AB} = \sum_{i,i',j,j'} \rho_{ii';jj'}\ |a_i\rangle\langle a_{i'}| \otimes |b_j\rangle\langle b_{j'}|\,, \tag{1.98}$$

one finds

$$\mathrm{Tr}_B(\hat{\rho}_{AB}) = \sum_{i,i',j} \rho_{ii';jj}\ |a_i\rangle\langle a_{i'}|\,. \tag{1.99}$$

Therefore,

$$\mathrm{Tr}_A(\hat{\rho}_A \hat{A}) = \sum_{i,i',j} \rho_{ii';jj}\ \langle a_{i'}|\hat{A}|a_i\rangle\,, \tag{1.100}$$

while

$$\mathrm{Tr}_{AB}\Big[\hat{\rho}_{AB}(\hat{A} \otimes \mathbb{1}_B)\Big] = \sum_{i,i',j,j'} \rho_{ii';jj'}\ \langle a_{i'}|\hat{A}|a_i\rangle\ \langle b_{j'}|b_j\rangle = \sum_{i,i',j} \rho_{ii';jj}\ \langle a_{i'}|\hat{A}|a_i\rangle\,. \tag{1.101}$$

Hence,

$$\mathrm{Tr}_A(\hat{\rho}_A \hat{A}) = \mathrm{Tr}_{AB}\Big[\hat{\rho}_{AB}(\hat{A} \otimes \mathbb{1}_B)\Big]\,. \tag{1.102}$$

Thus, the reduced density operator contains all information relevant to local measurements on subsystem A.
As a simple example, if the global state is a product state $\hat{\rho}_{AB} = \hat{\rho}_A \otimes \hat{\rho}_B$, then

$$\mathrm{Tr}_B(\hat{\rho}_A \otimes \hat{\rho}_B) = \hat{\rho}_A \cdot \mathrm{Tr}(\hat{\rho}_B) = \hat{\rho}_A, \tag{1.103}$$

since $\mathrm{Tr}(\hat{\rho}_B) = 1$.

1.6.3 Bell basis and parity sectors

We now introduce the Bell basis, which plays a central role in two-qubit quantum information. It consists of the four maximally entangled states

$$\left|\Phi^{\pm}\right\rangle = \frac{1}{\sqrt{2}}\,(|00\rangle \pm |11\rangle)\,, \qquad \left|\Psi^{\pm}\right\rangle = \frac{1}{\sqrt{2}}\,(|01\rangle \pm |10\rangle)\,. \tag{1.104}$$

These states form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ and they explicit a simple parity structure. Defining the parity operator

$$\Pi_z = \sigma_z \otimes \sigma_z, \tag{1.105}$$

one finds

$$\Pi_z \left|\Phi^\pm\right\rangle = + \left|\Phi^\pm\right\rangle, \qquad \Pi_z \left|\Psi^\pm\right\rangle = - \left|\Psi^\pm\right\rangle. \tag{1.106}$$

Hence, the two-qubit Hilbert space decomposes as

$$\mathcal{H}_2 = \mathcal{H}_{\text{even}} \oplus \mathcal{H}_{\text{odd}}, \tag{1.107}$$

with

$$\mathcal{H}_{\text{even}} = \text{span}\{|00\rangle, |11\rangle\}, \qquad \mathcal{H}_{\text{odd}} = \text{span}\{|01\rangle, |10\rangle\}. \tag{1.108}$$

State	Computational form	Parity subspace	Eigenvalue of Π_z
$\|\Phi^+\rangle$	$\frac{1}{\sqrt{2}}(\|00\rangle + \|11\rangle)$	even	$+1$
$\|\Phi^-\rangle$	$\frac{1}{\sqrt{2}}(\|00\rangle - \|11\rangle)$	even	$+1$
$\|\Psi^+\rangle$	$\frac{1}{\sqrt{2}}(\|01\rangle + \|10\rangle)$	odd	-1
$\|\Psi^-\rangle$	$\frac{1}{\sqrt{2}}(\|01\rangle - \|10\rangle)$	odd	-1

Table 1.1: Bell basis and parity sectors for two qubits.

The Bell basis is useful because it makes maximally entangled states explicit, diagonalizes relevant two-qubit observables, and naturally separates even and odd parity sectors. These notions will reappear later in the thesis in connection with parity measurements and monitored dynamics.

1.6.4 Separable and entangled states: definition and methods

We may now introduce the notion of separability. A pure bipartite state $|\Psi\rangle_{AB} \in \mathcal{H}_A \otimes \mathcal{H}_B$ is called separable if it can be written as $|\Psi\rangle_{AB} = |\phi\rangle_A \otimes |\chi\rangle_B$. If no such decomposition exists, the state is entangled. For mixed states, the definition is more general: a bipartite state $\hat{\rho}_{AB}$ is separable if it can be written as a convex combination of product states,

$$\hat{\rho}_{AB} = \sum_k p_k\, \hat{\rho}_A^{(k)} \otimes \hat{\rho}_B^{(k)}, \qquad p_k \geq 0, \qquad \sum_k p_k = 1. \tag{1.109}$$

If no decomposition of the form (1.109) exists, the state is entangled [14].
This distinction is conceptually fundamental. In a separable state, all correlations may be understood as arising from classical ignorance about local preparations. In an entangled state, by contrast, the global state contains correlations that cannot be reduced to any classical mixture of local states. Entanglement is therefore not simply a stronger form of correlation, but a genuinely nonclassical feature of the tensor-product structure of quantum theory. It is precisely this possibility that makes composite quantum systems radically different from their classical counterparts and turns entanglement into a central resource in quantum information, quantum metrology, and many-body physics.

A simple example of separable pure state is $|0\rangle \otimes |1\rangle = |01\rangle$. A standard example of entangled state, see (1.104)

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right). \tag{1.110}$$

Indeed, if $|\Phi^+\rangle$ were a product state, one could write it as $(\alpha\,|0\rangle + \beta\,|1\rangle) \otimes (\gamma\,|0\rangle + \delta\,|1\rangle)$, which expands as

$$\alpha\gamma\,|00\rangle + \alpha\delta\,|01\rangle + \beta\gamma\,|10\rangle + \beta\delta\,|11\rangle\,. \tag{1.111}$$

Comparison with Eq. (1.110) would require

$$\alpha\delta = 0, \qquad \beta\gamma = 0, \qquad \alpha\gamma = \beta\delta = \frac{1}{\sqrt{2}}, \tag{1.112}$$

which is impossible. Hence, $|\Phi^+\rangle$ is entangled.
For pure bipartite states, the structure of entanglement is completely characterized by the Schmidt decomposition. Let

$$|\Psi\rangle_{AB} = \sum_{i,j} c_{ij}\,|a_i\rangle \otimes |b_j\rangle\,. \tag{1.113}$$

By applying the singular value decomposition to the coefficient matrix $C = (c_{ij})$, one can always find orthonormal sets $\{|u_n\rangle\} \subset \mathcal{H}_A$ and $\{|v_n\rangle\} \subset \mathcal{H}_B$ such that

$$|\Psi\rangle_{AB} = \sum_{n=1}^{r} \sqrt{\lambda_n}\,|u_n\rangle \otimes |v_n\rangle\,, \tag{1.114}$$

where

$$\lambda_n \geq 0, \qquad \sum_{n=1}^{r} \lambda_n = 1. \tag{1.115}$$

The integer r is called the Schmidt rank. The reduced density operators are then

$$\hat{\rho}_A = \mathrm{Tr}_B(|\Psi\rangle\langle\Psi|_{AB}) = \sum_{n=1}^{r} \lambda_n\,|u_n\rangle\langle u_n|\,, \tag{1.116}$$

and similarly

$$\hat{\rho}_B = \sum_{n=1}^{r} \lambda_n\,|v_n\rangle\langle v_n|\,. \tag{1.117}$$

Hence, $\hat{\rho}_A$ and $\hat{\rho}_B$ have the same nonzero spectrum.
The Schmidt decomposition immediately implies that a pure bipartite state is separable if and only if its Schmidt rank is equal to one:

$$|\Psi\rangle_{AB} \text{ separable} \iff r = 1. \tag{1.118}$$

Equivalently, a pure bipartite state is entangled if and only if $r > 1$. The Schmidt form also makes clear why entanglement is directly connected with the mixedness of the reduced states. If the global state is pure and separable, then the reduced states are pure.

By contrast, if more than one Schmidt coefficient is nonzero, both reduced states are mixed. For instance, for the Bell state $|\Phi^+\rangle$ one has

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left|0\right\rangle \otimes \left|0\right\rangle + \frac{1}{\sqrt{2}}\left|1\right\rangle \otimes \left|1\right\rangle, \tag{1.119}$$

so that $\hat{\rho}_A = \hat{\rho}_B = \mathbb{1}/2$.
A natural quantity associated with a quantum state is the von Neumann entropy,

$$S(\hat{\rho}) = -\operatorname{Tr}(\hat{\rho}\log\hat{\rho}). \tag{1.120}$$

If $\hat{\rho} = \sum_n \lambda_n |\varphi_n\rangle\langle\varphi_n|$, then

$$S(\hat{\rho}) = -\sum_n \lambda_n \log\lambda_n. \tag{1.121}$$

In particular, $S(\hat{\rho}) = 0$ if and only if $\hat{\rho}$ is pure. For a pure bipartite state $|\Psi\rangle\langle\Psi|_{AB}$, the reduced entropies satisfy

$$S(\hat{\rho}_A) = S(\hat{\rho}_B), \tag{1.122}$$

and this common value quantifies bipartite entanglement. In particular, separable pure states have zero reduced entropy, whereas Bell states have entropy $\log 2$.
For mixed states, deciding separabilty is in general much harder. One of the most important criteria is the partial-transposition criterion introduced by Peres [15]. Given a bipartite operator

$$\hat{\rho}_{AB} = \sum_{i,i',j,j'} \rho_{ii';jj'}\, |a_i\rangle\langle a_{i'}| \otimes |b_j\rangle\langle b_{j'}|, \tag{1.123}$$

its partial transpose with respect to subsystem B is defined as

$$\hat{\rho}_{AB}^{T_B} = \sum_{i,i',j,j'} \rho_{ii';jj'}\, |a_i\rangle\langle a_{i'}| \otimes |b_{j'}\rangle\langle b_j|. \tag{1.124}$$

If $\hat{\rho}_{AB}$ is separable and can be written as in Eq. (1.109), then

$$\hat{\rho}_{AB}^{T_B} = \sum_k p_k\, \hat{\rho}_A^{(k)} \otimes \left(\hat{\rho}_B^{(k)}\right)^T, \tag{1.125}$$

which is again positive, since transposition preserves positivity. Therefore,

$$\hat{\rho}_{AB} \text{ separable} \quad \Longrightarrow \quad \hat{\rho}_{AB}^{T_B} \geq 0. \tag{1.126}$$

For 2×2 and 2×3 systems, the Horodecki family [14] proved that this condition is also sufficient:

$$\hat{\rho}_{AB}^{T_B} \geq 0 \iff \hat{\rho}_{AB} \text{ separable}, \qquad d_A d_B \in \{4, 6\}. \tag{1.127}$$

In higher dimensions, however, PPT is only a necessary but not sufficient criterion.
A simple two-qubit example is again provided by the Bell state $|\Phi^+\rangle$. In the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$,

$$\hat{\rho}_{\Phi^+} = \left|\Phi^+\right\rangle\left\langle\Phi^+\right| = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \tag{1.128}$$

while its partial transpose is

$$\hat{\rho}_{\Phi^+}^{T_B} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{1.129}$$

whose eigenvalues are $\left\{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right\}$. Since one eigenvalue is negative, the state is entangled.
A complementary and experimentally useful approach is provided by entanglement witnesses. A Hermitian operator $\hat{W}$ is called an entanglement witness if

$$\mathrm{Tr}\left(\hat{W}\hat{\sigma}\right) \geq 0 \qquad \text{for all separable states } \hat{\sigma}, \tag{1.130}$$

while there exists at least one entangled state $\hat{\rho}$ such that

$$\mathrm{Tr}\left(\hat{W}\hat{\rho}\right) < 0. \tag{1.131}$$

Thus, a negative expectation value of $\hat{W}$ certifies entanglement. For the Bell state $|\Phi^+\rangle$, a simple witness is

$$\hat{W}_{\Phi^+} = \frac{\mathbb{1}}{2} - \left|\Phi^+\middle\rangle\!\middle\langle\Phi^+\right|, \tag{1.132}$$

for which

$$\mathrm{Tr}\left(\hat{W}_{\Phi^+}\left|\Phi^+\middle\rangle\!\middle\langle\Phi^+\right|\right) = -\frac{1}{2} < 0. \tag{1.133}$$

1.6.5 Concurrence for two-qubit states

For two-qubit systems, one of the most widely used entanglement measures is the concurrence. This quantity is especially important because, unlike the general mixed-state separability problem, it admits a closed analytical expression for arbitrary two-qubit states [16, 17]. It is therefore particularly useful whenever one needs an explicit and computable measure of entanglement, both for pure states and for generic mixed states.
We consider a two-qubit system with Hilbert space $\mathcal{H}_{AB} = \mathbb{C}^2 \otimes \mathbb{C}^2$ and computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Let

$$|\psi\rangle = a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle, \qquad |a|^2+|b|^2+|c|^2+|d|^2 = 1 \tag{1.134}$$

be a pure state. The spin-flipped state is defined as

$$\left|\tilde{\psi}\right\rangle = (\sigma_y \otimes \sigma_y)\,|\psi^*\rangle, \tag{1.135}$$

where the complex conjugation is taken in the computational basis. The concurrence is then

$$C(|\psi\rangle) = \left|\left\langle\psi\middle|\tilde{\psi}\right\rangle\right| = 2|ad - bc|. \tag{1.136}$$

This expression already shows the basic behavior of the measure. One has

$$0 \leq C(|\psi\rangle) \leq 1, \tag{1.137}$$

with

$$C(|\psi\rangle) = 0 \iff |\psi\rangle \text{ is separable,} \qquad C(|\psi\rangle) = 1 \iff |\psi\rangle \text{ is maximally entangled.} \tag{1.138}$$

In particular, $C(|\Phi^+\rangle) = 1$, whereas every product state has vanishing concurrence because $ad - bc = 0$.
The quantity $ad-bc$ is the determinant of the 2×2 coefficient matrix associated with the pure state. Its vanishing is therefore equivalent to the factorization of the state vector, which makes the relation between concurrence and separability particularly transparent in the pure-state case. The spin-flip operation entering Eq. (1.135) is also natural from the two-qubit point of view, since it maps each local basis vector to its orthogonal counterpart up to a phase and allows one to construct a scalar quantity sensitive to entanglement.
The concurrence of a pure two-qubit state can also be expressed in terms of the reduced state:

$$C(|\psi\rangle) = 2\sqrt{\det \hat{\rho}_A} = \sqrt{2\,(1 - \mathrm{Tr}(\hat{\rho}_A^2))}, \qquad \hat{\rho}_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|)\,. \tag{1.139}$$

Thus, for pure bipartite states, concurrence is directly related to the mixedness of the reduced state. This is fully consistent with the Schmidt-decomposition picture discussed above: a pure bipartite state is entangled precisely when its reduced states are mixed.
For mixed states, the situation is conceptually more subtle. A mixed state may admit many different pure-state decompositions, and an entanglement measure should not depend on a particular ensemble representation. For this reason, concurrence is defined through the convex-roof construction [16]:

$$C(\hat{\rho}) = \min_{\{p_k,|\psi_k\rangle\}} \sum_k p_k\, C(|\psi_k\rangle), \tag{1.140}$$

where the minimum is taken over all decompositions

$$\hat{\rho} = \sum_k p_k\, |\psi_k\rangle\langle\psi_k|\,. \tag{1.141}$$

This definition is natural but, in general, difficult to evaluate explicitly. The remarkable result proved by Wootters is that for two qubits the minimization can be solved in closed form [16].
Defining the spin-flipped density operator

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\hat{\rho}^*(\sigma_y \otimes \sigma_y), \tag{1.142}$$

and the operator

$$\hat{R} = \sqrt{\sqrt{\hat{\rho}}\,\tilde{\rho}\,\sqrt{\hat{\rho}}}, \tag{1.143}$$

let $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq 0$ be the eigenvalues of $\hat{R}$. Then

$$C(\hat{\rho}) = \max\{0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\}\,. \tag{1.144}$$

Equivalently, if μ_i are the eigenvalues of $\hat{\rho}\tilde{\rho}$, one may take $\lambda_i = \sqrt{\mu_i}$ ordered in decreasing way. Thus, given a generic two-qubit density matrix, the calculation proceeds through the chain

$$\hat{\rho} \longrightarrow \tilde{\rho} \longrightarrow \hat{\rho}\tilde{\rho} \longrightarrow \{\mu_i\} \longrightarrow \{\lambda_i\} \longrightarrow C(\hat{\rho}). \tag{1.145}$$

This provides a complete analytical measure of entanglement for arbitrary two-qubit states.
Several properties are worth emphasizing. First,

$$0 \leq C(\hat{\rho}) \leq 1. \tag{1.146}$$

Second, in the two-qubit case,

$$C(\hat{\rho}) = 0 \iff \hat{\rho} \text{ is separable.} \tag{1.147}$$

Thus, concurrence is not only an entanglement monotone, but also a complete separability criterion for two qubits. Third, concurrence is invariant under local unitary transformations,

$$C\Big[(\hat{U}_A \otimes \hat{U}_B)\hat{\rho}(\hat{U}_A^\dagger \otimes \hat{U}_B^\dagger)\Big] = C(\hat{\rho}), \tag{1.148}$$

as expected for any physically meaningful measure of bipartite entanglement.

1.6.6 Generic multipartite systems and collective spin observables

The generalization from bipartite to generic multipartite systems is conceptually straightforward at the level of Hilbert spaces, but it immediately leads to a much richer mathematical and physical structure. For N distinguishable subsystems with Hilbert spaces $\mathcal{H}_1, \dots, \mathcal{H}_N$, the global Hilbert space is the tensor product

$$\mathcal{H}^{(N)} = \mathcal{H}_1 \otimes \cdots \otimes \mathcal{H}_N. \tag{1.149}$$

In the case of N qubits, each local Hilbert space is $\mathbb{C}^2$, and therefore

$$\mathcal{H}^{(N)} = (\mathbb{C}^2)^{\otimes N}, \qquad \dim \mathcal{H}^{(N)} = 2^N. \tag{1.150}$$

A natural orthonormal basis is provided by the computational product basis

$$|x_1, \dots, x_N\rangle = |x_1\rangle \otimes \cdots \otimes |x_N\rangle, \qquad x_k \in \{0, 1\}. \tag{1.151}$$

These vectors are orthonormal, since

$$\langle x_1, \dots, x_N | y_1, \dots, y_N\rangle = \prod_{k=1}^{N} \langle x_k | y_k\rangle = \prod_{k=1}^{N} \delta_{x_k y_k}. \tag{1.152}$$

Accordingly, any pure state of the N-qubit system may be expanded as

$$|\Psi\rangle = \sum_{x_1,\dots,x_N \in \{0,1\}} c_{x_1 \dots x_N} |x_1, \dots, x_N\rangle, \tag{1.153}$$

with normalization condition

$$\sum_{x_1,\ldots,x_N} |c_{x_1\ldots x_N}|^2 = 1. \tag{1.154}$$

At the multipartite level, local and nonlocal operators must be carefully distinguished. If $\hat{A}$ is an operator acting on the Hilbert space of a single subsystem, say the k-th one, then the corresponding local operator on the full Hilbert space is defined as

$$\hat{A}^{(k)} = \mathbb{1}^{\otimes(k-1)} \otimes \hat{A} \otimes \mathbb{1}^{\otimes(N-k)}. \tag{1.155}$$

This means that $\hat{A}^{(k)}$ acts as $\hat{A}$ on the k-th factor and as the identity on all the others. Its action on a computational-basis vector is therefore

$$\hat{A}^{(k)} |x_1, \ldots, x_k, \ldots, x_N\rangle = |x_1\rangle \otimes \cdots \otimes \hat{A}\, |x_k\rangle \otimes \cdots \otimes |x_N\rangle\,. \tag{1.156}$$

For example, let $N = 3$ and consider the operator $\sigma_x^{(2)} = \mathbb{1} \otimes \sigma_x \otimes \mathbb{1}$. Since $\sigma_x |0\rangle = |1\rangle$ and $\sigma_x |1\rangle = |0\rangle$, one has

$$\sigma_x^{(2)} |0, 0, 1\rangle = |0, 1, 1\rangle\,, \qquad \sigma_x^{(2)} |1, 1, 0\rangle = |1, 0, 0\rangle\,. \tag{1.157}$$

Similarly, using the convention $\sigma_z |0\rangle = |0\rangle$ and $\sigma_z |1\rangle = -|1\rangle$, one finds

$$\sigma_z^{(3)} |0, 1, 0\rangle = |0, 1, 0\rangle\,, \qquad \sigma_z^{(3)} |0, 1, 1\rangle = -|0, 1, 1\rangle\,. \tag{1.158}$$

Thus, local operators act only on the corresponding slot of the tensor product, leaving the remaining factors unchanged.
Their action on a generic state follows by linearity. For instance, if

$$|\Psi\rangle = a\,|000\rangle + b\,|011\rangle + c\,|101\rangle + d\,|110\rangle\,, \tag{1.159}$$

then

$$\sigma_x^{(2)} |\Psi\rangle = a\,|010\rangle + b\,|001\rangle + c\,|111\rangle + d\,|100\rangle\,, \tag{1.160}$$

while

$$\sigma_z^{(1)} |\Psi\rangle = a\,|000\rangle + b\,|011\rangle - c\,|101\rangle - d\,|110\rangle\,. \tag{1.161}$$

These explicit examples are useful because they show that a local operator on a multipartite system still acts on the full Hilbert space, but in a factorized way.
More generally, if $\hat{A}$ acts on subsystem k and $\hat{B}$ acts on subsystem ℓ, with $k \neq \ell$, then one defines the corresponding two-body operator as

$$\hat{A}^{(k)}\hat{B}^{(\ell)} = \mathbb{1}^{\otimes(k-1)} \otimes \hat{A} \otimes \mathbb{1}^{\otimes(\ell-k-1)} \otimes \hat{B} \otimes \mathbb{1}^{\otimes(N-\ell)}, \tag{1.162}$$

assuming $k < \ell$. Such operators are central in interacting many-body models, correlation functions, and spin Hamiltonians. For example, for $N = 3$,

$$\sigma_z^{(1)}\sigma_z^{(3)} |x_1, x_2, x_3\rangle = \lambda_{x_1}\lambda_{x_3} |x_1, x_2, x_3\rangle\,, \tag{1.163}$$

where $\lambda_0 = +1$ and $\lambda_1 = -1$. Thus, $\sigma_z^{(1)}\sigma_z^{(3)}$ measures the relative parity of the first and third qubits.

Reduced states are also naturally generalized. If $I \subseteq \{1, \ldots, N\}$ denotes a subset of subsystems, one may define the reduced state of that subset by tracing out the complementary degrees of freedom:

$$\hat{\rho}_I = \mathrm{Tr}_{\bar{I}}(\hat{\rho}). \tag{1.164}$$

In this way, local properties, marginal statistics, and subsystem correlations may be studied inside the full multipartite state. As in the bipartite case, the reduced state contains all information relevant to observables acting only on the chosen subset.
The notion of separability becomes substantially richer in the multipartite setting. A pure state is said to be fully separable if it can be written as

$$|\Psi\rangle = |\phi_1\rangle \otimes \cdots \otimes |\phi_N\rangle\,. \tag{1.165}$$

Mixed states are fully separable if they are convex combinations of such product states. However, one may also encounter states that are separable with respect to some bipartitions but not with respect to others. This leads to several inequivalent notions of multipartite entanglement. In particular, a state is said to possess genuine multipartite entanglement if it cannot be written as a convex combination of states that are separable across some bipartition [18, 19]. Thus, the multipartite case is not merely a larger version of the bipartite one: it exhibits a genuinely more intricate hierarchy of correlation structures.
A simple example illustrates this point. For three qubits, the state

$$|\Psi\rangle = |0\rangle \otimes \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{1.166}$$

is entangled, but not genuinely tripartite entangled, since the first qubit is completely factorized from the remaining pair. By contrast, the GHZ state

$$|\mathrm{GHZ}_3\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}} \tag{1.167}$$

cannot be factorized with respect to any bipartition and is therefore genuinely tripartite entangled.
For the purposes of the present thesis, a particularly important family of observables is given by the collective spin operators

$$\hat{J}_\alpha = \frac{1}{2}\sum_{k=1}^{N} \sigma_\alpha^{(k)}, \qquad \alpha \in \{x, y, z\}. \tag{1.168}$$

These are sums of local operators, and therefore represent global observables obtained by coherently combining the contributions of all subsystems. Their action on a basis vector is determined term by term. For example,

$$\hat{J}_z\,|x_1, \ldots, x_N\rangle = \frac{1}{2}\sum_{k=1}^{N} \lambda_{x_k}\,|x_1, \ldots, x_N\rangle\,, \tag{1.169}$$

with $\lambda_0 = +1$ and $\lambda_1 = -1$. Thus, each computational-basis vector is an eigenvector of $\hat{J}_z$, with eigenvalue equal to one half of the difference between the number of qubits in $|0\rangle$ and the number of qubits in $|1\rangle$.

The operators $\hat{J}_x, \hat{J}_y, \hat{J}_z$ satisfy the angular-momentum commutation relations

$$[\hat{J}_x, \hat{J}_y] = i\hat{J}_z, \qquad [\hat{J}_y, \hat{J}_z] = i\hat{J}_x, \qquad [\hat{J}_z, \hat{J}_x] = i\hat{J}_y, \tag{1.170}$$

and therefore provide the natural language for the description of collective spin observables, permutation-symmetric states, and spin-squeezing properties in many-qubit systems.

It is worth stressing that collective operators are global, but not generic. They are built from sums of identical local operators and therefore respect the permutation symmetry of the subsystems. This will be especially important later, since many states of interest in quantum metrology and collective open-system dynamics naturally live in the symmetric sector of the full Hilbert space.

Finally, one may also construct collective two-body quantities, such as

$$\hat{J}_\alpha^2 = \frac{1}{4}\sum_{k,\ell=1}^{N} \sigma_\alpha^{(k)}\sigma_\alpha^{(\ell)} = \frac{N}{4}\mathbb{1} + \frac{1}{4}\sum_{k\neq\ell}^{N} \sigma_\alpha^{(k)}\sigma_\alpha^{(\ell)}, \tag{1.171}$$

where the identity contribution comes from $(\sigma_\alpha^{(k)})^2 = \mathbb{1}$. This expansion makes explicit the connection between collective observables and two-body correlations. It is precisely this link that will later make collective variances useful probes of multipartite entanglement and spin squeezing.

A central role in collective spin systems is played by Dicke states. These states form the natural basis for the symmetric subspace of N spin-1/2 particles, that is, the subspace invariant under particle exchange. Since spin squeezing is typically discussed in terms of collective operators and permutation-invariant states, Dicke states provide the most convenient language for describing many of the relevant multipartite states.

They are defined as simultaneous eigenstates of the collective operators $\hat{J}^2$ and $\hat{J}_z$,

$$\hat{J}^2 |j, m\rangle = j(j+1)|j, m\rangle, \qquad \hat{J}_z |j, m\rangle = m|j, m\rangle, \tag{1.172}$$

with $j = 0, \frac{1}{2}, 1, \ldots, \frac{N}{2}$ and $m = -j, -j+1, \ldots, j$. The fully symmetric Dicke states belong to the maximal-spin sector $j = N/2$ which is the sector most commonly encountered in collective-spin models and in the theory of spin squeezing. It follows that a basis state containing k qubits in the state $|1\rangle$ has $m = N/2 - k$.

The corresponding symmetric Dicke state is the equal superposition of all computational basis states with exactly k qubits in $|1\rangle$,

$$\left|D_N^{(k)}\right\rangle = \frac{1}{\sqrt{\binom{N}{k}}}\sum_{\text{perm}}\left|\underbrace{1\cdots1}_{k}\underbrace{0\cdots0}_{N-k}\right\rangle, \tag{1.173}$$

where the sum runs over all distinct permutations. In terms of the collective-spin basis, one may equivalently write

$$\left|D_N^{(k)}\right\rangle = |j = N/2, \, m = N/2 - k\rangle. \tag{1.174}$$

These states have a simple physical meaning: they describe k spin flips delocalized symmetrically over all qubits. For this reason, they naturally appear whenever the dynamics preserves exchange symmetry and is expressed in terms of collective operators. They are also closely related to the discussion of spin squeezing for states with fixed parity, since a state spanned only by Dicke states with even or odd values of k has a definite parity with respect to the collective spin along z. This is precisely the setting in which several simplifications arise in the analysis of spin-squeezing parameters.

1.6.7 Standard quantum limit vs Heisenberg limit

The quest for ever more precise parameter estimation has long provided one of the main motivations for quantum metrology. Historically, this problem became especially relevant in optical interferometry, where quantum fluctuations of the electromagnetic field set fundamental constraints on phase sensitivity.

A paradigmatic example is provided by kilometer-scale interferometers for gravitational-wave detection, such as LIGO and Virgo. These devices are designed to measure extremely small phase shifts, or equivalently differential arm-length variations, induced by the passage of a gravitational wave. The first direct detection of gravitational waves, reported in 2016 by the LIGO Scientific Collaboration, marked a major milestone in precision interferometry [20]. In such experiments, quantum noise is not a minor correction, but one of the fundamental mechanisms limiting the sensitivity over a significant portion of the detection band.

The standard quantum limit is the precision scaling obtained when the available resources are used in a classical or semiclassical way, namely when the probes are independent and no entanglement is exploited (see Appendix 4). If N denotes the relevant number of resources, such as the number of particles or photons, the corresponding uncertainty scales as

$$\Delta\theta_{\mathrm{SQL}} \sim \frac{1}{\sqrt{N}}. \tag{1.175}$$

This scaling is the analogue of the central-limit behavior and reflects the fact that independent measurements improve the signal-to-noise ratio only statistically.

By contrast, in the ideal noiseless quantum setting one may reach the Heisenberg limit,

$$\Delta\theta_{\mathrm{HL}} \sim \frac{1}{N}. \tag{1.176}$$

that is more favorable by a factor $\sqrt{N}$ with respect to the SQL. In this sense, the Heisenberg limit represents the ultimate scaling allowed by quantum mechanics for parameter estimation with N resources. The SQL is not an absolute bound, but rather the best scaling achievable with uncorrelated probes. The Heisenberg limit, instead, identifies the optimal scaling allowed in the ideal quantum regime. Reaching this regime requires the preparation of suitable non-classical states, whose quantum correlations amplify the response to the parameter while controlling the relevant fluctuations.

Different classes of quantum states can be used to approach or attain these scalings. Squeezed states reduce the fluctuations of the measured quadrature below the coherent-

state level and therefore allow one to surpass the SQL, as first pointed out in the context of interferometry by Caves [21]. Squeezed light can be generated and injected into large-scale interferometers: they are implemented in gravitational-wave detectors such as LIGO and Virgo, where yhese type of states are used to reduce quantum noise and improve the astrophysical reach of the instruments [22].
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To reach the Heisenberg scaling in its canonical form, one typically needs more strongly non-classical states. Prototypical examples are maximally path-entangled states, such as NOON states, or highly entangled collective states such as GHZ states and other suitable multipartite resources. More generally, the gain arises because entanglement allows the parameter to be encoded collectively on the whole probe, rather than independently on each constituent [23, 24].

1.7 Spin squeezing

Spin squeezing is one of the most important collective manifestations of nonclassical correlations in many-spin systems. It was originally introduced by Kitagawa and Ueda as the spin analogue of quadrature squeezing in bosonic systems [25], and was later reinterpreted by Wineland *et al.* as a directly metrological resource for phase estimation beyond the standard quantum limit [26]. The modern viewpoint, clearly reviewed in [27], is that spin squeezing lies at the intersection of three closely related notions: redistribution of collective quantum fluctuations, multipartite entanglement, and quantum-enhanced metrology.

1.7.1 Collective spin operators and coherent spin states

For a system of N spin-1/2 particles, the collective spin operators are

$$\hat{J}_\alpha = \frac{1}{2}\sum_{k=1}^{N} \sigma_\alpha^{(k)}, \qquad \alpha \in \{x, y, z\}, \tag{1.177}$$

and satisfy the usual angular-momentum commutation relations. For any unit vector $\mathbf{n}$, we write $\hat{J}_\mathbf{n} = \mathbf{n} \cdot \hat{\mathbf{J}}$.
The natural reference family is given by coherent spin states. In the present convention, a generic coherent spin state reads

$$|\theta, \phi\rangle = \bigotimes_{k=1}^{N} \left(e^{i\phi} \sin\frac{\theta}{2} |1\rangle_k + \cos\frac{\theta}{2} |0\rangle_k \right), \tag{1.178}$$

with mean-spin direction

$$\mathbf{n}_0 = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta). \tag{1.179}$$

For these states one has

$$\langle \hat{\mathbf{J}} \rangle = \frac{N}{2}\mathbf{n}_0, \tag{1.180}$$

so that the mean-spin length is maximal, $|\langle \hat{\mathbf{J}} \rangle| = N/2$. Moreover, for every direction $\mathbf{n}_\perp$ orthogonal to $\mathbf{n}_0$,

$$(\Delta \hat{J}_{\mathbf{n}_\perp})^2 = \frac{N}{4}. \tag{1.181}$$

Thus, coherent spin states define the standard quantum limit for collective-spin fluctuations.

1.7.2 Kitagawa-Ueda and Wineland definitions

Assuming $|\langle \hat{\mathbf{J}} \rangle| \neq 0$, we define the mean-spin direction as

$$\mathbf{n}_0 = \frac{\langle \hat{\mathbf{J}} \rangle}{|\langle \hat{\mathbf{J}} \rangle|}. \tag{1.182}$$

Spin squeezing is then characterized through the minimal variance in the plane orthogonal to $\mathbf{n}_0$:

$$(\Delta \hat{J}_\perp)^2_{\text{min}} = \min_{\mathbf{n}_\perp \perp \mathbf{n}_0} (\Delta \hat{J}_{\mathbf{n}_\perp})^2. \tag{1.183}$$

If the mean spin points along the z axis, one finds (see Appendix 4)

$$(\Delta \hat{J}_\perp)^2_{\text{min}} = \frac{1}{2} \left[\langle \hat{J}_x^2 + \hat{J}_y^2 \rangle - \sqrt{\langle \hat{J}_x^2 - \hat{J}_y^2 \rangle^2 + \langle \{ \hat{J}_x, \hat{J}_y \} \rangle^2} \right]. \tag{1.184}$$

The Kitagawa-Ueda parameter is defined as [25, 27]

$$\xi^2_{\text{KU}} = \frac{4}{N} (\Delta \hat{J}_\perp)^2_{\text{min}}. \tag{1.185}$$

A state is spin squeezed in the Kitagawa-Ueda sense if $\xi^2_{\text{KU}} < 1$. This criterion compares the minimal transverse variance with the coherent-spin reference value $N/4$, and therefore captures the purely fluctuation-based aspect of squeezing.

The Wineland parameter is instead motivated by Ramsey interferometry [26, 27]. For optimal small-angle phase estimation one obtains

$$\Delta \phi = \frac{(\Delta \hat{J}_\perp)_{\text{min}}}{|\langle \hat{\mathbf{J}} \rangle|}, \tag{1.186}$$

which, compared with the shot-noise limit $\Delta \phi_{\text{SQL}} = 1/\sqrt{N}$, leads to

$$\xi^2_{\text{W}} = \frac{N (\Delta \hat{J}_\perp)^2_{\text{min}}}{|\langle \hat{\mathbf{J}} \rangle|^2}. \tag{1.187}$$

The condition $\xi^2_{\text{W}} < 1$ indicates metrological gain beyond the standard quantum limit. The two parameters are related by

$$\xi^2_{\text{W}} = \frac{N^2}{4 |\langle \hat{\mathbf{J}} \rangle|^2} \xi^2_{\text{KU}}, \tag{1.188}$$

so that $\xi^2_{\text{KU}} \leq \xi^2_{\text{W}}$, since $|\langle \hat{\mathbf{J}} \rangle| \leq N/2$. Thus, Wineland squeezing is the stronger requirement.

For a coherent spin state, one has $(\Delta \hat{J}_\perp)^2_{\min} = N/4$ and $|\langle \hat{\mathbf{J}} \rangle| = N/2$, hence

$$\xi^2_{\mathrm{KU}} = \xi^2_{\mathrm{W}} = 1. \tag{1.189}$$

Thus, coherent spin states are unsqueezed and define the natural classical benchmark.
An instructive family is given by Dicke states $|J, m\rangle$ with $J = N/2$. For these states, $\langle \hat{J}_z \rangle = m$ and $\langle \hat{J}_x \rangle = \langle \hat{J}_y \rangle = 0$. For $m \neq 0$, the mean spin points along z, and one finds

$$(\Delta \hat{J}_x)^2 = (\Delta \hat{J}_y)^2 = \frac{1}{2}\left[J(J+1) - m^2\right], \tag{1.190}$$

so that

$$\xi^2_{\mathrm{KU}} = \frac{2}{N}\left[J(J+1) - m^2\right]. \tag{1.191}$$

For $m = \pm J$, one recovers $\xi^2_{\mathrm{KU}} = 1$, consistently with the fact that $|J, \pm J\rangle$ are coherent spin states, for $|m| < J$, one obtains $\xi^2_{\mathrm{KU}} > 1$.
A further caveat is that the original mean-spin-based construction (Wineland case!) becomes problematic when $|\langle \hat{\mathbf{J}} \rangle| = 0$. In that case the direction $\mathbf{n}_0$ is not defined, and the interpretation of both ξ^2_{KU} and ξ^2_{W} must be handled with care. This is one of the reasons why more general squeezing and entanglement criteria were later developed [27, 28].
Lastly, their original work, Kitagawa and Ueda also introduced two paradigmatic nonlinear Hamiltonians capable of generating spin squeezing from an initial coherent spin state [25]:

$$\hat{H}_{\mathrm{OAT}} = \chi \hat{J}_z^2, \qquad \hat{H}_{\mathrm{TAT}} = \chi(\hat{J}_x^2 - \hat{J}_y^2), \tag{1.192}$$

up to trivial relabelings of the axes. The nonlinear dynamics shears the initially isotropic fluctuation profile in the transverse plane, producing a squeezed axis and an anti-squeezed one. The one-axis twisting model is analytically tractable and experimentally widespread, while the two-axis counter-twisting model is more efficient and can approach Heisenberg scaling more closely, although it is harder to engineer.

1.7.3 Spin squeezing and symmetric states

We now discuss an important result due to Wang and Sanders [29], which connects the Kitagawa-Ueda squeezing parameter to negative pairwise correlations and pairwise entanglement for symmetric multiqubit states.
Let $\mathbf{n}_\perp$ be a direction orthogonal to the mean spin, and define the local Pauli operator

$$\sigma^{(i)}_{\mathbf{n}_\perp} = \mathbf{n}_\perp \cdot \vec{\sigma}^{(i)}. \tag{1.193}$$

For a symmetric state, all one-body expectations $\langle \sigma^{(i)}_{\mathbf{n}_\perp} \rangle$ are equal, and all two-body correlations $\langle \sigma^{(i)}_{\mathbf{n}_\perp} \otimes \sigma^{(j)}_{\mathbf{n}_\perp} \rangle$ with $i \neq j$ are identical.

Lemma 1: For a symmetric separable state of N qubits, the pairwise transverse correlation is nonnegative:

$$\left\langle \sigma^{(i)}_{\mathbf{n}_\perp} \otimes \sigma^{(j)}_{\mathbf{n}_\perp} \right\rangle \geq 0, \qquad i \neq j. \tag{1.194}$$

Proof. A symmetric separable state may be written as

$$\hat{\rho}_{\mathrm{sep}} = \sum_k p_k\, \hat{\rho}^{(k)} \otimes \hat{\rho}^{(k)} \otimes \cdots \otimes \hat{\rho}^{(k)}, \qquad p_k \geq 0, \qquad \sum_k p_k = 1. \tag{1.195}$$

Its two-qubit reduced state is therefore

$$\hat{\rho}_{ij} = \sum_k p_k \, \hat{\rho}^{(k)} \otimes \hat{\rho}^{(k)}. \tag{1.196}$$

Hence,

$$\left\langle \sigma^{(i)}_{\mathbf{n}_\perp} \otimes \sigma^{(j)}_{\mathbf{n}_\perp} \right\rangle = \sum_k p_k \, \mathrm{Tr}\left[(\hat{\rho}^{(k)} \otimes \hat{\rho}^{(k)})(\sigma_{\mathbf{n}_\perp} \otimes \sigma_{\mathbf{n}_\perp}) \right]. \tag{1.197}$$

Since the tensor factors are identical, this becomes

$$\left\langle \sigma^{(i)}_{\mathbf{n}_\perp} \otimes \sigma^{(j)}_{\mathbf{n}_\perp} \right\rangle = \sum_k p_k \left(\mathrm{Tr}\left[\hat{\rho}^{(k)} \sigma_{\mathbf{n}_\perp} \right] \right)^2 \geq 0. \tag{1.198}$$

This proves the claim.

Proposition: For an arbitrary symmetric multiqubit state, the condition $\xi^2_{\mathrm{KU}} < 1$ implies pairwise entanglement.

Proof. For a direction $\mathbf{n}_\perp$ orthogonal to the mean spin, one has $\left\langle \hat{J}_{\mathbf{n}_\perp} \right\rangle = 0$, and therefore

$$(\Delta \hat{J}_{\mathbf{n}_\perp})^2 = \left\langle \hat{J}^2_{\mathbf{n}_\perp} \right\rangle. \tag{1.199}$$

Now,

$$\hat{J}_{\mathbf{n}_\perp} = \frac{1}{2} \sum_{k=1}^{N} \sigma^{(k)}_{\mathbf{n}_\perp}, \tag{1.200}$$

hence

$$\hat{J}^2_{\mathbf{n}_\perp} = \frac{1}{4} \sum_{k=1}^{N} \left(\sigma^{(k)}_{\mathbf{n}_\perp} \right)^2 + \frac{1}{4} \sum_{\substack{k,\ell=1 \\ k \neq \ell}}^{N} \sigma^{(k)}_{\mathbf{n}_\perp} \sigma^{(\ell)}_{\mathbf{n}_\perp}. \tag{1.201}$$

Since $\left(\sigma^{(k)}_{\mathbf{n}_\perp} \right)^2 = \mathbb{1}$, taking the expectation value yields

$$\left\langle \hat{J}^2_{\mathbf{n}_\perp} \right\rangle = \frac{N}{4} + \frac{1}{4} \sum_{\substack{k,\ell=1 \\ k \neq \ell}}^{N} \left\langle \sigma^{(k)}_{\mathbf{n}_\perp} \sigma^{(\ell)}_{\mathbf{n}_\perp} \right\rangle. \tag{1.202}$$

By exchange symmetry, all pair correlations are equal, so for any $i \neq j$,

$$\left\langle \hat{J}^2_{\mathbf{n}_\perp} \right\rangle = \frac{N}{4} + \frac{N(N-1)}{4} \left\langle \sigma^{(i)}_{\mathbf{n}_\perp} \otimes \sigma^{(j)}_{\mathbf{n}_\perp} \right\rangle. \tag{1.203}$$

Substituting this into the Kitagawa-Ueda definition gives

$$\xi^2_{\mathrm{KU}} = \frac{4}{N} (\Delta \hat{J}_{\mathbf{n}_\perp})^2 = 1 + (N-1) \left\langle \sigma^{(i)}_{\mathbf{n}_\perp} \otimes \sigma^{(j)}_{\mathbf{n}_\perp} \right\rangle. \tag{1.204}$$

Therefore,

$$\xi^2_{\mathrm{KU}} < 1 \iff \left\langle \sigma^{(i)}_{\mathbf{n}_\perp} \otimes \sigma^{(j)}_{\mathbf{n}_\perp} \right\rangle < 0. \tag{1.205}$$

By Lemma 1, such a negative pairwise correlation is impossible for a symmetric separable two-qubit reduction. Hence, if $\xi^2_{\mathrm{KU}} < 1$, the reduced two-qubit state must be entangled. This proves pairwise entanglement.

This result is particularly important because it shows that, for symmetric states, Kitagawa-Ueda spin squeezing is not merely a fluctuation-based property: it is already a direct witness of nonclassical two-body correlations. At the same time, the implication is one-way in general. There exist symmetric entangled states, such as Dicke states, that are not spin squeezed according to the standard criterion.

2
Continuously Monitored Open Quantum Systems

The formulation presented in Chapter 1 provides the standard description of closed and isolated quantum systems. In that setting, the state is associated with a vector in Hilbert space, the dynamics is unitary, and measurements are described through the Born rule and the projection postulate. This framework is conceptually fundamental and mathematically consistent. At the same time, it relies on an idealization that is rarely realized in practice. Any realistic quantum system is, at least to some extent, coupled to additional degrees of freedom, whether they correspond to uncontrolled modes of the electromagnetic field, external reservoirs, measurement devices, or simply to the surrounding laboratory conditions [30, 31].
This observation is not merely of practical relevance. It marks a genuine conceptual extension of the standard theory. If the system under investigation interacts with external degrees of freedom, then its evolution is no longer described, in general, by a unitary operator acting only on its Hilbert space. The reduced dynamics may display decoherence, dissipation, irreversibility, and noise. These effects emerge naturally once one recognizes that the system of interest should be treated as a subsystem of a larger closed composite system.
For this reason, the theory of open quantum systems may be viewed as the natural continuation of the standard formulation of quantum mechanics. One distinguishes between the subsystem of interest, usually denoted by S, and an external environment E where the total system $S + E$ is still described within the ordinary postulates of quantum mechanics. However, the subsystem S is no longer isolated, and its state must be described through a reduced statistical operator.

2.1 Open quantum systems: formalism

To make this point more explicit, let us consider a quantum system S coupled to an environment E. The Hilbert space of the composite is

$$\mathcal{H}_{SE} = \mathcal{H}_S \otimes \mathcal{H}_E. \tag{2.1}$$

The total system $S+E$ is still assumed to be closed. Therefore, its dynamics is governed by a unitary evolution on $\mathcal{H}_{SE}$.
The total Hamiltonian is written as

$$\hat{H}_{SE} = \hat{H}_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \hat{H}_E + \hat{H}_{\text{int}}, \tag{2.2}$$

so the corresponding evolution of the global density operator is

$$\rho_{SE}(t) = \hat{U}_{SE}(t,t_0)\,\rho_{SE}(t_0)\,\hat{U}^{\dagger}_{SE}(t,t_0), \tag{2.3}$$

where $\hat{U}_{SE}(t,t_0)$ is unitary. Since the environment is not directly observed, the physically relevant description is obtained by tracing out its degrees of freedom:

$$\rho_S(t) = \mathrm{Tr}_E[\rho_{SE}(t)]\,. \tag{2.4}$$

By combining Eqs. (2.3) and (2.4), one obtains

$$\rho_S(t) = \mathrm{Tr}_E\left[\hat{U}_{SE}(t,t_0)\,\rho_{SE}(t_0)\,\hat{U}^{\dagger}_{SE}(t,t_0)\right]. \tag{2.5}$$

The total evolution is unitary, but not the reduced evolution of the subsystem. In the standard formulation for isolated systems, the evolution is described by a unitary operator acting on the Hilbert space of the system alone: such an evolution preserves purity, preserves the spectrum of the density operator, and is reversible. In particular, if the initial state is pure, it remains pure at all times.
By contrast, the reduced evolution in Eq. (2.5) is totally different, and it loses all these "good" properties. The reason is that the interaction term $\hat{H}_{\mathrm{int}}$ creates correlations between S and E. Even if the total state remains pure, the subsystem may become mixed because part of the information about S is now stored in the correlations with the environment. In this sense, the non-unitary character of the reduced dynamics comes from the fact that one is describing only a part of a larger quantum system.
A first important consequence is that an initially pure state of the subsystem may evolve into a mixed state. This is impossible under unitary dynamics on $\mathcal{H}_S$, but it is completely natural for an open system. A second consequence is that the reduced evolution need not be reversible. The partial trace is not an invertible operation: once the environmental degrees of freedom are discarded, the information encoded in system-environment correlations is no longer accessible from the subsystem alone. A third consequence is that quantities such as purity (1.70) and von Neumann entropy (1.120) are no longer conserved for the reduced state, even though they are conserved for the total isolated system.
It is therefore natural to replace the unitary description on $\mathcal{H}_S$ with a more general dynamical law. In the simplest and most relevant situation, one assumes that the initial global state is factorized,

$$\rho_{SE}(t_0) = \rho_S(t_0) \otimes \rho_E(t_0). \tag{2.6}$$

Under this assumption, the reduced evolution may be written as a map

$$\rho_S(t) = \Phi_{t,t_0}[\rho_S(t_0)]\,, \tag{2.7}$$

where Φ_{t,t_0} is, in general, not unitary. Rather, it is a quantum dynamical map acting on density operators [30, 31].
This observation already shows why the standard postulates of closed-system quantum mechanics are not sufficient in the present context. The appropriate framework is instead based on generalized measurements and quantum channels. Within this more general formulation, continuously monitored systems will arise as a further refinement, in which the open-system evolution is conditioned on the information extracted from the environment through a continuous measurement record.

2.1.1 Generalized measurements, the POVMs

The projective description of quantum measurements is sufficient when one deals with ideal measurements directly performed on the system Hilbert space. In many relevant situations, however, the measurement process is more involved. The system first interacts with additional degrees of freedom, such as an ancilla, a probe, or a measuring device, and only part of the total setup is eventually observed. In this case, the effective measurement acting on the system alone is no longer described, in general, by a family of orthogonal projectors [31, 30].

This is the natural setting in which generalized measurements arise. Their role is to describe the statistics of outcomes when the measurement is indirect, noisy, or only partially resolving. From this point of view, generalized measurements are not a modification introduced by hand. They are the effective description of standard measurements performed on a larger Hilbert space and then reduced to the subsystem of interest.

A more general framework is therefore provided by positive operator-valued measures, POVMs. A POVM is a set of positive operators $\{\hat{\Pi}_x\}$ satisfying

$$\hat{\Pi}_x \geq 0, \qquad \sum_x \hat{\Pi}_x = \mathbb{1}. \tag{2.8}$$

The probability of obtaining the outcome x when the system is in the state ρ is

$$p_x = \mathrm{Tr}\left[\rho\,\hat{\Pi}_x\right]. \tag{2.9}$$

Projective measurements are recovered as a special case by taking $\Pi_x = \hat{P}_x$ where these are orthogonal projectors.

To describe the post-measurement state one introduces a set of operators $\{\hat{M}_x\}$, usually called detection operators or measurement operators, such that

$$\hat{\Pi}_x = \hat{M}_x^\dagger \hat{M}_x, \qquad \sum_x \hat{M}_x^\dagger \hat{M}_x = \mathbb{1}. \tag{2.10}$$

In terms of these operators, the generalized Born rule may be written as

$$p_x = \mathrm{Tr}\left[\hat{M}_x \rho \hat{M}_x^\dagger\right] = \mathrm{Tr}\left[\rho\,\hat{M}_x^\dagger \hat{M}_x\right] = \mathrm{Tr}\left[\rho\,\hat{\Pi}_x\right]. \tag{2.11}$$

If the outcome x is recorded, the corresponding conditional state is

$$\rho_x = \frac{\hat{M}_x \rho \hat{M}_x^\dagger}{p_x} = \frac{\hat{M}_x \rho \hat{M}_x^\dagger}{\mathrm{Tr}\left[\rho\,\hat{\Pi}_x\right]}. \tag{2.12}$$

If the measurement is performed but the outcome is not read out, the post-measurement state is

$$\widetilde{\rho} = \sum_x p_x \rho_x = \sum_x \hat{M}_x \rho \hat{M}_x^\dagger. \tag{2.13}$$

This formulation contains the projective one, but is strictly more general. In particular, orthogonality of the POVM elements is not required, and the number of outcomes is not bounded by the dimension of the Hilbert space. Moreover, for a fixed POVM the detection operators are not unique. Since each $\hat{\Pi}_x$ is positive, its square root is well defined, and one may always write

$$\hat{M}_x = \hat{U}_x \sqrt{\hat{\Pi}_x}, \tag{2.14}$$

where $\hat{U}_x$ is unitary on the support of $\hat{\Pi}_x$. This is simply the polar decomposition of $\hat{M}_x$. It shows that the POVM elements determine the probabilities, whereas the unitary part of the detection operators determines how the state is transformed after the measurement.

The definition above extends naturally to continuous outcome spaces, where sums are replaced by operator-valued measures. For the present purposes, the discrete version is sufficient and already contains the essential ideas.

The relation between generalized measurements and ordinary projective measurements is clarified by the Naimark theorem [31]. First, we have to discuss about the ancilla. It is an auxiliary quantum system introduced in addition to the system of interest. It is typically used to model additional degrees of freedom involved in a physical process, such as a probe, a measurement apparatus, or an effective environment, and it allows one to describe indirect measurements and more general state transformations on an enlarged Hilbert space. Consider a system S coupled to an ancilla A, prepared in a reference state $|\omega_A\rangle$. Let the composite evolve unitarily, and let a projective measurement $\{\hat{P}_x\}$ be performed on the ancilla. Starting from the factorized state

$$\rho_{SA} = \rho_S \otimes |\omega_A\rangle \langle\omega_A| , \tag{2.15}$$

the probability of the outcome x can be written as

$$p_x = \mathrm{Tr}_S\left[\rho_S \hat{\Pi}_x\right], \tag{2.16}$$

where

$$\hat{\Pi}_x = \mathrm{Tr}_A\Big[(\mathbb{1}_S \otimes |\omega_A\rangle \langle\omega_A|)\, \hat{U}^\dagger \left(\mathbb{1}_S \otimes \hat{P}_x\right) \hat{U}\Big] . \tag{2.17}$$

The operators $\hat{\Pi}_x$ are positive and satisfy completeness, therefore they define a POVM on $\mathcal{H}_S$.

The Naimark theorem states that the converse is also true: every POVM on $\mathcal{H}_S$ can be realized as a projective measurement on a larger Hilbert space, obtained by introducing suitable ancillary degrees of freedom and a unitary interaction. Generalized measurements are the effective description viewed as ordinary projective measurements performed indirectly on an enlarged system.

(METTERE IMMAGINE STILE PARIS - SCHEMA MISURA GENERALIZZATA)

2.1.2 Quantum maps: CPTP transformations

As we have seen before, the interaction with unobserved degrees of freedom, as well as the possibility of conditioning on measurement outcomes, leads to transformations that need not preserve purity and need not be reversible. This motivates the introduction of quantum maps, namely linear maps acting on statistical operators [30].

A quantum map is a rule that associates to each input state ρ an output operator $\Phi[\rho]$. In the physical setting of interest, this rule must send density operators into density operators, so that positivity and trace are preserved. Moreover, since statistical mixtures must evolve consistently, the map has to be linear. In this way, quantum maps provide the natural generalization of the unitary transformation law of closed quantum systems. The unitary case is recovered when $\Phi[\rho] = \hat{U}\rho\hat{U}^\dagger$. In general, however, open-system transformations are not of this form: they may decrease coherences, transform pure states into mixed states, and describe irreversible processes. This is precisely the kind of behavior that cannot be captured within the closed-system framework alone. A simple example makes the difference clear. Consider a qubit and define the map

$$\Phi_p[\rho] = (1-p)\rho + p\,\sigma_z\rho\sigma_z, \qquad 0 \le p \le 1, \tag{2.18}$$

called the phase-flip channel. This map is trace preserving and sends states into states. However, it is not unitary unless $p = 0$ or $p = 1$. In the computational basis $\{|0\rangle, |1\rangle\}$ and for $\rho = |+\rangle\langle+|$ one has

$$\Phi_p[\rho_+] = \frac{1}{2}\begin{pmatrix} 1 & 1-2p \\ 1-2p & 1 \end{pmatrix}. \tag{2.19}$$

For $p = 1/2$ the output state is maximally mixed, and therefore a pure state is mapped into a mixed state. This is impossible under unitary evolution on the system Hilbert space alone.

The physically relevant class of transformations is selected by a few basic requirements. First, the map must be linear, in order to preserve the convex structure of statistical mixtures. Second, it must preserve positivity and trace, so that density operators are mapped into density operators. Most importantly, it must be completely positive. This means that the extension of the map to any larger Hilbert space, $\Phi \otimes \mathbb{1}_A$ must still preserve positivity for every ancillary system A. This requirement is essential because the system of interest may be correlated, or even entangled, with additional degrees of freedom. Positivity on the system alone is thus not sufficient.

A quantum physical transformation is thus described by a completely positive trace-preserving map, usually abbreviated as CPTP map. More generally, if one considers a selective transformation associated with a specific measurement outcome, the trace need not be preserved. In that case one deals with a completely positive trace-non-increasing map.

2.1.3 Stinespring theorem and Kraus' operator-sum representation

The general structure of quantum maps is fixed by two important theorems. In finite dimension, this result states that any CPTP map can be represented in two equivalent ways: either as a reduced unitary evolution on a larger Hilbert space [32], or through an operator-sum representation [33].

Let S be the system of interest and E an ancilla or effective environment, prepared in a reference state $|\omega_E\rangle\langle\omega_E|$. Let the composite system evolve under a unitary operator $\hat{U}$ on

$\mathcal{H}_S \otimes \mathcal{H}_E$. The reduced transformation of the system is then

$$\Phi[\rho] = \mathrm{Tr}_E\left[\hat{U}\left(\rho \otimes |\omega_E\rangle\langle\omega_E|\right)\hat{U}^\dagger\right]. \tag{2.20}$$

This is the Stinespring form of the map. It shows that every admissible reduced dynamics may be viewed as ordinary unitary evolution on an enlarged Hilbert space followed by a partial trace.
To obtain the operator-sum representation, let $\{|k_E\rangle\}$ be an orthonormal basis of $\mathcal{H}_E$, and define the operators

$$\hat{K}_j = \langle k_E|\,\hat{U}\,|\omega_E\rangle\,. \tag{2.21}$$

These operators act on $\mathcal{H}_S$. By expanding the partial trace in the basis $\{|k_E\rangle\}$, one finds

$$\Phi[\rho] = \sum_j \hat{K}_j\rho\hat{K}_j^\dagger \tag{2.22}$$

they also satisfy the normalization condition. Equation (2.22) is called the operator-sum representation or Kraus representation of the map.
The converse statement also holds: given any family of operators $\{\hat{K}_j\}$ satisfying Eq. (**??**), the map in Eq. (2.22) is completely positive and trace preserving. In this way, the two descriptions are equivalent.
Unitary evolution is recovered as a special case. If there is only one Kraus operator, $\hat{K}_1 = \hat{U}$, then Eq. (2.22) reduces to the usual unitary evolution of the generic density operator.
This formalism also unifies dynamics and measurement. In the measurement setting, the Kraus operators are precisely the measurement operators introduced in the previous subsection. A selective outcome corresponds to a trace-non-increasing completely positive map, while the sum over all outcomes yields a CPTP map. For this reason, the Kraus representation provides the natural language for both reduced dynamics and generalized measurements.
It is worth remarking that the Kraus representation is not unique, different sets of Kraus operators may describe the same map. This non-uniqueness reflects the fact that the same reduced transformation may arise from different Stinespring dilations, or equivalently, from different choices of basis in the ancillary Hilbert space.

2.1.4 Generalized quantum postulates

The formalism introduced in the previous sections leads to a natural extension of the standard quantum postulates where the usual formulation remains fully valid for isolated systems, ideal measurements, and unitary dynamics.
The first modification concerns the notion of state. In the standard formulation, pure states represented by vectors in Hilbert space play a central role but in the open-system setting, instead, the natural object is the density operator. This is the correct description of a subsystem that may be correlated with an external environment: as shown in Eq. (2.4), even a globally pure state may correspond to a mixed reduced state.

The second modification concerns measurements. In the standard case, a measurement is described by orthogonal projectors, and the same objects determine both the outcome probabilities and the post-measurement state. In the generalized setting, these two aspects are separated. The outcomes statistics is described by a POVM (2.8), while the corresponding state transformation is described by the measurement operators of Eq. (2.10). This extension is necessary whenever the measurement is indirect, noisy, or only partially resolving.

The third modification concerns dynamics. For closed systems, time evolution is unitary and therefore reversible. By contrast, reduced dynamics may transform pure states into mixed states and may describe irreversible processes. For this reason, the dynamical law is generalized into completely positive maps. In the deterministic case, one deals with trace-preserving maps. In the selective case, associated with a specific outcome, one generally obtains trace-non-increasing maps.

2.2 Markovian master equations

We now move from the general framework of open quantum systems introduced in the previous sections to a specific and particularly important class of reduced dynamics, namely Markovian dynamics. The main idea is that, although the total system made of the quantum system and its environment still evolves unitarily, the reduced evolution of the system alone may acquire dissipation, decoherence, and irreversibility once the environmental degrees of freedom are traced out. A central problem in the theory of open quantum systems is therefore to identify the conditions under which this reduced dynamics can be described by a closed differential equation acting only on the system state.

In the Markovian regime, the environment is assumed not to retain relevant memory of its past interaction with the system. As a consequence, the reduced evolution becomes time local, and its generator takes the well-known Gorini–Kossakowski–Sudarshan–Lindblad form. Our goal in this section is to derive this master equation in a physically transparent way by modeling the environment as a bosonic field and by exploiting the white-noise limit. This construction will also be the natural starting point for the next section, where the same system-bath interaction will be used to derive the stochastic unravellings associated with continuous monitoring [**Wi**, 30, 31, 34, 35].

We consider a quantum system S coupled to a bosonic environment. The total Hilbert space is

$$\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B. \tag{2.23}$$

The total Hamiltonian is written as

$$\hat{H}_{\text{tot}} = \hat{H}_S + \hat{H}_B + \hat{H}_{\text{int}}. \tag{2.24}$$

The free bath Hamiltonian is

$$\hat{H}_B = \int d\omega\, \omega\, \hat{b}^\dagger_\omega \hat{b}_\omega, \tag{2.25}$$

where the operators $\hat{b}_\omega$ satisfy the canonical bosonic commutation relations

$$[\hat{b}_\omega, \hat{b}^\dagger_{\omega'}] = \delta(\omega - \omega'). \tag{2.26}$$

We assume that the system couples linearly to the bath through a system operator $\hat{c}$. In the interaction picture with respect to the free bath dynamics, and after the rotating-wave approximation, the interaction Hamiltonian takes the form

$$\hat{H}_{\text{int}}(t) = i\sqrt{\frac{\gamma}{2\pi}} \int d\omega \left(\hat{c}\,\hat{b}^\dagger_\omega e^{i(\omega-\omega_0)t} - \hat{c}^\dagger \hat{b}_\omega e^{-i(\omega-\omega_0)t}\right), \tag{2.27}$$

where ω_0 is a characteristic system frequency and $\gamma \geq 0$ sets the coupling strength. At this stage, the relevant approximation is that the system-bath coupling is effectively flat over a frequency window much broader than the dynamical scales of the reduced system. This is the usual broadband or white-noise limit, and it is the basic assumption behind the Markovian description.

It is convenient to introduce time-domain bath operators

$$\hat{b}_t = \frac{1}{\sqrt{2\pi}} \int d\omega\, \hat{b}_\omega e^{-i(\omega-\omega_0)t}, \tag{2.28}$$

for which one finds

$$[\hat{b}_t, \hat{b}^\dagger_{t'}] = \delta(t - t'). \tag{2.29}$$

In terms of these operators, Eq. (2.27) becomes

$$\hat{H}_{\text{int}}(t) = i\sqrt{\gamma}\left(\hat{c}\,\hat{b}^\dagger_t - \hat{c}^\dagger \hat{b}_t\right). \tag{2.30}$$

The singular commutation relation in Eq. (2.29) expresses the fact that bath correlations are local in time. Physically, this means that the environment has no relevant memory on the timescale of the system evolution. In this sense, the reduced dynamics is Markovian: the future state depends only on the present state and not on the previous system-bath history.

The key simplification is to regard the bath as a continuous train of independent time bins [36, 37]. Over an infinitesimal interval $[t, t + dt]$, we introduce the quantum stochastic increments [38]

$$d\hat{B}_t = \hat{b}_t\, dt, \qquad d\hat{B}^\dagger_t = \hat{b}^\dagger_t\, dt. \tag{2.31}$$

Because of Eq. (2.29), these increments satisfy the Ito table

$$d\hat{B}_t\, d\hat{B}^\dagger_t = dt, \qquad d\hat{B}^\dagger_t d\hat{B}_t = 0, \qquad d\hat{B}^2_t = (d\hat{B}^\dagger_t)^2 = 0, \tag{2.32}$$

where the vacuum input has been assumed.

The Markovian approximation is encoded in the assumption that, at each infinitesimal step, the system interacts with a fresh environmental mode prepared in the same state. In the vacuum case, this means that at time t the joint state factorizes as

$$\hat{R}(t) = \rho(t) \otimes |0_t\rangle\langle 0_t|, \tag{2.33}$$

where $\rho(t)$ is the reduced system state and $|0\rangle_t$ is the vacuum state of the time bin centered at t. This factorization expresses the absence of memory between different collisions: the environment subsystem interacting with the system at time t is uncorrelated with the ones that interacted at earlier times.
Using Eq. (2.30), the infinitesimal interaction can be written as

$$\hat{H}_{\text{int}}(t)\,dt = i\sqrt{\gamma}\left(\hat{c}\,d\hat{B}_t^\dagger - \hat{c}^\dagger d\hat{B}_t\right). \tag{2.34}$$

The corresponding infinitesimal unitary reads

$$\hat{U}(t+dt,t) = \exp\left[-i\hat{H}_{\text{int}}(t)\,dt\right]. \tag{2.35}$$

Expanding up to order dt, and using Eq. (2.32), one obtains

$$\hat{U}(t+dt,t) = \mathbb{1} + \sqrt{\gamma}\left(\hat{c}\,d\hat{B}_t^\dagger - \hat{c}^\dagger d\hat{B}_t\right) - \frac{\gamma}{2}\hat{c}^\dagger\hat{c}\,dt + \mathcal{O}(dt^{3/2}). \tag{2.36}$$

Here one keeps all contributions up to order dt. Terms of order $dt^{3/2}$ or higher are neglected in the continuous-time limit. This is consistent because the stochastic increments scale as $\sqrt{dt}$.
When this operator acts on a system state tensored with the vacuum of the time bin, the only relevant relations are

$$d\hat{B}_t\,|0\rangle_t = 0, \qquad d\hat{B}_t^\dagger\,|0\rangle_t = \sqrt{dt}\;|1\rangle_t, \qquad d\hat{B}_t d\hat{B}_t^\dagger\,|0\rangle_t = dt\;|0\rangle_t. \tag{2.37}$$

Hence, for a generic system vector $|\psi\rangle$,

$$\hat{U}(t+dt,t)\,(|\psi\rangle\otimes|0\rangle_t) = \left(\mathbb{1} - \frac{\gamma}{2}\hat{c}^\dagger\hat{c}\,dt\right)|\psi\rangle\otimes|0\rangle_t + \sqrt{\gamma\,dt}\,\hat{c}\,|\psi\rangle\otimes|1\rangle_t + \mathcal{O}(dt^{3/2}). \tag{2.38}$$

This expression already displays the physical picture: either no excitation is emitted into the output time bin, or one excitation is created with amplitude proportional to $\sqrt{dt}$.
The reduced system state at time $t+dt$ is obtained by tracing out the time bin:

$$\rho(t+dt) = \mathrm{Tr}_t\left[\hat{U}(t+dt,t)\,\hat{R}(t)\,\hat{U}^\dagger(t+dt,t)\right]. \tag{2.39}$$

Substituting Eqs. (2.33) and (2.36), and keeping terms up to first order in dt, one finds

$$\rho(t+dt) = \rho(t) + \gamma\left(\hat{c}\rho(t)\hat{c}^\dagger - \frac{1}{2}\left\{\hat{c}^\dagger\hat{c},\rho(t)\right\}\right)dt. \tag{2.40}$$

If one also includes the coherent evolution generated by a system Hamiltonian $\hat{H}$, the differential equation becomes

$$\frac{d\rho(t)}{dt} = -i[\hat{H},\rho(t)] + \gamma\left(\hat{c}\rho(t)\hat{c}^\dagger - \frac{1}{2}\left\{\hat{c}^\dagger\hat{c},\rho(t)\right\}\right). \tag{2.41}$$

This is the hallmark of the Markovian approximation. The generator depends only on the instantaneous state $\rho(t)$, and no memory kernel appears.

Introducing the dissipator superoperator

$$\mathcal{D}[\hat{A}]\rho = \hat{A}\rho\hat{A}^\dagger - \frac{1}{2}\left\{\hat{A}^\dagger\hat{A}, \rho\right\}, \tag{2.42}$$

Eq. (2.41) is written as

$$\frac{d\rho}{dt} = -i[\hat{H}, \rho] + \gamma\mathcal{D}[\hat{c}]\rho. \tag{2.43}$$

For several independent output channels $\hat{c}_j$, one obtains the Gorini–Kossakowski––Sudarshan–Lindblad (GKSL) form [39]

$$\frac{d\rho(t)}{dt} = \mathcal{L}\rho(t) = -i[\hat{H}, \rho(t)] + \sum_j \gamma_j\mathcal{D}[\hat{c}_j]\rho(t). \tag{2.44}$$

The corresponding dynamical maps form a quantum dynamical semigroup,

$$\mathcal{E}_t = e^{t\mathcal{L}}, \qquad \mathcal{E}_{t+s} = \mathcal{E}_t \circ \mathcal{E}_s, \tag{2.45}$$

which makes explicit the time-homogeneous Markovian character of the evolution. The semigroup property reflects the absence of memory in the reduced dynamics, while the GKSL structure guarantees complete positivity and trace preservation [40].

2.3 Stochastic master equations

The master equation (2.44) describes the unconditional evolution of the system, namely the reduced dynamics obtained when the environmental degrees of freedom are ignored. In this description, the bath is traced out and no information is extracted from the output field. As a consequence, the state $\rho(t)$ represents the average evolution over all the possible ways in which the system may have interacted with the environment.
This point is conceptually important. The Lindblad master equation provides a complete description of the reduced dynamics at the unconditional level, but it does not keep track of the information that may be carried away by the environment. If one starts monitoring the output field, the situation changes. The environment becomes a carrier of information about the system: the evolution must then be conditioned on the specific measurement outcomes observed in time.
This leads to a more refined dynamical description. Instead of a deterministic equation for the averaged state, one obtains a stochastic equation for the conditional state [41], that is, for the state of the system given the measurement record accumulated up to time t. The corresponding equations are the stochastic master equations. Their role is to describe how the system state is continuously updated as information is progressively extracted from the output field through a measurement process [30, 34].
At this stage, the notion of unravelling becomes central. The unconditional master equation fixes the average reduced dynamics, but it does not determine a unique conditioned evolution. In fact, the same Lindblad equation could be associated with different stochastic processes, depending on how the environment is monitored. Each of these stochastic descriptions is called an **unravelling of the master equation**.

The unconditional state $\rho(t)$ is a mixed state that results from averaging over all the possible measurement records, whether these records are actually observed or not. An unravelling resolves this average evolution into individual conditioned trajectories. Each trajectory corresponds to one possible realization of the measurement record and therefore to one possible conditional evolution of the system. When one averages over all these trajectories, the unconditional state is recovered:

$$\rho_{traj}(t) = \mathbb{E}[\rho_c(t)]\,, \tag{2.46}$$

where $\rho_c(t)$ denotes the conditional state and $\mathbb{E}[\cdot]$ denotes the average over all records/-trajectories.

This relation shows that the stochastic evolution is not an alternative to the master equation, but rather a finer description of the same physical dynamics. The master equation gives the average behavior. The unravelling gives the trajectory-by-trajectory behavior associated with a specific monitoring scheme. For this reason, different unravellings correspond to different experimental ways of extracting information from the environment, and therefore to different conditioned dynamics, all compatible with the same unconditional Lindblad evolution.

Two paradigmatic examples are particularly important. If the output field is monitored through photon counting, the measurement record is discontinuous: one observes random detection events, and the conditional dynamics is described by quantum jumps. If the output field is instead monitored through homodyne or heterodyne detection, the measurement record is continuous but noisy, and the conditional dynamics is diffusive. These two cases look very different at the level of individual trajectories, yet they reproduce the same unconditional master equation after averaging over the measurement outcomes. This is precisely why the concept of unravelling is so useful: it connects one and the same reduced open-system dynamics with different ways of reading out the information stored in the environment.

From a physical point of view, continuously monitored quantum systems lie exactly at this interface between dissipation and information. On the one hand, the coupling to the environment produces decoherence and irreversible reduced dynamics. On the other hand, the same coupling opens the possibility of observing the output field and updating the system state conditionally on the acquired record. Stochastic master equations are the mathematical tool that captures this interplay between backaction and information gain.

2.3.1 Photodetection, the quantum-jump unravelling

We first consider the case in which the output field is continuously monitored by photodetection. In this measurement scheme, one counts the excitations emitted into the output channel. The corresponding measurement record is therefore discontinuous: at each infinitesimal time step, either the detector clicks or it does not. For this reason, the associated stochastic evolution is called a quantum-jump unravelling of the master equation.

The physical picture is simple. During a short time interval $[t, t+dt]$, the system interacts with a single output time bin. After this interaction, one measures the number of excitations contained in that bin. Since the interaction lasts only an infinitesimal time, the probability of emitting more than one excitation is of order dt^2. Therefore, to first order in dt, only two outcomes are relevant: either no excitation is detected, or exactly one excitation is detected. These two possibilities are associated with the projectors

$$\hat{\Pi}_0 = |0_t\rangle\langle 0_t| \,, \qquad \hat{\Pi}_1 = |1_t\rangle\langle 1_t| \,, \tag{2.47}$$

acting on the output time bin.
Using the infinitesimal evolution derived in Eq. (2.38), and including the coherent system evolution generated by $\hat{H}$, one obtains

$$\hat{U}(t+dt, t)\,(|\psi\rangle \otimes |0\rangle_t) = \left(\mathbb{1} - i\hat{H}\,dt - \frac{\gamma}{2}\hat{c}^\dagger\hat{c}\,dt\right)|\psi\rangle \otimes |0\rangle_t + \sqrt{\gamma\,dt}\,\hat{c}\,|\psi\rangle \otimes |1\rangle_t + \mathcal{O}(dt^{3/2}). \tag{2.48}$$

This expression already contains the two possible outcomes. The first term corresponds to no detected excitation in the output bin. The second term corresponds to the detection of one excitation, with amplitude proportional to $\sqrt{dt}$.
From Eq. (2.48), one immediately reads the measurement operators acting on the system:

$$\hat{M}_0 = \mathbb{1} - i\hat{H}\,dt - \frac{\gamma}{2}\hat{c}^\dagger\hat{c}\,dt, \qquad \hat{M}_1 = \sqrt{\gamma\,dt}\,\hat{c}. \tag{2.49}$$

They satisfy the normalization condition

$$\hat{M}_0^\dagger\hat{M}_0 + \hat{M}_1^\dagger\hat{M}_1 = \mathbb{1} + \mathcal{O}(dt^2), \tag{2.50}$$

as required for a consistent infinitesimal measurement scheme.
Let $\rho_c(t)$ denote the conditional state at time t. The probability of no click in the interval $[t, t+dt]$ is

$$p_0 = \mathrm{Tr}\left[\hat{M}_0\rho_c(t)\hat{M}_0^\dagger\right] = 1 - \gamma\ \mathrm{Tr}\left[\hat{c}^\dagger\hat{c}\,\rho_c(t)\right]dt + \mathcal{O}(dt^2), \tag{2.51}$$

while the probability of one click is

$$p_1 = \mathrm{Tr}\left[\hat{M}_1\rho_c(t)\hat{M}_1^\dagger\right] = \gamma\ \mathrm{Tr}\left[\hat{c}^\dagger\hat{c}\,\rho_c(t)\right]dt + \mathcal{O}(dt^2). \tag{2.52}$$

As expected, the click probability is proportional to dt, and the proportionality factor is the conditional emission rate.
If no click is observed, the state is updated according to

$$\rho_c(t+dt) = \frac{\hat{M}_0\rho_c(t)\hat{M}_0^\dagger}{p_0}. \tag{2.53}$$

Expanding up to first order in dt, one finds

$$\rho_c(t+dt) = \rho_c(t) + dt\left[-i[\hat{H}, \rho_c(t)] - \frac{\gamma}{2}\left\{\hat{c}^\dagger\hat{c}, \rho_c(t)\right\} + \gamma\ \mathrm{Tr}\left[\hat{c}^\dagger\hat{c}\,\rho_c(t)\right]\rho_c(t)\right]. \tag{2.54}$$

This is the smooth conditional evolution between clicks. It is nonlinear because the state is continuously renormalized under the condition that no detection event has occurred. If one click is observed, the state instead undergoes the jump

$$\rho_c(t+dt) = \frac{\hat{M}_1 \rho_c(t) \hat{M}_1^\dagger}{p_1} = \frac{\hat{c}\,\rho_c(t)\,\hat{c}^\dagger}{\mathrm{Tr}[\hat{c}^\dagger \hat{c}\,\rho_c(t)]}. \tag{2.55}$$

This expression makes the jump interpretation explicit. A detection event corresponds to the application of the jump operator $\hat{c}$, followed by normalization.
The two conditional updates can be combined into a single stochastic equation by introducing the counting increment dN_t, defined by

$$dN_t = \begin{cases} 1, & \text{if a click occurs in } [t, t+dt], \\ 0, & \text{otherwise.} \end{cases} \tag{2.56}$$

This is a Poisson increment satisfying

$$dN_t^2 = dN_t, \qquad \mathbb{E}[dN_t] = \gamma\ \mathrm{Tr}\big[\hat{c}^\dagger \hat{c}\,\rho_c(t)\big]\,dt. \tag{2.57}$$

In terms of dN_t, the conditional evolution can be written as

$$d\rho_c = dN_t \left(\frac{\hat{c}\,\rho_c\,\hat{c}^\dagger}{\mathrm{Tr}[\hat{c}^\dagger \hat{c}\,\rho_c]} - \rho_c \right) + dt \left[-i[\hat{H}, \rho_c] - \frac{\gamma}{2} \left\{ \hat{c}^\dagger \hat{c}, \rho_c \right\} + \gamma\ \mathrm{Tr}\big[\hat{c}^\dagger \hat{c}\,\rho_c\big]\,\rho_c \right]. \tag{2.58}$$

This is the normalized stochastic master equation for direct photodetection.
It is useful to comment on the structure of this equation. The term proportional to dt describes the continuous evolution between detection events. The term proportional to dN_t describes the discontinuous state update associated with a click. The stochasticity of the process is entirely encoded in the random increment dN_t, whose average is fixed by the conditional counting rate in Eq. (2.57). The evolution is therefore piecewise deterministic: the state evolves smoothly in the intervals between clicks and undergoes sudden jumps whenever a detection event occurs.
Finally, by averaging Eq. (2.58) over all realizations of the counting process, one recovers the unconditional master equation (2.44), this confirms that photodetection provides a jump unravelling of the Lindblad dynamics.

2.3.2 Homodyne detection, a diffusive unravelling

We now consider homodyne detection. In this case, the output field is not monitored by counting excitations, but by mixing it with a strong local oscillator and measuring the resulting photocurrent. The quantity returned by the detector is therefore not a discrete sequence of clicks, but a continuous and noisy measurement record. For this reason, the corresponing stochastic evolution is diffusive rather than discontinuous. The information extracted at each infinitesimal step is weak, but it is accumulated in time through a continuous photocurrent. The conditional state then evolves continuously, although in a stochastic way.

IN this situation, one projects the output time bin onto the quadrature basis selected by the phase ϕ of the local oscillator, described by the operator

$$\hat{x}_\theta = e^{-i\phi}\,\hat{B}_t + e^{i\phi}\,\hat{B}_t^\dagger. \tag{2.59}$$

We denote by $|x_\phi\rangle_t$ the generalized eigenstates of the measured quadrature operator. The corresponding unnormalized conditional state is

$$\tilde{\rho}_c(t+dt) = {}_t\langle x_\phi|\,\hat{U}(t+dt,t)\,(\rho_c(t)\otimes|0_t\rangle\langle 0_t|)\,\hat{U}^\dagger(t+dt,t)\,|x_\phi\rangle_t\,, \tag{2.60}$$

so the corresponding probability density is $p(x_\phi, t+dt) = \mathrm{Tr}[\tilde{\rho}_c(t+dt)]$.
The normalized conditional state is then

$$\rho_c(t+dt) = \frac{\tilde{\rho}_c(t+dt)}{p(x_\phi, t+dt)}. \tag{2.61}$$

At this stage, the relevant point is that, unlike photodetection, it is sufficient to expand the numerator and the denominator only up to order $\sqrt{\gamma\,dt}$. Indeed, after normalization, all the contributions of order $\gamma\,dt$ will be correctly recovered.
Before the interaction, the distribution of the quadrature outcome is the vacuum Gaussian

$$p(x_\phi, t) = |{}_t\langle x_\phi|0_t\rangle\,|^2 = \frac{1}{\sqrt{\pi}}e^{-x_\phi^2}. \tag{2.62}$$

Using the infinitesimal unitary evolution and the overlaps between the vacuum, the one-photon state, and the quadrature eigenstates, one finds

$$\tilde{\rho}_c(t+dt) = p(x_\phi,t)\left[\rho_c(t) + \left(\hat{c}\,\rho_c(t)e^{i\phi} + \rho_c(t)\hat{c}^\dagger e^{-i\phi}\right)x_\phi\sqrt{2\gamma\,dt}\right] + o\!\left(\sqrt{\gamma\,dt}\right). \tag{2.63}$$

Taking the trace, one obtains

$$p(x_\phi, t+dt) = p(x_\phi,t)\left[1 + x_\phi\sqrt{2\gamma\,dt}\,\left\langle \hat{c}\,e^{i\phi} + \hat{c}^\dagger e^{-i\phi}\right\rangle_t\right] + o\!\left(\sqrt{\gamma\,dt}\right), \tag{2.64}$$

where

$$\left\langle \hat{c}\,e^{i\phi} + \hat{c}^\dagger e^{-i\phi}\right\rangle_t = \mathrm{Tr}\!\left[\rho_c(t)\,\left(\hat{c}\,e^{i\phi} + \hat{c}^\dagger e^{-i\phi}\right)\right]. \tag{2.65}$$

Equation (2.64) shows that the measurement outcome is still Gaussian, but its mean is shifted by the conditional expectation value of the measured quadrature. This motivates the introduction of the infinitesimal stochastic record

$$dy_t = \sqrt{2\,dt}\,x_\phi. \tag{2.66}$$

Its statistics is

$$dy_t = I(t)\,dt = \sqrt{\gamma}\,\left\langle \hat{c}\,e^{i\phi} + \hat{c}^\dagger e^{-i\phi}\right\rangle_t dt + dw_t, \tag{2.67}$$

where dw_t is a Wiener increment satisfying

$$\mathbb{E}[dw_t] = 0, \qquad dw_t^2 = dt. \tag{2.68}$$

The stochastic variable that corresponds to the time-integral is called the integrated photocurrent while the corresponding instantaneous photocurrent is formally defined as $I(t) = \frac{dy_t}{dt}$. Equation (2.67) then takes the form

$$I(t) = \sqrt{\gamma}\, \left\langle \hat{c}\, e^{i\phi} + \hat{c}^\dagger e^{-i\phi} \right\rangle_t + \xi(t), \tag{2.69}$$

where

$$\xi(t) = \frac{dw_t}{dt} \tag{2.70}$$

is a white-noise term, heuristically characterized by $\mathbb{E}[\xi(t)\xi(t')] = \delta(t-t')$.
This expression is physically very important. The photocurrent is not equal to the measured quadrature expectation value alone. Rather, it consists of two contributions: a deterministic signal term, given by the conditional average of the measured quadrature, and an irreducible noise term. The latter reflects the vacuum fluctuations of the output field and is present even for ideal detection. Therefore, the information extracted by homodyne monitoring is necessarily noisy. At each infinitesimal step the detector only gains weak information on the system, but this information is accumulated continuously in time through the full photocurrent record.
By inserting Eqs. (2.63) and (2.64) into Eq. (2.61), and expanding the denominator, one finds

$$d\rho_c(t) = \sqrt{\gamma}\, \mathcal{H}\left[\hat{c}\, e^{i\phi}\right] \rho_c(t)\, dw_t + o(\gamma\, dt), \tag{2.71}$$

where

$$\mathcal{H}[\hat{A}]\rho = \hat{A}\rho + \rho\hat{A}^\dagger - \mathrm{Tr}\left[(\hat{A} + \hat{A}^\dagger)\rho\right] \rho. \tag{2.72}$$

Since averaging over the measurement outcomes must reproduce the unconditional evolution, the deterministic contribution of order dt is uniquely fixed by the Lindblad master equation. One finally obtains

$$d\rho_c(t) = -i[\hat{H}, \rho_c(t)]\, dt + \gamma\, \mathcal{D}[\hat{c}]\rho_c(t)\, dt + \sqrt{\gamma}\, \mathcal{H}\left[\hat{c}\, e^{i\phi}\right] \rho_c(t)\, dw_t. \tag{2.73}$$

Equation (2.73) is the homodyne stochastic master equation in Itô form. It is a diffusive unravelling of the unconditional Lindblad dynamics. The deterministic part coincides with the unconditional evolution, while the stochastic term accounts for the continuous update of the state induced by the noisy quadrature record.
It is often convenient to rewrite the same equation directly in terms of the observed photocurrent. Using Eq. (2.67), one has

$$dw_t = dy_t - \sqrt{\gamma}\, \left\langle \hat{c}\, e^{i\phi} + \hat{c}^\dagger e^{-i\phi} \right\rangle_t dt, \tag{2.74}$$

and therefore

$$d\rho_c(t) = -i[\hat{H}, \rho_c(t)]\, dt + \gamma\, \mathcal{D}[\hat{c}]\rho_c(t)\, dt + \sqrt{\gamma}\, \mathcal{H}\left[\hat{c}\, e^{i\phi}\right] \rho_c(t) \left[dy_t - \sqrt{\gamma}\, \left\langle \hat{c}\, e^{i\phi} + \hat{c}^\dagger e^{-i\phi} \right\rangle_t dt\right]. \tag{2.75}$$

In this form, the quantity in square brackets is the innovation process, namely the difference between the actual measurement outcome and its conditional expectation value. This is the genuinely new information acquired by the observer at time t.

The interpretation of Eq. (2.75) is now transparent. The state evolves deterministically according to the unconditional master equation, but this evolution is continuously corrected by the information contained in the photocurrent. Since the record is noisy, the conditioned state follows a continuous but stochastic trajectory.
A convenient way to include finite efficiency $\eta \in [0,1]$ is to split the output channel into a monitored and an unmonitored part,

$$\sqrt{\gamma}\,\hat{c} \;\longrightarrow\; \sqrt{\eta\gamma}\,\hat{c} \quad \text{and} \quad \sqrt{(1-\eta)\gamma}\,\hat{c}. \tag{2.76}$$

The homodyne unravelling is then performed only on the monitored branch $\sqrt{\eta\gamma}\,\hat{c}$, while the second branch contributes only through an unconditional dissipator. Summing the two contributions gives again the total Lindblad term $\gamma\mathcal{D}[\hat{c}]\rho_c$, while only the observed branch enters the stochastic part of the equation. The measured photocurrent then becomes

$$dy_t = \sqrt{\eta\gamma}\,\left\langle \hat{c}\,e^{i\phi} + \hat{c}^\dagger e^{-i\phi}\right\rangle_t dt + dw_t, \tag{2.77}$$

and the corresponding stochastic master equation reads

$$d\rho_c(t) = -i[\hat{H},\rho_c(t)]\,dt + \gamma\,\mathcal{D}[\hat{c}]\rho_c(t)\,dt + \sqrt{\eta\gamma}\,\mathcal{H}\left[\hat{c}\,e^{i\phi}\right]\rho_c(t)\,dw_t. \tag{2.78}$$

As expected, the unconditional dissipative term is unchanged, whereas the stochastic term is reduced by the factor $\sqrt{\eta}$. In the limit $\eta \to 0$, no information is extracted from the environment and one recovers the unconditional Lindblad evolution.
Finally, it is useful to write the same infinitesimal evolution in a completely positive form. The normalized update over a time step dt can be written as

$$\rho_c(t+dt) = \frac{M_{dy_t}\rho_c(t)M_{dy_t}^\dagger + \gamma(1-\eta)\,\hat{c}\,\rho_c(t)\hat{c}^\dagger\,dt}{\mathrm{Tr}\left[M_{dy_t}\rho_c(t)M_{dy_t}^\dagger + \gamma(1-\eta)\,\hat{c}\,\rho_c(t)\hat{c}^\dagger\,dt\right]}, \tag{2.79}$$

where the generalized Kraus operator is

$$M_{dy_t} = \mathbb{1} - i\hat{H}\,dt - \frac{\gamma}{2}\hat{c}^\dagger\hat{c}\,dt + \sqrt{\eta\gamma}\,\hat{c}\,e^{i\phi}\,dy_t. \tag{2.80}$$

This formulation makes the complete positivity of each infinitesimal conditioned step explicit. The first term describes the monitored part of the evolution associated with the observed homodyne record, while the second term accounts for the unobserved fraction of the output channel due to the finite detection efficiency.

2.4 Relevant single-qubit paradigms

Before discussing the multi-qubit scenarios of interest, it is useful to isolate the two elementary single-qubit channels that will underlie the rest of the analysis. These are spontaneous emission and dispersive coupling. They correspond, respectively, to the choice of collapse operator $\hat{\sigma}_-$ and $\hat{\sigma}_z$. Although both are described within the same Lindblad and stochastic framework, they represent two physically different situations.

Spontaneous emission is a dissipative process. It involves an exchange of energy with the environment and drives the qubit from the excited state to the ground state. Its unconditional evolution is the standard amplitude-damping channel, while its monitored version gives rise to the familiar jump or diffusive quantum trajectories of fluorescence. Dispersive coupling has a different character. In that case, the environment acquires information about the qubit without inducing transitions between the energy eigenstates. The dynamics is therefore dominated by dephasing rather than relaxation. Populations are left unchanged, whereas coherences are progressively suppressed. For this reason, these two channels provide a natural pair of reference models: the first describes irreversible energy relaxation, while the second describes measurement-induced loss of phase coherence.
In both cases, the single-qubit dynamics already contains the essential physical mechanisms that later reappear in the multi-qubit setting. For this reason, it is convenient to analyze them first at the one-qubit level, where their structure is more transparent.

2.4.1 Single-qubit spontaneous emission

We begin with spontaneous emission. In the convention adopted throughout this thesis, $\hat{\sigma}_z |0\rangle = +|0\rangle$ and $\hat{\sigma}_z |1\rangle = -|1\rangle$, so that $|0\rangle$ is the ground state and $|1\rangle$ is the excited state. The lowering and raising operators are therefore

$$\hat{\sigma}_- = |0\rangle\langle 1| , \qquad \hat{\sigma}_+ = |1\rangle\langle 0| . \tag{2.81}$$

If one neglects any additional coherent term, or equivalently moves to a rotating frame in which the free qubit Hamiltonian has been removed, the unconditional dynamics is described by:

$$\frac{d\rho}{dt} = \gamma \mathcal{D}[\hat{\sigma}_-]\rho. \tag{2.82}$$

Since

$$\hat{c}^\dagger \hat{c} = \hat{\sigma}_+ \hat{\sigma}_- = |1\rangle\langle 1| , \tag{2.83}$$

the dissipator acts only on the excited-state component of the density operator.
Let us write the qubit state as

$$\rho(t) = \begin{pmatrix} \rho_{00}(t) & \rho_{01}(t) \\ \rho_{10}(t) & \rho_{11}(t) \end{pmatrix} \tag{2.84}$$

in the ordered basis $\{|0\rangle, |1\rangle\}$. By direct substitution into Eq. (2.82), one finds

$$\frac{d\rho}{dt} = \gamma \begin{pmatrix} \rho_{11}(t) & -\frac{1}{2}\rho_{01}(t) \\ -\frac{1}{2}\rho_{10}(t) & -\rho_{11}(t) \end{pmatrix} . \tag{2.85}$$

Therefore,

$$\dot{\rho}_{11}(t) = -\gamma \rho_{11}(t), \qquad \dot{\rho}_{00}(t) = \gamma \rho_{11}(t), \tag{2.86}$$

and

$$\dot{\rho}_{01}(t) = -\frac{\gamma}{2}\rho_{01}(t), \qquad \dot{\rho}_{10}(t) = -\frac{\gamma}{2}\rho_{10}(t). \tag{2.87}$$

The solution is then

$$\rho(t) = \begin{pmatrix} 1-\rho_{11}(0)e^{-\gamma t} & \rho_{01}(0)e^{-\gamma t/2} \\ \rho_{10}(0)e^{-\gamma t/2} & \rho_{11}(0)e^{-\gamma t} \end{pmatrix} . \tag{2.88}$$

This solution shows the two characteristic features of spontaneous emission. First, the excited-state population decays exponentially with rate γ, while the ground-state population increases accordingly. Second, the off-diagonal coherences decay with rate $\gamma/2$. The channel is therefore dissipative and not purely dephasing: populations and coherences are both affected, but in different ways. We notice that the unique stationary state is $|0\rangle\langle 0|$.
It is also useful to express the same dynamics in terms of the Bloch components

$$x(t) = \langle \hat{\sigma}_x \rangle \, , \qquad y(t) = \langle \hat{\sigma}_y \rangle \, , \qquad z(t) = \langle \hat{\sigma}_z \rangle \, . \tag{2.89}$$

Using the convention adopted here, one obtains

$$x(t) = x(0)e^{-\gamma t/2}, \qquad y(t) = y(0)e^{-\gamma t/2}, \tag{2.90}$$

and

$$z(t) = 1 - [1 - z(0)]\, e^{-\gamma t}. \tag{2.91}$$

Hence the Bloch vector is contracted toward the north pole, corresponding to the ground state $|0\rangle$.
The monitored version of the same process is equally transparent. Over an infinitesimal interval dt, the amplitude-damping channel may be described by the two Kraus operators

$$\hat{M}_0 = |0\rangle\langle 0| + \sqrt{1-\gamma dt}\; |1\rangle\langle 1| \, , \qquad \hat{M}_1 = \sqrt{\gamma dt}\, \hat{\sigma}_-. \tag{2.92}$$

The operator $\hat{M}_0$ corresponds to the no-click event, namely to the case in which no emitted quantum is detected in the interval dt. The operator $\hat{M}_1$ corresponds instead to a detection event. If the output field is photodetected and a click occurs, the conditional state becomes

$$\rho_c(t+dt) = \frac{\hat{M}_1 \rho_c(t) \hat{M}_1^\dagger}{\mathrm{Tr}\left[\hat{M}_1 \rho_c(t) \hat{M}_1^\dagger\right]} = |0\rangle\langle 0| \, . \tag{2.93}$$

Thus, a click heralds the decay of the qubit to the ground state. If no click is observed, the state is updated according to

$$\rho_c(t+dt) = \frac{\hat{M}_0 \rho_c(t) \hat{M}_0^\dagger}{\mathrm{Tr}\left[\hat{M}_0 \rho_c(t) \hat{M}_0^\dagger\right]}, \tag{2.94}$$

which yields a smooth conditional evolution in between jumps. Averaging over the two outcomes reproduces the unconditional map,

$$\rho(t+dt) = \hat{M}_0 \rho(t) \hat{M}_0^\dagger + \hat{M}_1 \rho(t) \hat{M}_1^\dagger, \tag{2.95}$$

and, in the continuous-time limit, one recovers Eq. (2.82).

For this reason, spontaneous emission provides the paradigmatic example of a channel whose unconditional evolution is dissipative, while its monitored evolution admits different stochastic unravellings. Photodetection leads to quantum-jump trajectories associated with random emission events, whereas homodyne monitoring of the emitted field leads to diffusive trajectories. In both cases, however, the average evolution remains the same and is governed by the amplitude-damping master equation.

2.4.2 Single-qubit pure dephasing

We now consider the second paradigmatic single-qubit channel, namely dispersive coupling. In this case, the environment couples to the qubit through the operator $\hat{\sigma}_z$. The corresponding unconditional dynamics is therefore generated by the master equation

$$\frac{d\rho}{dt} = \gamma \mathcal{D}[\hat{\sigma}_z]\rho. \tag{2.96}$$

This channel has a clear physical interpretation. Unlike spontaneous emission, it does not induce transitions between the two energy eigenstates. The environment acquires information about the qubit state, but no excitation is exchanged. For this reason, the dynamics is purely dephasing: populations are preserved, while coherences decay in time. In this sense, dispersive coupling provides the simplest example of decoherence without dissipation.
Since $\hat{\sigma}_z^\dagger = \hat{\sigma}_z$ and $\hat{\sigma}_z^2 = \mathbb{1}$, Eq. (2.96) may be rewritten as

$$\frac{d\rho}{dt} = \gamma \left(\hat{\sigma}_z \rho \hat{\sigma}_z - \rho \right). \tag{2.97}$$

q One then finds

$$\hat{\sigma}_z \rho(t) \hat{\sigma}_z = \begin{pmatrix} \rho_{00}(t) & -\rho_{01}(t) \\ -\rho_{10}(t) & \rho_{11}(t) \end{pmatrix}, \tag{2.98}$$

and therefore

$$\frac{d\rho}{dt} = \gamma \begin{pmatrix} 0 & -2\rho_{01}(t) \\ -2\rho_{10}(t) & 0 \end{pmatrix}. \tag{2.99}$$

It follows that

$$\dot{\rho}_{00}(t) = 0, \qquad \dot{\rho}_{11}(t) = 0, \tag{2.100}$$

while the coherences obey

$$\dot{\rho}_{01}(t) = -2\gamma\rho_{01}(t), \qquad \dot{\rho}_{10}(t) = -2\gamma\rho_{10}(t). \tag{2.101}$$

Altogether,

$$\rho(t) = \begin{pmatrix} \rho_{00}(0) & \rho_{01}(0)e^{-2\gamma t} \\ \rho_{10}(0)e^{-2\gamma t} & \rho_{11}(0) \end{pmatrix}. \tag{2.102}$$

The populations remain unchanged at all times, while the off-diagonal elements are exponentially suppressed. The channel therefore destroys phase coherence in the computational basis without affecting the occupations of $|0\rangle$ and $|1\rangle$. In particular, the asymptotic state is not uniqu, indeed very density operator that is diagonal in the computational basis is stationary. This is a major difference with respect to spontaneous emission, where the dynamics selects a unique stationary state.

As we have seen in the fluorescence case, in terms of the Bloch components it displays as:

$$x(t) = x(0)e^{-2\gamma t}, \qquad y(t) = y(0)e^{-2\gamma t}, \qquad z(t) = z(0). \tag{2.103}$$

Hence the Bloch sphere is contracted toward the z axis. All coherences in the transverse plane are lost, whereas the population imbalance remains constant.
Over an infinitesimal interval dt, the corresponding Kraus operators for the jump unravelling are

$$\hat{M}_0 = \mathbb{1} - \frac{\gamma}{2}dt\,\hat{c}^\dagger\hat{c} = \mathbb{1} - \frac{\gamma}{2}dt\,\mathbb{1}, \qquad \hat{M}_1 = \sqrt{\gamma\,dt}\,\hat{\sigma}_z. \tag{2.104}$$

They satisfy

$$\hat{M}_0^\dagger\hat{M}_0 + \hat{M}_1^\dagger\hat{M}_1 = \mathbb{1} + \mathcal{O}(dt^2). \tag{2.105}$$

The key difference with respect to spontaneous emission is that the jump operator is now unitary and Hermitian. A detection event does not transfer the qubit to a preferred state. Instead, it applies the transformation

$$\rho_c(t+dt) = \frac{\hat{M}_1\rho_c(t)\hat{M}_1^\dagger}{\mathrm{Tr}\left[\hat{M}_1\rho_c(t)\hat{M}_1^\dagger\right]} = \hat{\sigma}_z\rho_c(t)\hat{\sigma}_z. \tag{2.106}$$

Thus, a click corresponds to a phase flip. Since $\hat{\sigma}_z$ leaves the diagonal entries unchanged and changes the sign of the coherences, the jump does not represent energy relaxation, but only a stochastic modification of the phase structure of the state.
By contrast, the no-click update is trivial after normalization. Indeed,

$$\rho_c(t+dt) = \frac{\hat{M}_0\rho_c(t)\hat{M}_0^\dagger}{\mathrm{Tr}\left[\hat{M}_0\rho_c(t)\hat{M}_0^\dagger\right]} = \rho_c(t), \tag{2.107}$$

up to higher-order corrections. Therefore, in the jump picture the conditional evolution is piecewise constant and is interrupted by random phase flips. Averaging over all trajectories reproduces the unconditional map,

$$\rho(t+dt) = \hat{M}_0\rho(t)\hat{M}_0^\dagger + \hat{M}_1\rho(t)\hat{M}_1^\dagger, \tag{2.108}$$

which yields Eq. (2.97) in the continuous-time limit.
The homodyne unravelling is equally important. Since $\hat{\sigma}_z$ is Hermitian, homodyne monitoring with phase $\phi = 0$ corresponds to a continuous weak measurement of $\hat{\sigma}_z$. The conditional state then obeys the diffusive stochastic master equation

$$d\rho_c = \gamma\mathcal{D}[\hat{\sigma}_z]\rho_c\,dt + \sqrt{\gamma}\,\mathcal{H}[\hat{\sigma}_z]\rho_c\,dW_t. \tag{2.109}$$

In this case, the measurement record continuously carries information about the qubit observable $\hat{\sigma}_z$. The conditional state is therefore gradually driven toward one of the two eigenstates of $\hat{\sigma}_z$, while the unconditional average over all records still reproduces the dephasing master equation.
For this reason, pure dephasing provides the paradigmatic example of decoherence without dissipation. At the unconditional level, it suppresses only the off-diagonal coherences. At the stochastic level, it admits different unravellings with very different interpretations: photon counting leads to random phase flips, whereas homodyne monitoring leads to a continuous weak measurement of $\hat{\sigma}_z$.

3

Two qubit independent dephasing system: looking for entanglement

3.1 Measurement-induced entanglement in the fluorescence scheme of Lewalle *et al.*

Before introducing our dephasing-based setup, it is useful to briefly review the fluorescence scheme discussed by Lewalle *et al.* in Ref. [**Lewalle21**]. Even though the microscopic dissipation mechanism is different, the operational logic is very close to the one employed in this thesis. In both cases, the key idea is to start from two independent local channels, interfere the corresponding output modes on a beam splitter, and then monitor suitable combinations of them. Entanglement is not generated by a direct interaction between the two qubits, but by the measurement backaction associated with a joint detection scheme.

The physical scenario considered in Ref. [**Lewalle21**] consists of two distant and non-interacting quantum emitters, each coupled to its own radiative environment. The emitted fluorescence from the two sites is collected and sent to a common optical stage. The central optical element is a balanced beam splitter, which mixes the two outgoing modes before they are measured. Additional phase plates can be placed after the beam splitter, so that the fields arriving at the two output ports acquire controllable phases. The final detection stage is implemented at the two output ports, labeled 3 and 4, either by photon counters or by quadrature measurements. In this way one does not monitor the two local emissions separately. One rather monitors their interferometrically mixed combinations. This is the essential physical ingredient, because it makes the measurement sensitive to non-local superpositions of decay processes rather than to distinguishable local emission events.

Lewalle *et al.* stress that this device geometry is closely related to the one commonly employed for Bell-state measurements. Its crucial feature is that it allows the creation of entanglement between distant qubits without any direct qubit-qubit coupling. The only resource is the information extracted from the optical measurement. In other words, coherent correlations are established entirely through the inferences that can be drawn from the joint monitoring of the emitted light. From this viewpoint, the setup realizes a measurement-induced entangling mechanism.

At the level of local channels, each qubit is associated with a spontaneous-emission operator,

$$\hat{\sigma}_-^{(A)}, \qquad \hat{\sigma}_-^{(B)}. \tag{3.1}$$

After the balanced beam splitter and the phase shifters, the measured optical modes correspond to linear combinations of the two original fluorescence outputs. The relevant point is not the explicit optical algebra itself, but its physical consequence: once the two fields have been mixed, a detector click or a homodyne current at one output port generally cannot be attributed to one emitter alone. The detection event is associated with a coherent superposition of the two local decay paths.

This is the origin of the connection with which-path information. If the measurement reveals which emitter produced the signal, then the two decay paths remain distinguishable and no entanglement is created. If, on the contrary, the optical mixing and the chosen measured quadratures erase that information, then the backaction acts jointly on the two-qubit state and may generate entanglement. In the review, this mechanism is repeatedly interpreted in terms of information erasure and, more conceptually, as a continuous-variable version of entanglement swapping.

3.1.1 Photodetection and average concurrence

The simplest monitored dynamics discussed in Ref. [**Lewalle21**] is joint photodetection of the two beam-splitter outputs. In that case one counts photons at the two output ports 3 and 4. If both qubits are initially prepared in $|ee\rangle$, the first detected photon projects the two-qubit state into the single-excitation sector. Because the two fluorescence channels have already been interferometrically mixed, this first detection event does not identify which qubit decayed. Instead, it prepares one of the entangled Bell-like states in the odd sector. The second detected photon then completes the relaxation towards $|gg\rangle$.

The corresponding average concurrence can be obtained by a simple statistical argument. Starting from $|ee\rangle$, entanglement is present precisely in the realizations in which exactly one of the two qubits has decayed up to time t. Since each emitter remains excited with probability $e^{-\gamma t}$, and decays with probability $1 - e^{-\gamma t}$, the probability that exactly one decay has occurred is

$$\overline{C}(t) = 2e^{-\gamma t}\left(1 - e^{-\gamma t}\right). \tag{3.2}$$

This quantity is identified in the tutorial with the trajectory-averaged concurrence. It vanishes at $t = 0$, reaches its maximum at intermediate times, and then decays back to zero as both emitters relax to the ground state. The result is particularly transparent: the concurrence tracks the probability that one, but not both, emission events has occurred.

This interpretation is very important for the discussion of the later results. In the ideal photodetection case, the entanglement is created in a discontinuous way. A click abruptly projects the system into an entangled state. The average behavior is therefore obtained from a simple competition between zero-click, one-click, and two-click realizations. This jump-based mechanism will later serve as the benchmark against which the diffusive homodyne trajectories are compared.

3.1.2 The role of homodyne phases and the meaning of Fig. 4

A more subtle situation arises when the two output ports are monitored by homodyne detection rather than by photon counting. In that case the measurement record is continuous and diffusive, and the two-qubit state evolves through stochastic trajectories that do not contain sharp emission times. The system diffuses from $|ee\rangle$ to $|gg\rangle$, and the entanglement is built gradually instead of appearing through isolated jumps.
The central issue is now the choice of the local-oscillator phases. Because the two beam-splitter outputs are measured in selected quadratures, the relative phases determine whether the measurement preserves or erases which-path information. Lewalle *et al.* show this very explicitly. If one chooses quadratures such that the two detection records still allow one to infer which qubit contributed to the signal, then entanglement generation is spoiled. If instead the quadratures are chosen so as to eliminate this distinguishability, then the conditional dynamics becomes entangling.
This is the precise content of the discussion preceding Fig. 4. The optimal condition is

$$|\theta - \vartheta| = \frac{\pi}{2}, \tag{3.3}$$

and a representative choice is

$$\theta = 0, \qquad \vartheta = \frac{\pi}{2}. \tag{3.4}$$

In this configuration the two homodyne detectors measure two quadratures that are 90° apart. In the language of the review, this is exactly the condition that ensures the erasure of which-path information. The corresponding measured channels acquire a relative factor i, and this phase difference is what makes the two local emission paths indistinguishable at the level of the diffusive measurement record.
Figure 4 in the tutorial summarizes the physical consequences of this choice. In the left panel, the authors plot the average concurrence together with a sample of individual stochastic trajectories, all initialized in $|ee\rangle$. For the optimal settings $\theta = 0$ and $\vartheta = 90°$, many trajectories exceed the average substantially, and some of them reach maximal concurrence $C = 1$. However, unlike the photodetection case, these highly entangled states are reached through a smooth stochastic evolution. The trajectories are diffusive, not jump-like, and there is no single identifiable emission time associated with entanglement generation.
At the level of averages, however, the result is remarkable. The review shows that, under the optimal homodyne choice, the average concurrence coincides with the photodetection result,

$$\overline{C}(t) = 2e^{-\gamma t}\left(1 - e^{-\gamma t}\right). \tag{3.5}$$

Thus, although the individual trajectories are qualitatively very different, the average entanglement yield is the same in the ideal case. Lewalle *et al.* interpret this as a manifestation of the fact that both schemes exploit the same underlying resource generated by fluorescence and simply rearrange it in different ways. In the language used in the tutorial, both measurements may be understood as implementing an optimal entanglement swap.

The right panel of Fig. 4 is equally important, because it shows how the concurrence is degraded when the relative phases are changed away from the optimal value. The average curves keep the same qualitative temporal profile as in the optimal case, but their amplitude is reduced. The review emphasizes that the suppression follows the distinguishability of the decay paths. In particular, changing the relation between θ and ϑ effectively modulates the entanglement yield by an overall factor of order $|\sin(\theta - \vartheta)|$. At the extreme non-optimal choice

$$\theta = \vartheta, \tag{3.6}$$

the measured quadratures maximize the which-path information, and entanglement generation disappears completely. This provides a very clear physical message: the role of the optical phases is not secondary. They directly control whether the measurement acts as an entangling joint probe or as a merely local, path-discriminating readout.
A further important point made in the review is that the optimal homodyne trajectories are still subject to a finite-time constraint. Even in the best case, one cannot reach a Bell state arbitrarily quickly. The fastest possible rise of the concurrence is bounded, and the corresponding idealized trajectory reaches maximal entanglement only after a finite time of order $T_1 \ln 2$. This is another qualitative difference with respect to photodetection: in the jump case, entanglement can appear immediately after the first click, whereas in the homodyne case even the best trajectories need a finite diffusive buildup.

3.1.3 Inefficient photodetection

For the purposes of the present thesis, the most relevant non-ideal extension discussed by Lewalle *et al.* is inefficient photodetection. The authors model detection inefficiency by inserting additional unbalanced beam splitters in the monitored channels after the main balanced beam splitter. The physical meaning is simple: after the two fluorescence modes have been mixed into the output ports 3 and 4, each of these outputs is split again into a transmitted part that reaches an otherwise ideal detector and a lost part that is irretrievably discarded. If $\eta \in [0, 1]$ denotes the efficiency, then a photon reaches the detector with probability η and is lost with probability $1 - \eta$.
This optical model is especially transparent because it makes clear what inefficiency means operationally. The observer does not merely see a weaker signal. Rather, part of the emitted information is genuinely inaccessible. As a consequence, the conditional state is no longer pure even when the underlying system-environment evolution is pure. After a long no-click interval, one cannot be certain that no photon has been emitted. One can only say that no photon has been *detected*. A missed emission remains possible, and therefore the observers state necessarily becomes mixed.
In the inefficient photodetection case, the review shows that two main effects occur. First, the average concurrence curve keeps essentially the same temporal shape as in the ideal case, but is reduced by an overall factor η. In practice, the numerical simulations are well fitted by

$$\overline{C}_\eta(t) \approx 2\eta \, e^{-\gamma t} \left(1 - e^{-\gamma t}\right). \tag{3.7}$$

Thus, the entanglement-generation profile remains the same, but the average yield is uniformly degraded by the fraction of information that is actually collected.

Second, and more importantly, the concurrence attainable in a single realization is no longer allowed to reach unity at arbitrary times. In the ideal case, a first detected photon may project the system onto a perfect Bell state. Under inefficient detection, instead, a detected click may correspond either to the first emitted photon or to a later one, while previous emissions may have been lost. The observer must therefore describe the two-qubit state by a mixture of an entangled branch and a separable branch. The review captures this effect through an upper bound on the concurrence obtainable after a detected jump. If one denotes by $C_\eta^{\max}(t)$ the maximum concurrence attainable at time t, one finds

$$C_\eta^{\max}(t) = \frac{1}{(1-\eta)e^{\gamma t} + \eta}. \tag{3.8}$$

This is one of the central quantitative results of the inefficient photodetection analysis. It shows that the concurrence ceiling decreases with time, even after a click, because the no-click evolution before that click progressively increases the weight of the separable contribution in the observers conditional state.

The physical interpretation is very clear and should be emphasized in the thesis. Suppose that one waits a long time before the first registered count. In the ideal case, that registered count is certainly the first emitted photon, and the associated state update can still produce a perfect Bell state. In the inefficient case, this is no longer true. After a long waiting time, it becomes increasingly likely that one or more earlier photons have already been emitted but lost. The detected click then no longer certifies that the system has just entered the one-excitation Bell sector. Instead, the observer must attribute some probability also to the possibility that the system is already close to $|gg\rangle$. This ignorance is exactly what lowers the maximum concurrence.

The review illustrates these effects in Fig. 8. Different efficiencies are considered, and one clearly sees that the jump trajectories remain qualitatively similar to the ideal ones, but their peaks lie below $C = 1$, with lower efficiencies producing stronger suppression. At the level of the average concurrence, the simple attenuated law in Eq. (3.7) reproduces the simulations well. At the level of single trajectories, the bound in Eq. (3.8) provides a tight upper envelope.

This analysis is conceptually important beyond the fluorescence model itself. It shows that the success of measurement-induced entanglement protocols depends not only on how the output channels are mixed, but also on how much of the leaked information is actually collected. The interferometric erasure of which-path information is a necessary ingredient, but it is not sufficient if too much of the signal is lost before reaching the detectors. The observers ability to infer non-local correlations from the measurement record is degraded by inefficiency in a very direct and quantitative way.

3.1.4 Summary of the lessons relevant for the present work

The review by Lewalle *et al.* provides three conceptual lessons that are directly relevant for the dephasing-based setup studied in this thesis.

First, entanglement can be generated from independent local environments provided that their outputs are interferometrically mixed before detection. The optical apparatus

is therefore not a technical detail, but the core of the protocol.
Second, the detection basis matters. Figure 4 makes this especially clear in the homodyne case: the relative phases of the measured quadratures determine whether the monitoring erases or preserves which-path information, and therefore whether entanglement can or cannot be generated.
Third, imperfect detection does not merely reduce the signal strength. It changes the conditional state itself by forcing the observer to average over undetected events. In the photodetection case, this leads both to a reduced average concurrence and to a time-dependent upper bound on the concurrence achievable after a detected click.
For our purposes, the importance of this tutorial lies precisely in this operational viewpoint. Although in the following sections we will replace spontaneous-emission channels with dephasing channels, the logic of the protocol remains the same: one starts from independent local dissipative channels, mixes them interferometrically, and studies how the chosen unraveling may convert environmental monitoring into a resource for conditional entanglement generation.

3.2 Interferometric monitoring of two independent dephasing channels

We now introduce the two-qubit setup studied in this work. The aim is to apply the same interferometric logic discussed above to a different dissipative mechanism. Instead of spontaneous emission, we consider two independent local dephasing channels acting on the two qubits. This difference is physically important: dephasing does not induce energy relaxation and does not transfer population across different excitation sectors. At the unconditional level, it only destroys phase coherence in the local computational basis. As a consequence, if one only looks at the master equation, the dynamics appears fully local and no entangling mechanism is visible.
The central question is then whether a non-trivial conditional dynamics may nevertheless emerge once the environmental outputs are monitored in a suitable basis. More precisely, we ask whether two independent local dephasing channels can give rise to collective effects when their outputs are first mixed interferometrically and only afterwards detected. The dissipative part of the dynamics remains local, whereas the measurement basis on the environment is changed.
We start from two local collapse operators, one for each qubit, $\hat{L}_1 = \sqrt{\gamma}\,\sigma_z^{(1)}$ and $\hat{L}_2 = \sqrt{\gamma}\,\sigma_z^{(2)}$. In the following, $\hat{L}_k$ denotes the local collapse operators before the beam-splitter mixing, while $\hat{C}_k$ labels the interferometrically mixed collapse operators at the output ports. The unconditional master equation is

$$\frac{d\rho}{dt} = \sum_{k=1}^{2} \mathcal{D}[\hat{L}_k]\rho = \gamma\,\mathcal{D}[\sigma_z^{(1)}]\rho + \gamma\,\mathcal{D}[\sigma_z^{(2)}]\rho. \tag{3.9}$$

If the two channels were monitored separately, the corresponding record would only resolve the two local environments, and no collective information would be extracted from the system.

The non-trivial step is to mix the two output modes on a balanced beam splitter before detection. The monitored channels are then no longer the original local ones, but their symmetric and antisymmetric combinations:

$$\begin{pmatrix} \hat{C}_+ \\ \hat{C}_- \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \hat{L}_1 \\ \hat{L}_2 \end{pmatrix}. \tag{3.10}$$

Explicitly,

$$\hat{C}_+ = \frac{\hat{L}_1 + \hat{L}_2}{\sqrt{2}} = \sqrt{\frac{\gamma}{2}} \left(\sigma_z^{(1)} + \sigma_z^{(2)}\right), \qquad \hat{C}_- = \frac{\hat{L}_1 - \hat{L}_2}{\sqrt{2}} = \sqrt{\frac{\gamma}{2}} \left(\sigma_z^{(1)} - \sigma_z^{(2)}\right). \tag{3.11}$$

These two operators define the natural interferometric measurement basis of the problem. By introducing $J_z = \frac{1}{2}\left(\sigma_z^{(1)} + \sigma_z^{(2)}\right)$, one immediately finds

$$\hat{C}_+ = \sqrt{2\gamma}\, J_z, \qquad \hat{C}_- = \sqrt{\frac{\gamma}{2}} \left(\sigma_z^{(1)} - \sigma_z^{(2)}\right). \tag{3.12}$$

The $+$ output port therefore realizes a collective measurement channel: it probes the symmetric combination of the two local dephasing signals and does not distinguish which qubit contributed individually. The $-$ port probes instead the differential mode between the two local outputs.
If the two output modes are not mixed before detection, the dynamics remains fully local not only at the level of the master equation, but also at the level of the corresponding quantum channel. This immediately rules out any measurement-induced entanglement mechanism. To see this, let's write the unconditional generator as

$$\mathcal{L} = \mathcal{L}^{(1)} + \mathcal{L}^{(2)}, \qquad \mathcal{L}^{(k)}[\rho] = \gamma\, \mathcal{D}[\sigma_z^{(k)}]\rho. \tag{3.13}$$

Since $\mathcal{L}^{(1)}$ and $\mathcal{L}^{(2)}$ act on different subsystems, they commute and the corresponding dynamical map therefore factorizes as

$$e^{t\mathcal{L}} = e^{t\mathcal{L}^{(1)}} e^{t\mathcal{L}^{(2)}} = \Lambda_t^{(1)} \otimes \Lambda_t^{(2)}, \tag{3.14}$$

where $\Lambda_t^{(k)}$ is the single-qubit dephasing channel acting on qubit k.
Consider now an arbitrary separable initial state,

$$\rho_{\text{sep}}(0) = \sum_j p_j\, \rho_j^{(1)} \otimes \rho_j^{(2)}, \qquad p_j \geq 0, \qquad \sum_j p_j = 1. \tag{3.15}$$

Using Eq. (3.14), its time-evolved state reads

$$\rho_{\text{sep}}(t) = (\Lambda_t^{(1)} \otimes \Lambda_t^{(2)})\rho_{\text{sep}}(0) = \sum_j p_j\, \Lambda_t^{(1)}[\rho_j^{(1)}] \otimes \Lambda_t^{(2)}[\rho_j^{(2)}]. \tag{3.16}$$

Each term in the sum is still a product state, and the whole state is still a convex combination of product states. Hence $\rho_{\text{sep}}(t)$ remains separable for all t.

This proves that, if the two dephasing channels are left local, the unconditional evolution cannot generate entanglement from separable initial states. In particular, any non-trivial entangling effect discussed in the following cannot originate from the local dissipators alone, but must come from the change of measurement basis induced by the interferometric mixing of the two output modes.
This is the key point of the protocol: the interferometric mixing does not modify the unconditional dynamics, since unitary transformations of the collapse operators leave the total Lindblad generator invariant (see Appendix 4). What changes is the unraveling, namely the way in which information is extracted from the environment and transferred to the conditional state through the measurement backaction. As a consequence, even though the master equation remains that of two independent dephasing channels, the conditioned dynamics may display collective features and generate non-trivial correlations such as entanglement.

3.2.1 No-jump evolution of two-qubits

In this section we derive the no-jump evolution of a two-qubit under the z-dephasing dynamics introduced above. Since the meaning of no-jump depends on the detection efficiency, we treat separately the cases $\eta = 1$, perfect photodetection, and $\eta < 1$ (inefficient photodetection, i.e. "no detected clicks").
For $\eta = 1$, no jump means that no click occurred in either monitored channel up to time t. In the quantum-jump picture, the unnormalized conditional pure state evolves under the non-Hermitian effective Hamiltonian

$$\hat{H}_{\text{eff}} = \hat{H} - \frac{i}{2}\sum_k \hat{C}_k^\dagger \hat{C}_k, \tag{3.17}$$

so that

$$\left|\tilde{\psi}(t)\right\rangle = e^{-i\hat{H}_{\text{eff}}t}\left|\psi(0)\right\rangle, \tag{3.18}$$

In our case (pure dephasing) we can work with the collective jump operators (**??**) and properties of σ_z to obtain

$$\hat{C}_+^\dagger \hat{C}_+ + \hat{C}_-^\dagger \hat{C}_- = \frac{\gamma}{2}\left[(\sigma_z^{(1)} + \sigma_z^{(2)})^2 + (\sigma_z^{(1)} - \sigma_z^{(2)})^2\right] = 2\gamma\,\mathbb{1}. \tag{3.19}$$

For $\hat{H} = 0$, Eq. (3.17) gives $\hat{H}_{\text{eff}} = -i\gamma\,\mathbb{1}$ and therefore

$$\left|\tilde{\psi}(t)\right\rangle = e^{-\gamma t}\left|\psi(0)\right\rangle. \tag{3.20}$$

After normalization, the global factor cancels and the conditional state is unchanged:

$$\left|\psi(t)\right\rangle_{\text{no jump}} = \left|\psi(0)\right\rangle \qquad (\eta = 1). \tag{3.21}$$

In particular, for any initial CSS one has

$$\left|\theta, \phi\right\rangle \xrightarrow{\text{no jump}} \left|\theta, \phi\right\rangle \qquad (\eta = 1). \tag{3.22}$$

For $\eta < 1$, conditioning on "no detected clicks" does not imply that no quantum jumps occurred: unobserved emissions contribute an additional unconditional evolution term. The conditional dynamics in the absence of detected counts is deterministic and coincides with the unmonitored part of the SME. With our dephasing model, the no-detected-click evolution is governed by

$$\frac{d\rho(t)}{dt} = \mathcal{L}^{(2)}_{(1-\eta)\gamma}[\rho(t)] = \frac{(1-\eta)\gamma}{2}\sum_{j=1}^{2}\Big(\sigma_z^{(j)}\rho(t)\sigma_z^{(j)} - \rho(t)\Big), \tag{3.23}$$

This is a local dephasing channel acting independently on each qubit. Since the initial CSS is a product state, the solution factorizes:

$$\rho^{(\text{no click})}_{\text{CSS},\theta}(t) = \rho^{(\text{no click})}_{1}(t) \otimes \rho^{(\text{no click})}_{1}(t), \tag{3.24}$$

where $\rho^{(\text{no click})}_{1}(t)$ is the single-qubit solution starting from the pure state $|\theta,\phi\rangle$. Writing the initial single-qubit density operator in the basis $\{|0\rangle, |1\rangle\}$,

$$\rho_1(0) = |\theta,\phi\rangle\langle\theta,\phi| = \begin{pmatrix} \cos^2\frac{\theta}{2} & e^{-i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2} \\ e^{i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2} & \sin^2\frac{\theta}{2} \end{pmatrix}, \tag{3.25}$$

one finds that the populations are constant and the coherences decay exponentially:

$$\rho^{(\text{no click})}_{1}(t) = \begin{pmatrix} \cos^2\frac{\theta}{2} & e^{-i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}\, e^{-(1-\eta)\gamma t} \\ e^{i\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}\, e^{-(1-\eta)\gamma t} & \sin^2\frac{\theta}{2} \end{pmatrix} \qquad (\eta < 1). \tag{3.26}$$

Equations (3.24) and (3.26) provide the full two-qubit no-detected-click state for any (θ,ϕ).
For later use, we report the explicit two-qubit matrix for $\phi = 0$ in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Introducing the shorthand $c \equiv \cos\frac{\theta}{2}$ and $s \equiv \sin\frac{\theta}{2}$ one obtains

$$\rho^{(\text{no click})}_{\text{CSS},\theta}(t) = \begin{pmatrix} c^4 & c^3 s\, e^{-(1-\eta)\gamma t} & c^3 s\, e^{-(1-\eta)\gamma t} & c^2 s^2\, e^{-2(1-\eta)\gamma t} \\ c^3 s\, e^{-(1-\eta)\gamma t} & c^2 s^2 & c^2 s^2\, e^{-2(1-\eta)\gamma t} & c s^3\, e^{-(1-\eta)\gamma t} \\ c^3 s\, e^{-(1-\eta)\gamma t} & c^2 s^2\, e^{-2(1-\eta)\gamma t} & c^2 s^2 & c s^3\, e^{-(1-\eta)\gamma t} \\ c^2 s^2\, e^{-2(1-\eta)\gamma t} & c s^3\, e^{-(1-\eta)\gamma t} & c s^3\, e^{-(1-\eta)\gamma t} & s^4 \end{pmatrix}. \tag{3.27}$$

3.2.2 First jump from $|{+}{+}\rangle$

We work in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, and we adopt the convention $\sigma_z|0\rangle = -|0\rangle$, $\sigma_z|1\rangle = +|1\rangle$. The single-qubit $|+\rangle$ state is $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$, hence

$$|{+}{+}\rangle = |+\rangle \otimes |+\rangle = \frac{1}{2}(\,|00\rangle + |01\rangle + |10\rangle + |11\rangle\,) = \frac{1}{2}(1,1,1,1)^{\mathrm{T}} \tag{3.28}$$

We define the two-qubit z-parity operator as

$$\Pi_z \equiv \sigma_z^{(1)}\sigma_z^{(2)}. \tag{3.29}$$

It is Hermitian and unitary: $\Pi_z^\dagger = \Pi_z$ and $\Pi_z^2 = \mathbb{1}$, hence its eigenvalues are ± 1. In the computational basis one has $\Pi_z |00\rangle = +|00\rangle$, $\Pi_z |11\rangle = +|11\rangle$, while $\Pi_z |01\rangle = -|01\rangle$, $\Pi_z |10\rangle = -|10\rangle$. Therefore its matrix representation is diagonal:

$$\Pi_z \equiv \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & +1 \end{pmatrix}. \tag{3.30}$$

The projectors onto the even/odd parity subspaces are $\Pi_{\text{even}} = \frac{1}{2}(\mathbb{1} + \Pi_z)$ and $\Pi_{\text{odd}} = \frac{1}{2}(\mathbb{1} - \Pi_z)$. A useful property for what follows is that $|{+}{+}\rangle$ (3.28) has equal weight on the even- and odd-parity computational sectors: the even sector is spanned by $\{|00\rangle, |11\rangle\}$, the odd one by $\{|01\rangle, |10\rangle\}$.
Since $\sigma_z^{(1)}$ and $\sigma_z^{(2)}$ commute and $(\sigma_z^{(i)})^2 = \mathbb{1}$,

$$[\sigma_z^{(1)}, \Pi_z] = \sigma_z^{(1)}\sigma_z^{(1)}\sigma_z^{(2)} - \sigma_z^{(1)}\sigma_z^{(1)}\sigma_z^{(2)} = 0, \tag{3.31}$$
$$[\sigma_z^{(2)}, \Pi_z] = \sigma_z^{(2)}\sigma_z^{(1)}\sigma_z^{(2)} - \sigma_z^{(1)}\sigma_z^{(2)}\sigma_z^{(2)} = \sigma_z^{(1)} - \sigma_z^{(1)} = 0. \tag{3.32}$$

Therefore $[\sigma_z^{(1)} \pm \sigma_z^{(2)}, \Pi_z] = 0$, and by (**??**) we have

$$[\hat{C}_\pm, \Pi_z] = 0. \tag{3.33}$$

This is a key result: a jump produced by $\hat{C}_\pm$ preserves the z-parity: if a pre-jump pure state $|\psi\rangle$ is a parity eigenstate, so is the post-jump normalized state : $|\psi'\rangle = \hat{C}_\pm |\psi\rangle / \left\| \hat{C}_\pm |\psi\rangle \right\|$.
Let us work on the collapsing operators, by calculating:

$$\hat{C}_\pm^\dagger \hat{C}_\pm = (\mathbb{1} \pm \Pi_z)\,. \tag{3.34}$$

we see that for pure-state jumps we use the standard update rule

$$p_\pm = \langle\psi|\hat{C}_\pm^\dagger \hat{C}_\pm|\psi\rangle \, \mathrm{d}t = \langle\psi|\mathbb{1} \pm \Pi_z|\psi\rangle = \Big(\mathbb{1} \pm \langle\Pi_z\rangle_\psi\Big)\, \mathrm{d}t \tag{3.35}$$

It is convenient to evaluate the direction of the unnormalized state $\hat{C}_\pm |{+}{+}\rangle$:

$$(\sigma_z^{(1)} + \sigma_z^{(2)})\,|{+}{+}\rangle = 2(\,|11\rangle - |00\rangle\,) = -2\sqrt{2}\,|\phi_-\rangle\,, \tag{3.36}$$
$$(\sigma_z^{(1)} - \sigma_z^{(2)})\,|{+}{+}\rangle = 2(\,|10\rangle - |01\rangle\,) = -2\sqrt{2}\,|\psi_-\rangle\,. \tag{3.37}$$

Hence the *normalized* post-jump states are

$$|{+}{+}\rangle \xrightarrow{C_+} -|\phi_-\rangle\,, \qquad |{+}{+}\rangle \xrightarrow{C_-} -|\psi_-\rangle\,. \tag{3.38}$$

Overall minus signs are global phases and carry no physical effect; we keep them to make the subsequent alternation visually clear.
Equations (3.38) show that the first click already fixes the z-parity: $\hat{C}_+$ lands in the even sector (Bell-ϕ states), while $\hat{C}_-$ lands in the odd sector (Bell-ψ states).

For the state $|{++}\rangle$, since $\langle\sigma_z\rangle_{|+\rangle} = 0$, we have $\langle\Pi_z\rangle_{|++\rangle} = \langle\sigma_z\rangle_{|+\rangle}\,\langle\sigma_z\rangle_{|+\rangle} = 0$, hence

$$p_+(|{++}\rangle) = p_-(|{++}\rangle) = \gamma\,\mathrm{d}t. \tag{3.39}$$

The two channels are equiprobable on the initial state, and the total jump rate is $2\gamma\,\mathrm{d}t$. For any even-parity state (eigenvalue $+1$ of Π_z; in particular $|\phi_\pm\rangle$),

$$p_+(\text{even}) = 2\gamma\,\mathrm{d}t, \qquad p_-(\text{even}) = 0. \tag{3.40}$$

For any odd-parity state (eigenvalue -1; in particular $|\psi_\pm\rangle$),

$$p_-(\text{odd}) = 2\gamma\,\mathrm{d}t, \qquad p_+(\text{odd}) = 0. \tag{3.41}$$

These two outcomes already show the key phenomenon: a single detected jump projects $|{++}\rangle$ onto a *definite* parity sector (even for $\hat{C}_+$, odd for $\hat{C}_-$), which we now formalize.

3.2.3 Subsequent jumps: two-state cyclicity

It follows that the action of the Lindbladians on the Bell states is:

$$\hat{C}_+\,|\phi_-\rangle = -\,|\phi_+\rangle\,, \qquad\qquad \hat{C}_+\,|\phi_+\rangle = -\,|\phi_-\rangle\,, \tag{3.42}$$

$$\hat{C}_-\,|\psi_-\rangle = -\,|\psi_+\rangle\,, \qquad\qquad \hat{C}_-\,|\psi_+\rangle = -\,|\psi_-\rangle\,. \tag{3.43}$$

After renormalization, this yields the deterministic cycles

$$-\,|\phi_-\rangle \xrightarrow{C_+} |\phi_+\rangle \xrightarrow{C_+} -\,|\phi_-\rangle \xrightarrow{C_+} \cdots, \qquad -\,|\psi_-\rangle \xrightarrow{C_-} |\psi_+\rangle \xrightarrow{C_-} -\,|\psi_-\rangle \xrightarrow{C_-} \cdots. \tag{3.44}$$

Because of $[C_\pm, \Pi_z] = 0$, the “wrong” channel is forbidden inside each parity subspace.

3.2.4 Dynamics of CSS states

We now specialize the discussion to two-qubit spin coherent states (CSS) (**??**) and describe how the photodetection dynamics acts on them. We set $\phi = 0$.
By applying the Lindbladian operators (**??**), one obtains

$$\hat{C}_+|\text{CSS},\theta\rangle = \sin^2\!\left(\frac{\theta}{2}\right)|11\rangle - \cos^2\!\left(\frac{\theta}{2}\right)|00\rangle\,, \tag{3.45}$$

$$\hat{C}_-|\text{CSS},\theta\rangle = \sin\frac{\theta}{2}\cos\frac{\theta}{2}\left(|10\rangle - |01\rangle\right). \tag{3.46}$$

After normalization, the conditional post-click pure states are

$$\left|\tilde{\phi}_\theta\right\rangle = \frac{\sin^2\!\left(\frac{\theta}{2}\right)|11\rangle - \cos^2\!\left(\frac{\theta}{2}\right)|00\rangle}{\sqrt{\sin^4\!\left(\frac{\theta}{2}\right) + \cos^4\!\left(\frac{\theta}{2}\right)}}, \qquad |\psi_-\rangle = \frac{|10\rangle - |01\rangle}{\sqrt{2}}. \tag{3.47}$$

The “$-$” click therefore projects onto a maximally entangled Bell state independently of θ, while the “$+$” click yields an even-parity state whose entanglement depends on θ.

After the first click, parity is fixed. Indeed,

$$\Big(|10\rangle\langle 10| - |01\rangle\langle 01|\Big)\left|\tilde{\phi}_\theta\right\rangle = 0, \qquad \Big(|11\rangle\langle 11| - |00\rangle\langle 00|\Big)|\psi_-\rangle = 0. \tag{3.48}$$

Within each parity sector the jump action induces a two-state cyclicity. In the even sector we define

$$\left|\tilde{\phi}_{\pm,\theta}\right\rangle = \frac{\sin^2\left(\frac{\theta}{2}\right)|11\rangle \pm \cos^2\left(\frac{\theta}{2}\right)|00\rangle}{\sqrt{\sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right)}}, \tag{3.49}$$

so it follows that we retrieve the more general deterministic cycles

$$|\mathrm{CSS},\theta\rangle \xrightarrow{\hat{C}_+} \left|\tilde{\phi}_{-,\theta}\right\rangle \xrightarrow{\hat{C}_+} \left|\tilde{\phi}_{+,\theta}\right\rangle \xrightarrow{\hat{C}_+} \left|\tilde{\psi}_{-,\theta}\right\rangle \qquad |\mathrm{CSS},\theta\rangle \xrightarrow{\hat{C}_-} |\psi_-\rangle \xrightarrow{\hat{C}_-} |\psi_+\rangle \xrightarrow{\hat{C}_-} |\psi_-\rangle \tag{3.50}$$

3.2.5 Analyzing evolution and jumps: even-parity channel

Now we want to prove that evolving $|{+}{+}\rangle$ with liouvillian as (**??**) is totally equivalent in evolving in time the first click state for both channel. We show it explicitly for the even-parity case, meaning:

$$\hat{C}_+ \cdot e^{\mathcal{L}^{(2)}_{(1-\eta)\gamma}t}[|{+}{+}\rangle\langle + +|] \cdot \hat{C}_+ = e^{\mathcal{L}^{(2)}_{(1-\eta)\gamma}t}[|\psi_-\rangle\langle\psi_-|], \tag{3.51}$$

where $\mathcal{L}^{(2)}_{(1-\eta)\gamma}$ is the liouvillian in the ME.
Using the collective jump operators $\hat{C}_\pm$, we write

$$\mathcal{L}^{(2)}_{(1-\eta)\gamma}[\rho_t] = (1-\eta)\frac{\gamma}{2}\Big(\mathcal{D}[\hat{C}_+]\rho_t + \mathcal{D}[\hat{C}_-]\rho_t\Big). \tag{3.52}$$

Since the unmonitored dephasing channel acts locally on each qubit, for product inputs the evolution factorizes. In particular,

$$e^{\mathcal{L}^{(2)}_{(1-\eta)\gamma}t}[|{+}{+}\rangle\langle{+}{+}|] = \Big(e^{\mathcal{L}^{(1)}_{(1-\eta)\gamma}t}[|+\rangle\langle +|]\Big) \otimes \Big(e^{\mathcal{L}^{(1)}_{(1-\eta)\gamma}t}[|+\rangle\langle +|]\Big), \tag{3.53}$$

where $\mathcal{L}^{(1)}_{(1-\eta)\gamma}$ denotes the corresponding single-qubit Liouvillian, the "single-qubit dephasing" channel.
We vectrize the single-qubit problem in the computational basis $\{|0\rangle, |1\rangle\}$. Let $\rho(t)$ be a single-qubit density operator written as

$$\rho(t) = \begin{pmatrix} \alpha_{00}(t) & \alpha_{01}(t) \\ \alpha_{10}(t) & \alpha_{11}(t) \end{pmatrix}, \tag{3.54}$$

and its equation of motion

$$\dot{\rho}(t) = \mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)]\,. \tag{3.55}$$

In the computational basis, the superoperator $\mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)]$ can be represented through its matrix elements as

$$\mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)] \;\equiv\; \begin{pmatrix} \langle 0|\,\mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)]\,|0\rangle & \langle 0|\,\mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)]\,|1\rangle \\ \langle 1|\,\mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)]\,|0\rangle & \langle 1|\,\mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)]\,|1\rangle \end{pmatrix}. \tag{3.56}$$

Hence, to obtain the explicit differential equations for the coefficients $\alpha_{00}(t), \alpha_{01}(t), \alpha_{10}(t)$, one computes $\mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)]$ and projects it onto the computational basis.
We consider the single-qubit dephasing generator (unmonitored part)

$$\dot{\rho}(t) = \mathcal{L}^{(1)}_{\Gamma}[\rho(t)] = \frac{\Gamma}{2}\Big(\sigma_z\,\rho(t)\,\sigma_z - \rho(t)\Big), \tag{3.57}$$

Then

$$\sigma_z\rho(t)\sigma_z = \begin{pmatrix} \alpha_{00}(t) & -\alpha_{01}(t) \\ -\alpha_{10}(t) & \alpha_{11}(t) \end{pmatrix}, \tag{3.58}$$

and therefore

$$\mathcal{L}^{(1)}_{(1-\eta)\gamma}[\rho(t)] = \frac{(1-\eta)\gamma}{2}\begin{pmatrix} 0 & -2\alpha_{01}(t) \\ -2\alpha_{10}(t) & 0 \end{pmatrix}. \tag{3.59}$$

Hence the populations are constants, while coherences decay exponentially:

$$\dot{\alpha}_{00}(t) = \dot{\alpha}_{11}(t) = 0, \tag{3.60}$$

$$\dot{\alpha}_{01}(t) = \dot{\alpha}_{10}(t) = -(1-\eta)\gamma \tag{3.61}$$

so that

$$\rho(t) = \begin{pmatrix} \alpha_{00}(0) & \alpha_{01}(0)e^{-(1-\eta)\gamma t} \\ \alpha_{10}(0)e^{-(1-\eta)\gamma t} & \alpha_{11}(0) \end{pmatrix}. \tag{3.62}$$

In particular, for $\rho(0) = |+\rangle\langle+|$ one has $\alpha_{jk} = 1/2\,\forall j,k$, hence

$$\rho_+(t) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2}e^{-(1-\eta)\gamma t} \\ \frac{1}{2}e^{-(1-\eta)\gamma t} & \frac{1}{2} \end{pmatrix}, \qquad \rho_+(t\to\infty) = \frac{\mathbb{1}_2}{2}. \tag{3.63}$$

Since the channel acts independently on the two qubits,

$$\rho_{++}(t) = e^{\mathcal{L}^{(2)}_{(1-\eta)\gamma}t}[|{+}{+}\rangle\langle{+}{+}|] = \rho_+(t)\otimes\rho_+(t), \tag{3.64}$$

where $\rho_+(t)$ is given in Eq. (3.62). In the computational basis ordered as $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, this yields

$$\rho_{++}(t) = \frac{1}{4}\begin{pmatrix} 1 & e^{-(1-\eta)\gamma t} & e^{-(1-\eta)\gamma t} & e^{-2(1-\eta)\gamma t} \\ e^{-(1-\eta)\gamma t} & 1 & e^{-2(1-\eta)\gamma t} & e^{-(1-\eta)\gamma t} \\ e^{-(1-\eta)\gamma t} & e^{-2(1-\eta)\gamma t} & 1 & e^{-(1-\eta)\gamma t} \\ e^{-2(1-\eta)\gamma t} & e^{-(1-\eta)\gamma t} & e^{-(1-\eta)\gamma t} & 1 \end{pmatrix}. \tag{3.65}$$

We now apply the collective operator $\hat{C}_+ = \mathrm{diag}(-\sqrt{2}, 0, 0, \sqrt{2})$, used to model the + jump channel. Only the $\{|00\rangle, |11\rangle\}$ block survives, and one finds

$$C_+\,\rho_{++}(t)\,C_+^\dagger = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & -e^{-2(1-\eta)\gamma t} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -e^{-2(1-\eta)\gamma t} & 0 & 0 & 1 \end{pmatrix} \tag{3.66}$$

which is supported on the even subspace.
At $t = 0$,

$$C_+\,\rho_{++}(0)\,C_+^\dagger = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix} = |\phi_-\rangle\langle\phi_-|\,. \tag{3.67}$$

We now compute the right-hand side

$$e^{\mathcal{L}_\gamma^{(2)} t}[|\phi_-\rangle\langle\phi_-|]\,, \tag{3.68}$$

and show that it matches the expression obtained for the LHS.
The corresponding Liouvillian act as:

$$\dot\rho(t) = \mathcal{L}_\gamma^{(2)}[\rho(t)] = \frac{\gamma}{2}\Big(\mathcal{D}[\hat C_+]\rho(t) + \mathcal{D}[\hat C_-]\rho(t)\Big), \tag{3.69}$$

Since $|\phi_-\rangle$ has even z-parity, one has $\hat C_-\,|\phi_-\rangle = 0$, hence

$$\mathcal{D}[\hat C_-](\,|\phi_-\rangle\langle\phi_-|\,) = 0, \tag{3.70}$$

and the dynamics reduces to the + channel.
Using (3.29) the dissipator can be rewritten as

$$\begin{aligned}\mathcal{D}[\hat C_\pm]\rho &= \hat C_\pm\rho\hat C_\pm - \frac{1}{2}\{(\mathbb{1}\pm\Pi_z),\rho\} \\ &= \hat C_\pm\rho\hat C_\pm - \rho \mp \frac{1}{2}(\Pi_z\rho + \rho\Pi_z).\end{aligned} \tag{3.71}$$

For parity-preserving states (block-diagonal with respect to even/odd parity) one has $[\rho,\Pi_z] = 0$. In the even subspace span$\{|00\rangle\,,|11\rangle\}$, $\Pi_z = +\mathbb{1}$, hence $\Pi_z\rho + \rho\Pi_z = 2\rho$ and therefore

$$\mathcal{D}[\hat C_+]\rho = \hat C_+\rho\hat C_+ - 2\rho \qquad \text{(even subspace)}. \tag{3.72}$$

Restricting to the even basis $\{|00\rangle\,,|11\rangle\}$, define the reduced 2×2 state

$$\tilde\rho(t) = \begin{pmatrix} \alpha_{00,00}(t) & \alpha_{00,11}(t) \\ \alpha_{11,00}(t) & \alpha_{11,11}(t) \end{pmatrix}. \tag{3.73}$$

In this basis,

$$\tilde C_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} -2 & 0 \\ 0 & +2 \end{pmatrix} = \sqrt{2}\,\tilde\sigma_z, \qquad \tilde\sigma_z = \begin{pmatrix} -1 & 0 \\ 0 & +1 \end{pmatrix}. \tag{3.74}$$

Using Eq. (3.72), the reduced equation of motion becomes

$$\begin{aligned}\dot{\tilde\rho}(t) &= \frac{\gamma}{2}\Big(\tilde C_+\tilde\rho(t)\tilde C_+ - 2\tilde\rho(t)\Big) \\ &= \frac{\gamma}{2}\Big(2\,\tilde\sigma_z\tilde\rho(t)\tilde\sigma_z - 2\tilde\rho(t)\Big) = \gamma\Big(\tilde\sigma_z\tilde\rho(t)\tilde\sigma_z - \tilde\rho(t)\Big).\end{aligned} \tag{3.75}$$

Explicitly,

$$\tilde\sigma_z\tilde\rho(t)\tilde\sigma_z = \begin{pmatrix} \alpha_{00,00}(t) & -\alpha_{00,11}(t) \\ -\alpha_{11,00}(t) & \alpha_{11,11}(t) \end{pmatrix}, \tag{3.76}$$

so that

$$\dot{\tilde{\rho}}(t) = \gamma \begin{pmatrix} 0 & -2\alpha_{00,11}(t) \\ -2\alpha_{11,00}(t) & 0 \end{pmatrix}. \tag{3.77}$$

Hence the populations are constants and the coherences decay as $e^{-2\gamma t}$:

$$\tilde{\rho}(t) = \begin{pmatrix} \alpha_{00,00}(0) & \alpha_{00,11}(0)e^{-2\gamma t} \\ \alpha_{11,00}(0)e^{-2\gamma t} & \alpha_{11,11}(0) \end{pmatrix}. \tag{3.78}$$

For $\tilde{\rho}(0) = |\phi_-\rangle\langle\phi_-|$ one has

$$\tilde{\rho}(0) = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \tag{3.79}$$

thus

$$\tilde{\rho}(t) = \frac{1}{2}\begin{pmatrix} 1 & -e^{-2\gamma t} \\ -e^{-2\gamma t} & 1 \end{pmatrix}. \tag{3.80}$$

Embedding back into the full computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ yields

$$e^{\mathcal{L}_\gamma^{(2)} t}[|\phi_-\rangle\langle\phi_-|] = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & -e^{-2\gamma t} \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -e^{-2\gamma t} & 0 & 0 & 1 \end{pmatrix} \tag{3.81}$$

which matches the LHS expression upon the replacement $\gamma \mapsto (1-\eta)\gamma$ when interpreting $\mathcal{L}^{(2)}$ as the unmonitored part of the inefficient photodetection dynamics.

3.2.6 Analyzing evolution and jumps: odd-parity channel case

We now repeat the same computation for the negative channel. Explicitly,

$$\hat{C}_- \cdot e^{\mathcal{L}^{(2)}_{(1-\eta)\gamma} t}[|{+}{+}\rangle\langle{+}{+}|] \cdot \hat{C}_-^\dagger \tag{3.82}$$

The unmonitored generator can be written in terms of collective operators as

$$\mathcal{L}^{(2)}_{(1-\eta)\gamma}[\rho_t] = \frac{(1-\eta)\gamma}{2}\Big(\mathcal{D}[\hat{C}_+]\rho_t + \mathcal{D}[\hat{C}_-]\rho_t\Big), \tag{3.83}$$

By $\hat{C}_- = \mathrm{diag}(0, -\sqrt{2}, \sqrt{2}, 0)$, one obtains

$$\hat{C}_-\, \rho_{++}(t)\, \hat{C}_-^\dagger = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -e^{-2(1-\eta)\gamma t} & 0 \\ 0 & -e^{-2(1-\eta)\gamma t} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{3.84}$$

which is supported on the odd subspace $\mathrm{span}\{|01\rangle, |10\rangle\}$.
At $t = 0$,

$$\hat{C}_-\, \rho_{++}(0)\, \hat{C}_-^\dagger = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} = |\psi_-\rangle\langle\psi_-|. \tag{3.85}$$

We now compute the evolution starting from $|\psi_-\rangle$ under the same Liouvillian and show that it reproduces Eq. (3.84).
Since $|\psi_-\rangle$ has odd z-parity, one has $\hat{C}_- |\psi_-\rangle = 0$, hence

$$\mathcal{D}[\hat{C}_+](\,|\psi_-\rangle\langle\psi_-|\,) = 0, \tag{3.86}$$

and the dynamics reduces to the minus channel. Restricting to the odd basis $\{|01\rangle, |10\rangle\}$, define the reduced 2×2 state

$$\tilde{\rho}(t) = \begin{pmatrix} \alpha_{01,01}(t) & \alpha_{01,10}(t) \\ \alpha_{10,01}(t) & \alpha_{10,10}(t) \end{pmatrix}. \tag{3.87}$$

For odd-parity states, the identity $\hat{C}_-^\dagger \hat{C}_- = \mathbb{1} - \Pi_z$ reduces to $\hat{C}_-^\dagger \hat{C}_- = 2 \cdot \mathbb{1}$ on the odd subspace. Therefore,

$$\mathcal{D}[\hat{C}_-]\rho = \hat{C}_-\rho\hat{C}_- - 2\rho \qquad \text{(odd subspace)}. \tag{3.88}$$

The reduced equation of motion becomes

$$\dot{\tilde{\rho}}(t) = (1-\eta)\gamma\Big(\tilde{\sigma}_z\tilde{\rho}(t)\tilde{\sigma}_z - \tilde{\rho}(t)\Big). \tag{3.89}$$

Explicitly,

$$\dot{\tilde{\rho}}(t) = (1-\eta)\gamma \begin{pmatrix} 0 & -2\alpha_{01,10}(t) \\ -2\alpha_{10,01}(t) & 0 \end{pmatrix}. \tag{3.90}$$

Hence the populations are constants and the coherences decay as $e^{-2(1-\eta)\gamma t}$:

$$\tilde{\rho}(t) = \begin{pmatrix} \alpha_{01,01}(0) & \alpha_{01,10}(0)e^{-2(1-\eta)\gamma t} \\ \alpha_{10,01}(0)e^{-2(1-\eta)\gamma t} & \alpha_{10,10}(0) \end{pmatrix}. \tag{3.91}$$

For $\tilde{\rho}(0) = |\psi_-\rangle\langle\psi_-|$ one has

$$\tilde{\rho}(t) = \frac{1}{2}\begin{pmatrix} 1 & -e^{-2(1-\eta)\gamma t} \\ -e^{-2(1-\eta)\gamma t} & 1 \end{pmatrix}. \tag{3.92}$$

Embedding back into the full computational basis yields

$$e^{\mathcal{L}^{(2)}_{(1-\eta)\gamma}t}[|\psi_-\rangle\langle\psi_-|] = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -e^{-2(1-\eta)\gamma t} & 0 \\ 0 & -e^{-2(1-\eta)\gamma t} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \tag{3.93}$$

which matches Eq. (3.84). Hence Eq. (3.82) is proven.

3.2.7 Ensemble-averaged quantities over trajectories

We now compute expectation values over the photodetection trajectories. Let $P_0(t)$ denote the probability of observing no click up to time t. Since total rate is 2γ, the waiting-time distribution is exponential and

$$P_0(t) = e^{-2\gamma t}, \qquad P_{\text{click}}(t) = 1 - P_0(t) = 1 - e^{-2\gamma t}. \tag{3.94}$$

The ensemble-averaged concurrence can be written as

$$\mathbb{E}[\mathcal{C}(t)] = \mathbb{E}[\mathcal{C}(t)\,|\,\text{no-click}]\,P_0(t) + \mathbb{E}[\mathcal{C}(t)\,|\,\text{click}]\,P_{\text{click}}(t). \tag{3.95}$$

For the present dynamics, $\mathcal{C} = 0$ along the no-click branch (the state remains separable), while $\mathcal{C} = 1$ after the first click (the state is projected onto a Bell state). Hence

$$\mathbb{E}[\mathcal{C}(t)] = 1 - e^{-2\gamma t}. \tag{3.96}$$

We proceed similarly for the variance $\Delta J_z^2(t)$. Splitting the trajectories into the no-click part and the click part, one has

$$\mathbb{E}\big[\Delta J_z^2(t)\big] = \mathbb{E}\big[\Delta J_z^2(t)\,|\,\text{no-click}\big]\,P_0(t) + \mathbb{E}\big[\Delta J_z^2(t)\,|\,\text{click}\big]\,P_{\text{click}}(t). \tag{3.97}$$

For an initial CSS state we see that the weights of even/odd sectors are

$$P_+(\theta) = \sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right), \qquad P_-(\theta) = 2\sin^2\left(\frac{\theta}{2}\right)\cos^2\left(\frac{\theta}{2}\right), \qquad P_+ + P_- = 1. \tag{3.98}$$

Along the no-click branch the state stays equal (up to normalization) to the initial CSS, hence

$$\Delta J_z^2(\text{no-click}) = \frac{\sin^2\theta}{2}. \tag{3.99}$$

After the first click, the state is projected onto the even-parity Bell sector with weight P_+ or onto the odd-parity Bell sector with weight P_-; using

$$\Delta J_z^2(|\phi_\pm\rangle) = 1, \qquad \Delta J_z^2(|\psi_\pm\rangle) = 0, \tag{3.100}$$

one obtains

$$\mathbb{E}\big[\Delta J_z^2(t)\,|\,\text{click}\big] = P_+(\theta)\cdot 1 + P_-(\theta)\cdot 0 = P_+(\theta). \tag{3.101}$$

Therefore

$$\mathbb{E}\big[\Delta J_z^2(t)\big] = e^{-2\gamma t}\,\frac{\sin^2\theta}{2} + \left(1 - e^{-2\gamma t}\right)\left[\sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right)\right]. \tag{3.102}$$

3.2.8 Spin-squeezing

We want to evaluate the spin-squezing paramter. We start from the mean spin direction (MSD) of the post-click states generated from $|\text{CSS},\theta\rangle$. We define the collective spin operators (for $N = 2$)

$$\hat{J}_k \equiv \frac{1}{2}\left(\sigma_i^{(1)} + \sigma_i^{(2)}\right), \qquad i \in \{x, y, z\}, \tag{3.103}$$

and the mean spin direction as

$$\mathbf{n}_0 \equiv \frac{\langle\hat{\mathbf{J}}\rangle}{\left|\langle\hat{\mathbf{J}}\rangle\right|}, \qquad \langle\hat{\mathbf{J}}\rangle \equiv \left(\left\langle\hat{J}_x\right\rangle, \left\langle\hat{J}_y\right\rangle, \left\langle\hat{J}_z\right\rangle\right). \tag{3.104}$$

For $N = 2$, the Kitagawa–Ueda parameter is

$$\xi^2_{KU} \equiv \frac{4}{N} \min_{\mathbf{n}_\perp} (\Delta \hat{J}_{\mathbf{n}_\perp})^2 = 2 \min_{\mathbf{n}_\perp} (\Delta \hat{J}_{\mathbf{n}_\perp})^2, \qquad \hat{J}_{\mathbf{n}} \equiv \mathbf{n} \cdot \hat{\mathbf{J}}, \tag{3.105}$$

where the minimization is taken over directions orthogonal to the mean spin direction (MSD). If the MSD is not defined, i.e. $\langle \hat{\mathbf{J}} \rangle = \mathbf{0}$, we adopt the natural extension given by minimizing over all directions $\mathbf{n}$.
We start from the trajectory decomposition. Along the no-click branch, the normalized conditional state remains equal to the initial CSS, hence

$$\xi^2_{KU}(\text{no-click}) = 1. \tag{3.106}$$

This follows from the defining property of CSS states, for which the minimal transverse variance equals $N/4$, i.e. $\min_{\mathbf{n}_\perp} (\Delta \hat{J}_{\mathbf{n}_\perp})^2 = N/4 = 1/2$ when $N = 2$.
We now compute ξ^2_{KU} for the post-click states generated from $|\text{CSS},\theta\rangle$. We use the notation introduced in Eq. (3.50):

$$\left|\tilde{\psi}_{\pm,\theta}\right\rangle = \frac{\sin^2\left(\frac{\theta}{2}\right)|11\rangle \pm \cos^2\left(\frac{\theta}{2}\right)|00\rangle}{\sqrt{\sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right)}}, \qquad |\psi_\pm\rangle = \frac{|10\rangle \pm |01\rangle}{\sqrt{2}}. \tag{3.107}$$

We now consider the odd-parity Bell states, related to the $\hat{C}_-$ channel. Since every Bell state has vanishing expectation value for all spin components,$\mathbf{n}_0$ is not defined for any post-click state. Therefore we have to minimize the variance all over the directions, and it is known that it is 0. Hence the odd-parity click channel yields $\xi^2_{KU} = 0$, independently of θ, and subsequent clicks do not change this value.

We start from the even-parity post-click family $\left|\tilde{\psi}_{\pm,\theta}\right\rangle$ as defined in (3.49). First, we compute $\left\langle \hat{J}_z \right\rangle$, by setting

$$A \equiv \sin^2\left(\frac{\theta}{2}\right), \qquad B \equiv \cos^2\left(\frac{\theta}{2}\right), \qquad \mathcal{N} \equiv A^2 + B^2, \tag{3.108}$$

we obtain

$$\left\langle \hat{J}_z \right\rangle_{\tilde{\psi}_{\pm,\theta}} = \left\langle \tilde{\psi}_{\pm,\theta} \right| \hat{J}_z \left| \tilde{\psi}_{\pm,\theta} \right\rangle = \frac{1}{\mathcal{N}} \left(A^2 \langle 11| \hat{J}_z |11\rangle + B^2 \langle 00| \hat{J}_z |00\rangle \right) = \frac{A^2 - B^2}{A^2 + B^2}. \tag{3.109}$$

Since $A^2 - B^2 = \sin^4(\theta/2) - \cos^4(\theta/2) = \sin^2(\theta/2) - \cos^2(\theta/2) = -\cos\theta$, we can rewrite Eq. (3.109) as

$$\boxed{\left\langle \hat{J}_z \right\rangle_{\tilde{\psi}_{\pm,\theta}} = \frac{-\cos\theta}{\sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right)}.} \tag{3.110}$$

We now compute $\left\langle \hat{J}_x \right\rangle$ and $\left\langle \hat{J}_y \right\rangle$. They both map the even subspace span$\{|00\rangle, |11\rangle\}$ into the odd subspace span$\{|01\rangle, |10\rangle\}$ and viceversa. Since our state has support only on the even sector, both expectation values vanish. Hence the MSD is along the z axis whenever $\cos\theta \neq 0$:

$$\boxed{\mathbf{n}_0(\tilde{\psi}_{\pm,\theta}) = \text{sgn}(-\cos\theta)\,\hat{\mathbf{z}}, \qquad \text{for } \theta \neq \frac{\pi}{2}.} \tag{3.111}$$

At $\theta = \pi/2$ one has $\left\langle \hat{J}_z \right\rangle = 0$ and therefore $\left| \langle \hat{\mathbf{J}} \rangle \right| = 0$, so $\mathbf{n}_0$ is not defined.
In the xy plane we introduce the rotated operator

$$\hat{J}_\alpha = \cos\alpha\, \hat{J}_x + \sin\alpha\, \hat{J}_y, \qquad \alpha \in [0, 2\pi). \tag{3.112}$$

Its variance can be written in terms of the 2×2 covariance matrix in the (x, y) sector. Explicitly,

$$(\Delta J_\alpha)^2 = \cos^2\alpha\, (\Delta J_x)^2 + \sin^2\alpha\, (\Delta J_y)^2 + \sin\alpha\cos\alpha \left(\frac{1}{2} \left\langle \{\hat{J}_x, \hat{J}_y\} \right\rangle - \left\langle \hat{J}_x \right\rangle \left\langle \hat{J}_y \right\rangle - \left\langle \hat{J}_y \right\rangle \left\langle \hat{J}_x \right\rangle \right), \tag{3.113}$$

that is

$$(\Delta J_\alpha)^2 = \cos^2\alpha\, (\Delta J_x)^2 + \sin^2\alpha\, (\Delta J_y)^2 + \sin\alpha\cos\alpha \left(\frac{1}{2} \left\langle \{\hat{J}_x, \hat{J}_y\} \right\rangle - 2 \left\langle \hat{J}_x \right\rangle \left\langle \hat{J}_y \right\rangle \right). \tag{3.114}$$

In general, the minimization over α corresponds to taking the smallest eigenvalue of the transverse covariance matrix. By parity considerations one has $\left\langle \hat{J}_x \right\rangle = \left\langle \hat{J}_y \right\rangle = 0$ on each post-click branch. Hence Eq. (3.114) reduces to

$$(\Delta J_\alpha)^2 = \cos^2\alpha\, (\Delta J_x)^2 + \sin^2\alpha\, (\Delta J_y)^2. \tag{3.115}$$

The minimum is then attained at one of the axes:

$$\min_\alpha (\Delta J_\alpha)^2 = \min\{(\Delta J_x)^2, (\Delta J_y)^2\}. \tag{3.116}$$

Therefore, for the present post-click states it is sufficient to compare $(\Delta J_x)^2$ and $(\Delta J_y)^2$, without scanning all transverse directions.
We use the identities

$$\hat{J}_x^2 = \frac{1}{4}\Big[(\sigma_x^{(1)})^2 + (\sigma_x^{(2)})^2 + 2\sigma_x^{(1)}\sigma_x^{(2)} \Big] = \frac{1}{2}\Big(\mathbb{1} + \sigma_x^{(1)}\sigma_x^{(2)} \Big), \tag{3.117}$$

and similarly

$$\hat{J}_y^2 = \frac{1}{2}\Big(\mathbb{1} + \sigma_y^{(1)}\sigma_y^{(2)} \Big). \tag{3.118}$$

Since $\left| \tilde{\psi}_{\pm,\theta} \right\rangle$ is supported on $\mathrm{span}\{|00\rangle, |11\rangle\}$, we only need the action of $\sigma_x^{(1)}\sigma_x^{(2)}$ and $\sigma_y^{(1)}\sigma_y^{(2)}$ on that subspace. One has

$$\sigma_x^{(1)}\sigma_x^{(2)} |00\rangle = |11\rangle, \qquad \sigma_x^{(1)}\sigma_x^{(2)} |11\rangle = |00\rangle, \tag{3.119}$$

and

$$\sigma_y^{(1)}\sigma_y^{(2)} |00\rangle = -|11\rangle, \qquad \sigma_y^{(1)}\sigma_y^{(2)} |11\rangle = -|00\rangle. \tag{3.120}$$

Hence

$$\langle \sigma_x^{(1)}\sigma_x^{(2)} \rangle_{\tilde{\psi}_{\pm,\theta}} = \frac{\pm 2AB}{P_+}, \qquad \langle \sigma_y^{(1)}\sigma_y^{(2)} \rangle_{\tilde{\psi}_{\pm,\theta}} = \frac{\mp 2AB}{P_+}. \tag{3.121}$$

Since $\langle \hat{J}_x \rangle = \langle \hat{J}_y \rangle = 0$, we obtain

$$\Delta \hat{J}_x^2(\tilde{\psi}_{\pm,\theta}) = \langle \hat{J}_x^2 \rangle = \frac{1}{2}\left(1 \pm \frac{2AB}{P_+} \right) = \frac{(A \pm B)^2}{2P_+}, \tag{3.122}$$

and

$$\Delta \hat{J}_y^2(\tilde{\psi}_{\pm,\theta}) = \langle \hat{J}_y^2 \rangle = \frac{1}{2}\left(1 \mp \frac{2AB}{P_+}\right) = \frac{(A \mp B)^2}{2P_+}. \tag{3.123}$$

Therefore the minimal transverse variance is the same for both $\left|\tilde{\psi}_{-,\theta}\right\rangle$ and $\left|\tilde{\psi}_{+,\theta}\right\rangle$, because it is always given by the smaller between $(A+B)^2$ and $(A-B)^2$. Since $A, B \geq 0$, one has $(A-B)^2 \leq (A+B)^2$, hence

$$\min_{\mathbf{n}_\perp} (\Delta \hat{J}_{\mathbf{n}_\perp})^2_{\tilde{\psi}_{\pm,\theta}} = \frac{(A-B)^2}{2P_+}. \tag{3.124}$$

For $N = 2$ this yields

$$\xi^2_{KU}(\tilde{\psi}_{\pm,\theta}) = 2 \cdot \frac{(A-B)^2}{2P_+} = \frac{(A-B)^2}{P_+}. \tag{3.125}$$

Using $A - B = \sin^2(\theta/2) - \cos^2(\theta/2) = -\cos\theta$ and $P_+ = \sin^4(\theta/2) + \cos^4(\theta/2)$, we obtain the compact form

$$\xi^2_{KU}(\tilde{\psi}_{\pm,\theta}) = \frac{\cos^2\theta}{\sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right)}. \tag{3.126}$$

We now compute the ensemble-averaged spin squeezing, following the same splitting used previously. We write

$$\mathbb{E}\left[\xi^2_{KU}(t)\right] = \mathbb{E}\left[\xi^2_{KU}(t) \,|\, \text{no-click}\right] P_0(t) + \mathbb{E}\left[\xi^2_{KU}(t) \,|\, \text{click}\right] P_{\text{click}}(t), \qquad P_{\text{click}}(t) = 1 - P_0(t). \tag{3.127}$$

In the present dynamics the no-click branch yields $\xi^2_{KU} = 1$, cf. Eq. (3.106). Moreover, a click projects either onto the even sector with probability P_+ or onto the odd sector with probability P_-, where

$$P_+ = \sin^4\left(\frac{\theta}{2}\right) + \cos^4\left(\frac{\theta}{2}\right), \qquad P_- = \frac{1}{2}\sin^2\theta, \qquad P_+ + P_- = 1. \tag{3.128}$$

In the odd sector we have $\xi^2_{KU} = 0$ while in the even sector we have Eq. (3.126). Therefore

$$\mathbb{E}\left[\xi^2_{KU}(t) \,|\, \text{click}\right] = P_+\, \xi^2_{KU}(\tilde{\psi}_{\pm,\theta}) + P_- \cdot 0 = P_+ \frac{\cos^2\theta}{P_+} = \cos^2\theta. \tag{3.129}$$

Finally,

$$\boxed{\mathbb{E}\left[\xi^2_{KU}(t)\right] = P_0(t) \cdot 1 + (1 - P_0(t))\cos^2\theta.} \tag{3.130}$$

Using the same no-click probability as in the previous sections, $P_0(t) = e^{-2\gamma t}$, we obtain

$$\boxed{\mathbb{E}\left[\xi^2_{KU}(t)\right] = e^{-2\gamma t} + (1 - e^{-2\gamma t})\cos^2\theta.} \tag{3.131}$$

For the special case $\theta = \pi/2$, i.e. $|\text{CSS}, \theta\rangle = |{+}{+}\rangle$, one has $\cos^2\theta = 0$ and

$$\mathbb{E}\left[\xi^2_{KU}(t)\right] = e^{-2\gamma t}, \tag{3.132}$$

which decays to zero as $t \to \infty$, consistently with the fact that after the first click the state is confined to Bell states with vanishing minimal collective-spin variance.

3.3 Homodyne detection

We now consider a diffusive unravelling of the same Markovian dynamics, obtained via homodyne detection. In this scheme the environment is continuously monitored and the measurement record is a real-valued photocurrent. The conditional state evolves stochastically, yet its ensemble average reproduces the GKSL master equation.
We start from a generic set of collapse operators $\{\hat{C}_k\}$. We denote by $\rho^{(c)}(t)$ the normalized conditional state. We introduce independent Wiener increments $\{dW_k(t)\}$ such that

$$\mathbb{E}[dW_k(t)] = 0, \qquad dW_k(t)\, dW_\ell(t) = \delta_{k\ell}\, dt. \tag{3.133}$$

The detector efficiency is $\eta \in [0, 1]$. The local-oscillator (LO) phase θ selects which field quadrature is measured.
For each monitored channel k, the infinitesimal homodyne record can be written as

$$dY_k(t) = \sqrt{\eta}\, \left\langle e^{-i\theta}\hat{C}_k + e^{i\theta}\hat{C}_k^\dagger \right\rangle_t dt + dW_k(t). \tag{3.134}$$

The innovation process is therefore $dW_k(t) = dY_k(t) - \sqrt{\eta}\, \left\langle e^{-i\theta}\hat{C}_k + e^{i\theta}\hat{C}_k^\dagger \right\rangle_t dt$.
The corresponding Itô stochastic master equation (SME) reads

$$d\rho^{(c)} = -i[\hat{H}, \rho^{(c)}]\, dt + \sum_k \mathcal{D}[\hat{C}_k]\rho^{(c)}\, dt + \sqrt{\eta} \sum_k \mathcal{H}\left[e^{i\theta}\hat{C}_k\right] \rho^{(c)}\, dW_k(t), \tag{3.135}$$

where

$$\mathcal{H}[\hat{A}]\rho \equiv \hat{A}\rho + \rho\hat{A}^\dagger - \mathrm{Tr}\left[(\hat{A} + \hat{A}^\dagger)\rho\right] \rho. \tag{3.136}$$

By construction, the average $\mathbb{E}[\, d\rho^{(c)}\,]$ reproduces the unconditional GKSL equation.
We now specialize to our two-qubit dephasing model. We use the collective jump operators introduced in Eq. (**??**),

$$\hat{C}_\pm = \sqrt{\gamma}\, \frac{\sigma_z^{(1)} \pm \sigma_z^{(2)}}{\sqrt{2}}. \tag{3.137}$$

We monitor the two output channels associated with $\hat{C}_+$ and $\hat{C}_-$. The homodyne SME becomes

$$\begin{aligned} d\rho^{(c)} &= -i[\hat{H}, \rho^{(c)}]\, dt + \mathcal{D}[\hat{C}_+]\rho^{(c)}\, dt + \mathcal{D}[\hat{C}_-]\rho^{(c)}\, dt \\ &\quad + \sqrt{\eta}\, \mathcal{H}\left[e^{i\theta}\hat{C}_+\right] \rho^{(c)}\, dW_+(t) + \sqrt{\eta}\, \mathcal{H}\left[e^{i\theta}\hat{C}_-\right] \rho^{(c)}\, dW_-(t), \end{aligned} \tag{3.138}$$

with measurement records

$$\begin{aligned} dY_+(t) &= \sqrt{\eta}\, \left\langle e^{-i\theta}\hat{C}_+ + e^{i\theta}\hat{C}_+^\dagger \right\rangle_t dt + dW_+(t), \\ dY_-(t) &= \sqrt{\eta}\, \left\langle e^{-i\theta}\hat{C}_- + e^{i\theta}\hat{C}_-^\dagger \right\rangle_t dt + dW_-(t). \end{aligned} \tag{3.139}$$

A crucial simplification occurs in our model because $\hat{C}_\pm$ are Hermitian: $\hat{C}_\pm^\dagger = \hat{C}_\pm$. Then Eq. (3.134) reduces to

$$dY_\pm(t) = 2\sqrt{\eta}\, \cos\theta\, \left\langle \hat{C}_\pm \right\rangle_t dt + dW_\pm(t). \tag{3.140}$$

Therefore, the amount of information in the record is controlled by $\cos\theta$.
If $\theta = 0$, the record depends on $\left\langle \hat{C}_\pm \right\rangle_t$. The SME is nonlinear through the innovation term $\mathcal{H}[\hat{C}_\pm]\rho^{(c)}$. This corresponds to an informational measurement of the collective observables $\hat{C}_\pm$.
If $\theta = \pi/2$, one has $\cos\theta = 0$ and hence

$$dY_\pm(t) = dW_\pm(t). \tag{3.141}$$

The record contains only noise and no information about the system state. In this case the innovation superoperator becomes a commutator. Indeed, for Hermitian $\hat{C}$ one has

$$\mathcal{H}[i\hat{C}]\rho = i[\hat{C}, \rho], \tag{3.142}$$

since $(i\hat{C})^\dagger = -i\hat{C}$ and $\mathrm{Tr}\left[(i\hat{C} - i\hat{C})\rho\right] = 0$. Therefore, at $\theta = \pi/2$ Eq. (3.138) becomes

$$\begin{aligned} d\rho^{(c)} &= -i[\hat{H}, \rho^{(c)}]\, dt + \mathcal{D}[\hat{C}_+]\rho^{(c)}\, dt + \mathcal{D}[\hat{C}_-]\rho^{(c)}\, dt \\ &\quad + i\sqrt{\eta}\, [\hat{C}_+, \rho^{(c)}]\, dW_+(t) + i\sqrt{\eta}\, [\hat{C}_-, \rho^{(c)}]\, dW_-(t). \end{aligned} \tag{3.143}$$

This is a random-unitary backaction driven by Wiener noise, with a state-independent measurement record. It is the diffusive analogue of the photodetection case with unitary jumps: monitoring does not reveal information, yet it induces stochastic phase kicks.
Equation (3.138) provides the starting point for the analysis of homodyne-based entangling unravellings in the two-qubit dephasing model, by selecting the monitored quadrature through θ and by exploiting collective channels $\hat{C}_\pm$.
In our model $\hat{C}_\pm^\dagger = \hat{C}_\pm$. Hence

$$\hat{A}_\pm + \hat{A}_\pm^\dagger = \left(e^{i\theta} + e^{-i\theta}\right) \hat{C}_\pm = 2\cos\theta\, \hat{C}_\pm. \tag{3.144}$$

This yields to the following decomposition:

$$\boxed{\mathcal{H}[e^{i\theta}\hat{C}]\rho = \cos\theta\Big(\{\hat{C}, \rho\} - 2\left\langle \hat{C} \right\rangle \rho\Big) + i\sin\theta\, [\hat{C}, \rho], \qquad (\hat{C}^\dagger = \hat{C}).} \tag{3.145}$$

Two limits are particularly relevant.
If $\theta = 0$, one obtains the informational (nonlinear) term

$$\mathcal{H}[\hat{C}]\rho = \{\hat{C}, \rho\} - 2\left\langle \hat{C} \right\rangle \rho. \tag{3.146}$$

If $\theta = \pi/2$, one obtains a purely commutator form

$$\mathcal{H}[i\hat{C}]\rho = i[\hat{C}, \rho], \tag{3.147}$$

which is linear in ρ.
If $[\rho, \Pi_z] = 0$ at some time, then

$$[\mathcal{D}[\hat{C}_\pm]\rho, \Pi_z] = 0, \qquad [\mathcal{H}[e^{i\theta}\hat{C}_\pm]\rho, \Pi_z] = 0. \tag{3.148}$$

Hence the homodyne SME preserves the block structure in the even/odd parity sectors.

3.3.1 No-information configurations in homodyne detection

We now discuss the notion of no-information unraveling in the homodyne scheme for the two-qubit case. As already shown above, after the unitary transformation of the local dephasing channels, the monitored operators are

$$\hat{C}_{\pm} = \sqrt{\frac{\gamma}{2}}\left(\sigma_z^{(1)} \pm \sigma_z^{(2)}\right). \tag{3.149}$$

Since $\hat{C}_{\pm} = \hat{C}_{\pm}^{\dagger}$, the corresponding homodyne currents are

$$dY_{\pm}(t) = \sqrt{\eta}\left\langle e^{-i\phi_{\pm}}\hat{C}_{\pm} + e^{i\phi_{\pm}}\hat{C}_{\pm}^{\dagger}\right\rangle_t dt + dW_{\pm}(t), \tag{3.150}$$

which simplify to

$$dY_{\pm}(t) = 2\sqrt{\eta}\,\cos\phi_{\pm}\left\langle \hat{C}_{\pm}\right\rangle_t dt + dW_{\pm}(t). \tag{3.151}$$

Equation (3.151) gives a direct operational meaning to the informational content of the homodyne signal. Indeed, the state-dependent part of the current is proportional to $\cos\phi_{\pm}$. Hence, if $\phi_{\pm} = 0$, the corresponding current contains a deterministic drift proportional to $\left\langle \hat{C}_{\pm}\right\rangle_t$ and is therefore informative. On the contrary, if $\phi_{\pm} = \pi/2$, one has $\cos\phi_{\pm} = 0$, and the current reduces to

$$dY_{\pm}(t) = dW_{\pm}(t). \tag{3.152}$$

In this case, the record is pure noise and carries no information on the conditional state. This is precisely the no-information condition in the homodyne scheme.
The same conclusion can be obtained directly from the stochastic term in the master equation. For $\phi_{\pm} = \pi/2$, one finds

$$\mathcal{H}\left[-i\hat{C}_{\pm}\right]\rho = -i\hat{C}_{\pm}\rho + i\rho\hat{C}_{\pm} = -i[\hat{C}_{\pm}, \rho]. \tag{3.153}$$

Therefore, in the no-information case the stochastic contribution becomes purely commutator-like. The corresponding backaction is then purely random-unitary. The state is continuously perturbed by the measurement, but the observed current does not reveal any information on the conditional expectation values of the monitored operators.
For $N = 2$, and restricting to the four phase choices

$$(\phi_+, \phi_-) \in \left\{(0,0), \left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, 0\right), \left(\frac{\pi}{2}, \frac{\pi}{2}\right)\right\}, \tag{3.154}$$

one obtains four distinct informational configurations.
We first consider the case $(\phi_+, \phi_-) = (0, 0)$. In this configuration both currents are informative,

$$dY_+(t) = 2\sqrt{\eta}\left\langle \hat{C}_+\right\rangle_t dt + dW_+(t), \qquad dY_-(t) = 2\sqrt{\eta}\left\langle \hat{C}_-\right\rangle_t dt + dW_-(t). \tag{3.155}$$

Hence, both detectors extract information on the conditional state. The first current monitors the collective combination $\sigma_z^{(1)} + \sigma_z^{(2)}$, while the second one monitors the difference $\sigma_z^{(1)} - \sigma_z^{(2)}$.

We next consider $(\phi_+, \phi_-) = (0, \pi/2)$. In this case, the $+$ current remains informative, while the $-$ current becomes pure noise:

$$dY_+(t) = 2\sqrt{\eta}\, \left\langle \hat{C}_+ \right\rangle_t dt + dW_+(t), \qquad dY_-(t) = dW_-(t). \tag{3.156}$$

Therefore, the collective channel is continuously monitored and provides state-dependent information, whereas the orthogonal channel contributes only through stochastic backaction.

The opposite situation is realized for $(\phi_+, \phi_-) = (\pi/2, 0)$. One now has

$$dY_+(t) = dW_+(t), \qquad dY_-(t) = 2\sqrt{\eta}\, \left\langle \hat{C}_- \right\rangle_t dt + dW_-(t). \tag{3.157}$$

Hence, the collective channel is no-information, while the difference channel remains informative. The measurement then extracts information only on the antisymmetric combination of the two local operators.

Finally, for $(\phi_+, \phi_-) = (\pi/2, \pi/2)$ both currents reduce to pure Wiener noise,

$$dY_+(t) = dW_+(t), \qquad dY_-(t) = dW_-(t). \tag{3.158}$$

This is the fully no-information configuration. The system is still affected by stochastic measurement backaction, but the measurement records carry no state-dependent drift.

The four phase configurations can thus be classified according to the number of informative currents. The case $(0, 0)$ is fully informative. The cases $(0, \pi/2)$ and $(\pi/2, 0)$ are partially informative, since only one of the two channels carries information. Finally, the case $(\pi/2, \pi/2)$ is fully no-information. This makes the physical meaning of the no-information condition particularly transparent. In homodyne detection, no-information does not mean that the monitoring is absent. Rather, it means that the selected quadrature does not contain any state-dependent average signal. As a consequence, the detector still induces stochastic backaction on the system, but the observed current does not reveal any information on the conditional quantum state.

3.3.2 Gram–Schmidt construction of the unraveling unitary

As discussed above, the unconditional master equation is invariant under unitary transformations acting on the vector of collapse operators. Therefore, once the local dephasing channels $\{\sigma_z^{(k)}\}_{k=1}^N$ are fixed, one can generate different unravelings by introducing an arbitrary unitary matrix $U \in \mathbb{C}^{N\times N}$ and defining

$$\hat{C}_i^{(U)} = \sqrt{\gamma} \sum_{j=1}^{N} U_{ij}\, \sigma_z^{(j)}, \qquad i = 1, \dots, N. \tag{3.159}$$

By construction, all these choices lead to the same unconditional GKSL dynamics after averaging over the measurement outcomes. However, they generally induce different conditional evolutions. For this reason, the choice of U becomes a relevant degree of freedom when searching for unravelings that enhance entanglement generation or spin squeezing.

In the present work we impose that the first monitored channel coincides with the collective spin component along the z direction. This amounts to fixing the first row of U as

$$u_1 = \frac{1}{\sqrt{N}}(1, 1, \ldots, 1), \tag{3.160}$$

so that

$$\hat{C}_1^{(U)} = \sqrt{\frac{\gamma}{N}} \sum_{j=1}^{N} \sigma_z^{(j)}. \tag{3.161}$$

The remaining $N-1$ rows must then be chosen in such a way that U remains unitary. Equivalently, the rows of U must form an orthonormal basis of $\mathbb{C}^N$.
For $N = 2$, this construction is essentially trivial. Once the first row is fixed to

$$u_1 = \frac{1}{\sqrt{2}}(1, 1), \tag{3.162}$$

the second row must be orthogonal to u_1 and normalized. Therefore one immediately obtains

$$u_2 = \frac{1}{\sqrt{2}}(1, -1), \tag{3.163}$$

which corresponds to the difference channel already discussed in the two-qubit case. Hence, for $N = 2$ the freedom in the choice of the unraveling unitary is essentially exhausted by fixing the collective channel.
For $N > 2$, the construction is no longer unique. A natural way to generate a unitary matrix with prescribed first row is to use the Gram–Schmidt orthonormalization procedure. In our case, we start from the ordered family

$$v_1 = \frac{1}{\sqrt{N}}(1, 1, \ldots, 1), \qquad v_2 = e_1, \qquad v_3 = e_2, \qquad \ldots, \qquad v_N = e_{N-1}, \tag{3.164}$$

where $\{e_j\}$ denotes the canonical basis of $\mathbb{C}^N$. Explicitly,

$$e_1 = (1, 0, \ldots, 0), \quad e_2 = (0, 1, \ldots, 0), \quad \ldots, \quad e_N = (0, 0, \ldots, 1). \tag{3.165}$$

The first vector already encodes the desired collective channel, while the remaining vectors are introduced only as auxiliary vectors and are then orthonormalized with respect to v_1 and to each other.
We now briefly recall the Gram–Schmidt construction. Given a linearly independent family $\{v_m\}_{m=1}^N$, one defines

$$u_1 = \frac{v_1}{\|v_1\|}, \tag{3.166}$$

and recursively, for $m \geq 2$,

$$w_m = v_m - \sum_{j=1}^{m-1} \langle u_j | v_m \rangle \, u_j. \tag{3.167}$$

If $w_m \neq 0$, one then normalizes it and sets

$$u_m = \frac{w_m}{\|w_m\|}. \tag{3.168}$$

By construction, the family $\{u_m\}_{m=1}^N$ is orthonormal,

$$\langle u_i | u_j \rangle = \delta_{ij}, \tag{3.169}$$

and can therefore be used as the set of rows of a unitary matrix,

$$U = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_N \end{pmatrix}. \tag{3.170}$$

In this way, Eq. (4.13) defines a valid unraveling of the same unconditional dynamics, with the first channel fixed to the collective one.
Let us now make this construction explicit for $N = 3$. We start from

$$v_1 = \frac{1}{\sqrt{3}}(1, 1, 1), \qquad v_2 = (1, 0, 0), \qquad v_3 = (0, 1, 0). \tag{3.171}$$

Since v_1 is already normalized, we simply set

$$u_1 = v_1 = \frac{1}{\sqrt{3}}(1, 1, 1). \tag{3.172}$$

We then construct the second orthogonal vector by subtracting from v_2 its projection along u_1:

$$w_2 = v_2 - \langle u_1 | v_2 \rangle \; u_1. \tag{3.173}$$

Since

$$\langle u_1 | v_2 \rangle = \frac{1}{\sqrt{3}}, \tag{3.174}$$

one finds

$$w_2 = (1, 0, 0) - \frac{1}{3}(1, 1, 1) = \left(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3} \right). \tag{3.175}$$

Its norm is

$$\|w_2\|^2 = \frac{4}{9} + \frac{1}{9} + \frac{1}{9} = \frac{2}{3}, \tag{3.176}$$

hence

$$u_2 = \frac{w_2}{\|w_2\|} = \frac{1}{\sqrt{6}}(2, -1, -1). \tag{3.177}$$

We proceed analogously for the third vector:

$$w_3 = v_3 - \langle u_1 | v_3 \rangle \; u_1 - \langle u_2 | v_3 \rangle \; u_2. \tag{3.178}$$

The relevant overlaps are

$$\langle u_1 | v_3 \rangle = \frac{1}{\sqrt{3}}, \qquad \langle u_2 | v_3 \rangle = -\frac{1}{\sqrt{6}}. \tag{3.179}$$

Substituting the explicit expressions of u_1 and u_2, one obtains

$$w_3 = (0, 1, 0) - \frac{1}{3}(1, 1, 1) + \frac{1}{6}(2, -1, -1) = \left(0, \frac{1}{2}, -\frac{1}{2} \right). \tag{3.180}$$

Its norm is

$$\|w_3\|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}, \tag{3.181}$$

thus

$$u_3 = \frac{w_3}{\|w_3\|} = \frac{1}{\sqrt{2}}(0, 1, -1). \tag{3.182}$$

We finally obtain the explicit unitary matrix

$$U_{N=3} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}. \tag{3.183}$$

The first row generates the collective channel, while the remaining two rows define orthogonal non-collective combinations of the local dephasing operators.
The extension to arbitrary dimension is now straightforward. Once the first row is fixed to the collective vector in Eq. (3.160), one applies the Gram–Schmidt procedure to a convenient set of $N-1$ auxiliary vectors, for instance the canonical basis vectors. This yields an orthonormal family

$$\left\{ \frac{1}{\sqrt{N}}(1, \ldots, 1),\, u_2, \ldots, u_N \right\}, \tag{3.184}$$

which defines a unitary matrix U and therefore a valid family of transformed collapse operators. In this way one obtains a systematic parametrization of unravelings with fixed collective first channel. The remaining freedom, encoded in the choice of the orthogonal complement to the collective direction, can then be exploited in order to search for the unraveling that optimizes the generation of metrologically useful spin squeezing.

4
Conclusions

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Appendix

Derivation of the minimal transverse variance for MSD along z

In this appendix, we derive Eq. (1.184). We assume that the MSD points along the z axis. Therefore,

$$\langle \hat{J}_x \rangle = \langle \hat{J}_y \rangle = 0, \qquad \langle \hat{J}_z \rangle = |\langle \hat{\mathbf{J}} \rangle|. \tag{4.1}$$

A generic spin component in the plane orthogonal to the mean spin direction may then be written as

$$\hat{J}_\perp(\phi) = \hat{J}_x \cos\phi + \hat{J}_y \sin\phi. \tag{4.2}$$

Since (4.1) holds, one has $(\Delta \hat{J}_\perp)^2 = \langle \hat{J}_\perp(\phi)^2 \rangle$.
We now expand the square of Eq. (4.2):

$$\hat{J}_\perp(\phi)^2 = \hat{J}_x^2 \cos^2\phi + \hat{J}_y^2 \sin^2\phi + \left(\hat{J}_x\hat{J}_y + \hat{J}_y\hat{J}_x\right)\cos\phi\sin\phi. \tag{4.3}$$

Taking the expectation value, we obtain

$$(\Delta \hat{J}_\perp)^2 = \langle \hat{J}_x^2 \rangle \cos^2\phi + \langle \hat{J}_y^2 \rangle \sin^2\phi + \langle \{\hat{J}_x, \hat{J}_y\} \rangle \cos\phi\sin\phi, \tag{4.4}$$

where $\{\hat{J}_x, \hat{J}_y\} = \hat{J}_x\hat{J}_y + \hat{J}_y\hat{J}_x$ is the usual definition of the anticommutator.
Using the trigonometric identities

$$\cos^2\phi = \frac{1+\cos 2\phi}{2}, \qquad \sin^2\phi = \frac{1-\cos 2\phi}{2}, \qquad \cos\phi\sin\phi = \frac{\sin 2\phi}{2}, \tag{4.5}$$

Eq. (4.4) becomes

$$(\Delta \hat{J}_\perp)^2 = \frac{1}{2}\langle \hat{J}_x^2 + \hat{J}_y^2 \rangle + \frac{1}{2}\langle \hat{J}_x^2 - \hat{J}_y^2 \rangle \cos 2\phi + \frac{1}{2}\langle \{\hat{J}_x, \hat{J}_y\} \rangle \sin 2\phi. \tag{4.6}$$

For simplicity, let us define

$$A = \frac{1}{2}\langle \hat{J}_x^2 + \hat{J}_y^2 \rangle, \qquad B = \frac{1}{2}\langle \hat{J}_x^2 - \hat{J}_y^2 \rangle, \qquad C = \frac{1}{2}\langle \{\hat{J}_x, \hat{J}_y\} \rangle. \tag{4.7}$$

Then Eq. (4.6) reads

$$(\Delta \hat{J}_\perp)^2 = A + B\cos 2\phi + C\sin 2\phi. \tag{4.8}$$

At this stage, the minimization is given by a reparametrization. Indeed, one may write

$$B\cos 2\phi + C\sin 2\phi = \sqrt{B^2 + C^2}\,\cos(2\phi - \phi_0), \tag{4.9}$$

for a suitable angle ϕ_0. Therefore, the minimum value of Eq. (4.8) is obtained when

$$\cos(2\phi - \phi_0) = -1, \tag{4.10}$$

and is equal to

$$(\Delta \hat{J}_\perp)^2_{\text{min}} = A - \sqrt{B^2 + C^2}. \tag{4.11}$$

By substituting the definitions in Eq. (4.7), one finally finds

$$(\Delta \hat{J}_\perp)^2_{\text{min}} = \frac{1}{2}\left[\langle \hat{J}_x^2 + \hat{J}_y^2 \rangle - \sqrt{\langle \hat{J}_x^2 - \hat{J}_y^2 \rangle^2 + \langle \{\hat{J}_x, \hat{J}_y\} \rangle^2}\right]. \tag{4.12}$$

This is gives us Eq. (1.184).
It is worth observing that this result is simply the smallest eigenvalue of the covariance matrix in the transverse plane. The assumption that the mean spin points along z reduces the minimization problem to the x-y plane, and this leads directly to the closed expression above.

Unitary invariance of Lindblad operators

Let U be an $n \times n$ unitary matrix acting on the vector of collapse operators $\{\hat{C}_i\}$. We prove the identity

$$\sum_{i=1}^{n} \mathcal{D}[\hat{C}_i]\rho = \sum_{i=1}^{n} \mathcal{D}\left[\sum_{j=1}^{n} U_{ij}\hat{C}_j\right]\rho, \tag{4.13}$$

where $\mathcal{D}[A]\rho = A\rho A^\dagger - \frac{1}{2}\{A^\dagger A, \rho\}$.
We expand the right-hand side:

$$\begin{aligned}\sum_i \mathcal{D}\left[\sum_j U_{ij}\hat{C}_j\right]\rho &= \sum_i \left[\left(\sum_j U_{ij}\hat{C}_j\right)\rho\left(\sum_k U_{ik}\hat{C}_k\right)^\dagger - \frac{1}{2}\left\{\left(\sum_j U_{ij}\hat{C}_j\right)^\dagger\left(\sum_k U_{ik}\hat{C}_k\right), \rho\right\}\right] \\ &= \sum_i \left[\sum_{j,k} U_{ij}U_{ik}^*\,\hat{C}_j\rho\hat{C}_k^\dagger - \frac{1}{2}\left\{\sum_{j,k} U_{ij}^*U_{ik}\,\hat{C}_j^\dagger\hat{C}_k, \rho\right\}\right].\end{aligned} \tag{4.14}$$

We interchange the sums and group the coefficients:

$$\sum_i \mathcal{D}\left[\sum_j U_{ij}\hat{C}_j\right]\rho = \sum_{j,k}\left[\left(\sum_i U_{ij}U_{ik}^*\right)\hat{C}_j\rho\hat{C}_k^\dagger - \frac{1}{2}\left\{\left(\sum_i U_{ij}^*U_{ik}\right)\hat{C}_j^\dagger\hat{C}_k, \rho\right\}\right]. \tag{4.15}$$

Since U is unitary, $U^\dagger U = \mathbb{1}$, hence

$$\sum_i U_{ij}^*U_{ik} = \delta_{jk}, \qquad \sum_i U_{ij}U_{ik}^* = \delta_{jk}. \tag{4.16}$$

Substituting in Eq. (4.15) gives

$$\sum_i \mathcal{D}\left[\sum_j U_{ij}\hat{C}_j\right]\rho = \sum_j \left[\hat{C}_j\rho\hat{C}_j^\dagger - \frac{1}{2}\left\{\hat{C}_j^\dagger\hat{C}_j, \rho\right\}\right] = \sum_j \mathcal{D}[\hat{C}_j]\rho. \tag{4.17}$$

This proves Eq. (4.13).
Applying this invariance to our system, the Lindblad master equation can be equivalently written as

$$\begin{aligned} d\varrho &= -i[H,\varrho]\,dt + \gamma\,\mathcal{D}\big[\sigma_z^{(1)}\big]\,\varrho\,dt + \gamma\,\mathcal{D}\big[\sigma_z^{(2)}\big]\,\varrho\,dt \\ &= -i[H,\varrho]\,dt + \gamma\,\mathcal{D}\left[\frac{\sigma_z^{(1)} + \sigma_z^{(2)}}{\sqrt{2}}\right]\varrho\,dt + \gamma\,\mathcal{D}\left[\frac{\sigma_z^{(1)} - \sigma_z^{(2)}}{\sqrt{2}}\right]\varrho\,dt. \end{aligned} \tag{4.18}$$

The last equality follows by applying the Hadamard unitary

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{4.19}$$

to the operator vector $(\sigma_z^{(1)}, \sigma_z^{(2)})^T$.

CSS states reach Standard Quantum Limit2

We now show that coherent spin states reach the standard quantum limit.
For definiteness, we consider the coherent spin state polarized along the x direction, namely $|\mathrm{CSS}\rangle_x = |+\rangle^{\otimes N}$.
For a single qubit in the state $|+\rangle$, one has

$$\langle \sigma_x \rangle = 1, \qquad \langle \sigma_y \rangle = 0, \qquad \langle \sigma_z \rangle = 0. \tag{4.20}$$

and it follows immediately that

$$\langle J_x \rangle = N/2, \qquad \langle J_y \rangle = 0, \qquad \langle J_z \rangle = 0. \tag{4.21}$$

Therefore, the mean spin points along the x axis and has maximal length $N/2$.
We now evaluate the transverse fluctuations. Since $|\mathrm{CSS}\rangle_x$ is a product state, different qubits are uncorrelated. Hence, for $i \neq j$,

$$\left\langle \sigma_y^{(i)} \sigma_y^{(j)} \right\rangle = \left\langle \sigma_y^{(i)} \right\rangle \left\langle \sigma_y^{(j)} \right\rangle = 0, \tag{4.22}$$

and analogously for σ_z. Using $(\sigma_y^{(i)})^2 = \mathbb{1}$, one finds

$$\left\langle J_y^2 \right\rangle = \frac{1}{4} \sum_{i,j=1}^{N} \left\langle \sigma_y^{(i)} \sigma_y^{(j)} \right\rangle = \frac{1}{4} \sum_{i=1}^{N} \left\langle (\sigma_y^{(i)})^2 \right\rangle = \frac{N}{4}. \tag{4.23}$$

Since (4.21) this gives $(\Delta J_y)^2 = (\Delta J_z)^2 = N/4$, thus the transverse fluctuations are isotropic and equal to

$$(\Delta J_\perp)^2 = \frac{N}{4}. \tag{4.24}$$

Let us now consider the estimation of a small phase θ encoded through the unitary transformation

$$U(\theta) = e^{-i\theta J_z}. \tag{4.25}$$

A natural observable for phase readout is J_y. In the Heisenberg picture,

$$J_y(\theta) = U^\dagger(\theta) J_y U(\theta) = J_y \cos\theta + J_x \sin\theta. \tag{4.26}$$

Taking the expectation value one obtains

$$\langle J_y(\theta) \rangle = \langle J_y \rangle \cos\theta + \langle J_x \rangle \sin\theta = \frac{N}{2} \sin\theta. \tag{4.27}$$

Therefore,

$$\frac{d}{d\theta} \langle J_y(\theta) \rangle \bigg|_{\theta=0} = \frac{N}{2}. \tag{4.28}$$

Using the error-propagation formula at $\theta = 0$, together with Eqs. (4.28), one finds

$$\Delta\theta_{\mathrm{CSS}} = \frac{\sqrt{N}/2}{N/2} = \frac{1}{\sqrt{N}}. \tag{4.29}$$

This is precisely the standard quantum limit.
The same conclusion can be expressed in terms of the Wineland squeezing parameter. Since for a coherent spin state one has $\| \langle \mathbf{J} \rangle \| = N/2$ and $(\Delta J_{\perp})^2 = N/4$, it follows that

$$\xi_{\mathrm{W}}^2 = \frac{N(\Delta J_{\perp})^2}{\| \langle \mathbf{J} \rangle \|^2} = \frac{N(N/4)}{(N/2)^2} = 1. \tag{4.30}$$

Hence coherent spin states do not provide any metrological enhancement beyond the standard quantum limit. They exactly saturate it.
Finally, although the derivation above was carried out for the state polarized along x, the result is completely general. Any coherent spin state is obtained from $|\mathrm{CSS}\rangle_x$ by a collective rotation. Such a rotation changes the orientation of the mean spin, but it does not change its length nor the isotropic structure of the transverse fluctuations. Therefore, every coherent spin state reaches the standard quantum limit.

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